

**INTEGRATING ERC-20 ALLOWANCE EDGES INTO
PAGERANK FOR ENHANCED ON-CHAIN WALLET
REPUTATION SCORING**

by

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INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

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Manuscripts submitted from this thesis.

[M1] Kwon, T., Mnkandla, E., Koulla Moulla, D., & Attipoe, D. S. *Allowance edges or transfer edges? Evaluating on-chain wallet reputation out of window against degree baselines*. Submitted to *PLOS ONE*, 2 November 2026.

- [M2] Kwon, T., Mnkandla, E., Koulla Moulla, D., & Attipoe, D. S. *Coupling authorization and payment layers in a PageRank walk: A pre-specified, hash-gated out-of-window test on an Ethereum rollup*. Submitted to *IEEE Access*, 2 November 2026.

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Abstract

Decentralized finance (DeFi) protocols use on-chain wallet reputation to screen counterparties and set risk parameters. Adaptive Weighted PageRank (AWP) ranks wallets on token transfers, but a transfer records a payment, not trust. An ERC-20 allowance records an owner authorizing a spender to move its tokens. This thesis asks what allowance edges add to PageRank as a reputation score. EndorseRank is PageRank on the latest non-zero allowances, with AWP’s edge weights and uniform restarts; it ranks spenders, not the owners approving them. A coupled PageRank (C-PR) walks both edge types. All scores, raw degrees included, are computed for 5,521 GMX V2 traders on Arbitrum One and compared with Kendall’s τ and paired bootstrap intervals. Scores frozen on 28 February 2026 are tested on 1,335 spenders, and a June–August 2026 window was registered before extraction.

Out of window no PageRank ranks a label as well as the raw degree of its edge type: the approval count ranks later approvals at $\tau = 0.711$, EndorseRank at 0.394, and the transfer in-degree beats AWP on new senders. A closed form explains the allowance gap: on this one-hop ego-network sample EndorseRank reduces to an owner-normalized in-strength; multi-hop propagation is untested. C-PR meets neither its first nor its registered rule. In pre-specified post hoc analyses of later liquidation avoidance, a trader outcome built from neither edge type, the approval count ranked traders above the transfer count in both windows ($\Delta\tau = +0.150$ and $+0.153$, 97.5% intervals above zero); the PageRank versions are not resolved. Most approval-receiving traders are contract accounts, an unresolved confound, and the signal is lower on the profitable and non-losing shares: it ranks forced-liquidation avoidance, not trading success. EndorseRank solves about nine times faster than AWP, on a ninefold smaller graph.

The thesis contributes the allowance as a reputation edge, an out-of-window evaluation of every score against the counts it smooths, a closed form of EndorseRank on a depth-one graph, and a

coupled walk whose rules, fixed in advance, failed. A ten-address farm can lift its target to 7.7 times the cohort's highest EndorseRank, so no score should decide rewards without Sybil controls, and none measures creditworthiness.

Keywords— DeFi, EndorseRank, Adaptive Weighted PageRank, coupled PageRank, on-chain reputation, ERC-20 allowance, temporal holdout, pre-registration, Arbitrum One, Kendall's tau

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Abbreviations and Notation

The abbreviations and mathematical notation used throughout this research are listed below. Each symbol is also defined where it first appears in the main chapters.

Abbreviations

Table I: Abbreviations used in this research.

Abbreviation	Meaning
AWP	Adaptive Weighted PageRank
C-PR	Coupled PageRank (per-node mixture of the raw-amount allowance and transfer layers, weight λ)
DeFi	Decentralized Finance
DEX	Decentralized Exchange
EIP	Ethereum Improvement Proposal
ER	EndorseRank (in table headings)
ERC-20	Ethereum Request for Comments 20 (fungible token standard)
FC	Financial Cryptography and Data Security
PR	PageRank
RQ	Research Question
SoK	Systematization of Knowledge
S-PR	Endorsement-Seeded PageRank (transfer layer walked with restarts drawn from the allowance walk)
USENIX	USENIX Security Symposium

Mathematical notation

Table II: Mathematical notation used in this research.

Symbol	Meaning
G_{approve}	Directed endorsement graph from latest ERC-20 allowances (owner $\rightarrow$ spender)
G_{transfer}	Directed transfer graph (sender $\rightarrow$ receiver; AWP input)
$ V $	Number of nodes in a reputation graph
$ E $	Number of directed edges in a reputation graph
w_{ij}	Weight of the directed edge from node i to node j
o_i	Out-strength of node i , $o_i = \sum_j w_{ij}$ (written $o(u)$, or $o_A(u)$ and $o_T(u)$ per layer, in Chapter 3)
P	Transition matrix, $P_{ij} = w_{ij}/o_i$ for $o_i > 0$; a dangling row is uniform in Equation (2.1) and zero in Equation (3.2), where its mass is redistributed separately
G_{PR}	Google matrix $dP^\top + (1-d)\mathbf{1}\mathbf{1}^\top/ V $ (Equation (2.2))
$\mathbf{r}, r_i$	PageRank vector and its entry for node i ; $\mathbf{r}^{(t)}$ is the iterate after t power-iteration steps
π	Teleportation (restart) distribution: uniform for EndorseRank and C-PR, proportional to activeness X_u for AWP, and the normalized allowance-walk score for S-PR
d	PageRank damping factor (default 0.85); the teleport probability is $1-d$
ε	Convergence tolerance of power iteration (10^{-8} , on the $\ \cdot\ _1$ change between iterates)
I_{max}	Iteration cap of power iteration (300)
λ	Allowance-layer weight of the coupled operator C-PR ($\lambda = 1$ and $\lambda = 0$: one layer walked alone)
$P^{(\lambda)}$	Coupled transition matrix of C-PR with allowance-layer weight λ (Equation (3.9))
$\alpha_A(u), \alpha_T(u)$	Weights of the allowance row and the transfer row in the coupled transition row of wallet u
k	Logistic time-decay steepness
t_0	Logistic time-decay midpoint (days)
Δt	Age in days of a transfer or approval
$\sigma(\Delta t)$	Logistic time weight of a transfer (AWP) or latest approval (EndorseRank)
$V(z)$	Bounded value weight of AWP and EndorseRank, $2/(1+e^{-bz})-1$ with $b=1$
X_u	Activeness of address u , AWP's restart weight
n	Number of wallets in a correlation (cohort size: matched $n = 5,521$, spender $n = 1,335$)
N	Expanded wallet-pool size (scaling: $N = 99,080$)
ρ	Spearman's rank correlation coefficient
τ	Kendall's rank correlation coefficient; every reported value is τ_b
t_1, t_2	Holdout dates: score freeze t_1 (28 February 2026, 23:59:59 UTC) and end of the label window t_2 (31 May 2026)
$\mathcal{W}$	Set of wallets in the evaluation cohort
$s(w)$	Reputation score of wallet w under a given method
$p(w)$	Validation-proxy value for wallet w

1 Introduction and Research Problem

Decentralized Finance (DeFi) has transformed how capital is allocated on public blockchains: lending, borrowing, and leveraged trading can now run without traditional intermediaries ^[1]. Most credit-like transactions are still heavily collateralized, though, because protocols cannot draw on off-chain identity frameworks or existing credit data ^[2]. Under pseudonymity, the central issue is how to allocate capital efficiently among unknown parties.

This thesis asks what ERC-20 allowance edges add to PageRank as a measure of on-chain wallet reputation. The allowance edges come from the Approval events that approve and permit emit, and EndorseRank is PageRank computed on them with the edge weights of the transfer-based Adaptive Weighted PageRank (AWP) of Do, Do and Nguyen^[3]. Both are run on the same Arbitrum One wallets, with a coupled walk over both edge types and the raw degrees that the walks smooth, and validated by the rank-correlation method used with AWP^[3], extended with paired contrasts, raw-degree baselines and temporal holdouts. In this thesis, an on-chain reputation metric is a value derived from on-chain activity and interactions without any off-chain identity information ^[4].

1.1 Background: DeFi, collateralization, and pseudonymity

Public blockchains make programmable settlement possible without a central authority ^[5]. A smart contract can encode the terms of a deal ^[6], and a DeFi platform that uses such contracts clears trades on the blockchain instead of routing them through a custodian ^[7]. Protocols then stack these contracts into decentralized markets for lending, automated market making and perpetual futures ^[8]. On Arbitrum One and other L2 networks, fees are low enough that ordinary users transact often, and every transfer, approval and protocol event can be inspected wallet by wallet.

Even with this visibility, DeFi credit markets remain inherently over-collateralized. Most protocols

make a borrower or trader post more collateral than the position is worth, because the counterparty is a pseudonymous account whose owner has not been verified ^{[9],[10]}. Pseudonymity is a basic property of public ledgers. Anyone can create addresses at almost no cost, and one entity can own more than one wallet ^[11]. Conventional credit ratings depend on personal histories that translate poorly to this setting ^[2], so protocols control risk through collateral reserves, liquidation mechanisms, and protocol rules instead of personalized reputation.

Collateralization protects protocols, but it uses capital inefficiently. Traders and borrowers have to tie up funds they could otherwise invest, and the platforms offer none of the under-collateralized products that traditional finance extends to people with a credit record. Researchers have therefore turned to on-chain signals as a substitute for off-chain credit histories: transfer-graph centrality ^[4], network risk propagation ^[12], prospect theory and blockchain-based credit scoring ^[13], and machine learning pipelines over lending protocols ^[14]. Each of these lines of work assumes that public ledger data contains information usable for risk assessment without deanonymizing users.

Graph ranking algorithms appear well suited to this problem. PageRank and its variants rank nodes by recursive endorsement, so that a node's score depends on the endorsements it receives from other nodes ^[15]. Earlier web trust algorithms such as EigenTrust ^[16] and TrustRank ^[17] apply similar ideas to peer-to-peer trust and to web spam. Do and Do^[4] use both PageRank and HodgeRank on a graph of transfers to judge how trustworthy Ethereum users are, and call the result a form of social credit. AWP^[3] is another transfer-graph ranking algorithm. It weights each transfer by a bounded function of its value and a logistic function of its age, and it restarts the walk at addresses that sent recently; its authors validate it by showing that it correlates with inbound transfer count and value, taken as proxies for economic importance. These studies establish transfer-graph reputation as a credible baseline, but they leave out explicit spending authorization, a separate on-chain action recorded as ERC-20 allowances ^{[18],[19]}.

1.2 Motivation for on-chain reputation

Four main factors drive the need for on-chain reputation research in leveraged DeFi.

- Capital efficiency: protocols are seeking ordinal ranking signals that could support tiered services (more frequent limit increases, lower fees, or higher priority in monitoring, for example), and perhaps even differentiated collateral requirements in future controlled experiments, while keeping settlement on chain ^[1].
- Auditability: off-chain credit scores are opaque, but graph-based scores computed from public event data can be recomputed and audited, which would help to ease some of the concerns about black-box decentralized credit scoring ^[2].
- Operational cost: a score that is refreshed often, before each leverage adjustment for instance, should be cheap to recompute.
- Standardization: other markets use screening ratios to tell counterparties apart, but pseudonymous DeFi has no standard ordinal language for doing so at the wallet level. Chapter 5 discusses where the scores studied here could serve that role and where they cannot.

Transfer-based AWP is public and auditable, but its graph grows with every payment ^[3]. Allowance graphs are a structurally different alternative. Whenever a user approves an ERC-20 token, the contract records how much of that token a spender may transfer ^{[18],[19]}, and each new approval supersedes the older ones, so the latest approval of each owner–spender pair and token stands for the history of that pair. Approvals are also rarer than transfers. On the same wallets the endorsement graph G_{approve} therefore has an order of magnitude fewer edges than the transfer graph G_{transfer} , and the PageRank solve, whose cost grows linearly with $|E|$, finishes proportionally faster (Table 4.1). The same solver runs in both cases, so the speedup comes from the sparser graph and is not an algorithmic improvement; Claim C1 treats it as an operational property of the representation. Efficiency aside, allowances also mean something different. A transfer might reflect routing, market making, or a passive receipt, whereas an allowance is an active grant of permission to spend

funds. Treating allowances as endorsement edges carries PageRank’s citation reading ^[15] over to a trust relationship that protocols already depend on. Whether this produces different rankings, and whether those rankings agree differently with authorization and transfer checks, is examined on a matched sample of $n = 5,521$ GMX V2 trading wallets on Arbitrum One.¹

1.3 Research problem

AWP infers trust from transfer history through weighted sums and logistic time-decay ^[3]. Its economic interpretation is rich, yet it (i) must aggregate the whole transfer history, (ii) models a high-value wallet as one that receives many inbound transfers, and (iii) assigns no role to spending authorization. Our proposal, EndorseRank, keeps AWP’s edge weights and solver but builds the directed graph from the latest allowance of each owner–spender pair and token. The resulting graph is much sparser, encodes a different signal and ranks a different side of each relationship. An allowance points from the owner to the spender, so EndorseRank scores the spenders that receive approvals, such as routers and vaults, and gives every owner that receives none the same value.

The main research question is: what do ERC-20 allowance edges add to PageRank as a reputation score for Arbitrum One wallets, compared with the transfer walk and with the raw degrees that both walks smooth? Research questions RQ1–RQ5 (Section 1.6) break it down. Because both edge types live on the same addresses, a single walk can also mix them node by node with a fixed weight λ , which defines a Coupled PageRank (C-PR); at $\lambda = 1$ and $\lambda = 0$ it walks one layer alone. The mixture is worth testing because of the unit of analysis: most traders are owners that an allowance score cannot tell apart, and one walk over both layers could keep the transfer layer’s coverage of them while carrying the allowance signal for spenders.

Agreement with statistics of a score’s own edges, or prediction of more of the same edges, does not yet show that a score measures reputation in a substantive sense. A post hoc analysis, which adds no

¹Matched means that every method scores the same wallets: the GMX V2 wallets with at least three closes between 1 December 2025 and 31 May 2026, a list fixed from GMX data before any approval or transfer was extracted (Section 3.2).

research question, therefore also uses an outcome built from neither edge type and observed after the freeze: the matched traders’ later liquidation avoidance (Section 3.7.3).

The scores appear in Table 1.1. Table 1.2 lists hypotheses (H1–H3), research questions (RQ1–RQ5), and empirical claims (C1–C6) from Chapter 4.

Table 1.1: The scores evaluated in this thesis (matched wallet cohort, $n = 5,521$).

Score	Input graph	Role
EndorseRank	Latest ERC-20 allowances (G_{approve}), weighted as in AWP	Allowance delegation; the object of study in this research
AWP	Transfers (G_{transfer})	Transfer flow; the published transfer-graph method of Do, Do and Nguyen ^[3] (Section 3.6.1)
C-PR	Both layers, raw amounts, mixed per node with weight λ	Coupled operator used to test whether the two edge types combine (RQ5); at $\lambda = 1$ and $\lambda = 0$ it walks one layer alone. Its ablation is S-PR, a transfer walk restarted from the allowance walk
Raw degrees	G_{approve} and G_{transfer} , counted per node	In-approve degree and transfer in-degree, the local counts that the PageRanks smooth: proxies in the same window and baselines out of window

The main terms are defined as follows:

- *DeFi (Decentralized Finance)*: Financial services such as lending and borrowing, implemented as smart contracts on public blockchains and generally run without a central operator in charge of custody or settlement ^[1].
- *Smart contract*: Self-executing program code that runs on a blockchain and carries out an agreement without a trusted intermediary ^[6]. In this thesis, the allowance and transfer logs are emitted by smart contracts.
- *Wallet / Address*: A blockchain account that holds funds and interacts with smart contracts; it is the object whose reputation we want to rank ^[7]. Each wallet is identified by its address, and no off-chain identity is assumed or recovered.
- *On-chain reputation score*: A number computed from on-chain data that expresses a wallet’s relative position for financial activity under pseudonymity ^[4]. Scores are ordinal and meant

for ranking; their absolute scale has no meaning for credit limits.

- *PageRank*: A graph ranking algorithm that spreads influence through a directed graph with a damping factor ^[15]. Adaptive Weighted PageRank (AWP) is PageRank on the transfer graph, with edges weighted by the value and age of each transfer and restarts at recently active senders ^[3]. EndorseRank applies AWP’s edge weights to the latest-allowance graph and restarts uniformly (Section 3.6).
- *Coupled PageRank (C-PR)*: One random walk over both the allowance and the transfer layer, which at each wallet mixes the two rows with weights λ and $1 - \lambda$ over the layers that wallet occupies (Section 3.6.3). Endorsement-seeded PageRank (S-PR) keeps the transfer walk but restarts it from the allowance walk, as TrustRank ^[17] restarts from a seed set.
- *ERC-20 allowance / Approve / Permit*: Contract methods that record how many of the owner’s tokens a spender may spend on the owner’s behalf ^{[18],[19]}. Approve is an ordinary on-chain call; EIP-2612 permit makes the same state change through a signature. For each owner–spender pair and token, EndorseRank uses the latest non-zero allowance.
- *Endorsement graph*: A directed graph whose edges denote an explicit delegation of spending authority, as opposed to an actual transfer of assets ^[18]. An edge $u \rightarrow v$ means that wallet u has approved wallet or contract v to spend tokens belonging to u .
- *Domain alignment*: The degree to which a reputation ranking agrees with an observable proxy ranking, measured with Spearman’s ρ ^[20] and Kendall’s τ ^[21] and interpreted as construct validity in the sense of Cronbach and Meehl^[22]. A method can align closely with the graph of its own construct and weakly with a different check.

1.4 Temporal holdout

The contemporaneous allowance Kendall τ (Table 4.3) is an intended-construct check: PageRank on G_{approve} should agree with spender-side approval degree, because it smooths that degree globally. To test whether the score conveys information beyond the same-window graph, the scores are frozen

at 28 February 2026 and compared with two labels that only become available in March–May 2026: new owner–spender approval pairs (the endorsement construct) and new transfer senders (the flow construct). The holdout cohort ($n = 1,335$) consists of every spender that holds at least one positive latest allowance at t_1 (28 February 2026, 23:59:59 UTC) in the extracted approval data. Each PageRank is also compared with the raw degree it smooths at t_1 , so the gain of the propagated score over the local count can be seen directly.

A second window was designed on 28 September 2026, after the results of this spring holdout were known, and registered on 1 October 2026. Its scores are frozen on 31 May 2026 and its labels are counted from June to August 2026, months whose logs had not been extracted when the registration was committed. It repeats the two spender labels and adds one for the matched traders: the share of their GMX closes that did not end in liquidation (Section 3.7.2).

1.5 Research objectives

The research has five objectives, one for each of RQ1–RQ5. Each is stated with the measurement used to operationalize it and with the criterion that Chapter 5 applies when judging whether the objective was met.

- O1. Establish whether EndorseRank and AWP order the wallets of the same cohort differently (RQ1). Measurement: Kendall τ between the two score vectors ($n = 5,521$) with a 95% bootstrap interval. Criterion: the interval contains neither 0 (unrelated orderings) nor 1 (identical orderings).
- O2. Assess how closely each score agrees with each construct in the same window (RQ2). Measurement: family-level Kendall τ with 95% intervals for every score and construct. Criterion: a score agrees with a construct when its interval on that family excludes 0; agreement with the construct of the score’s own edges is read as an intended-construct check, because those proxies come from the same events as the graph.
- O3. Measure what each score costs to compute on the same wallets, and how runtime responds

Table 1.2: Summary of hypotheses, research questions, and empirical claims.

Item	Addresses	Reported in
H1 / RQ1	Do EndorseRank and AWP produce different wallet orderings?	§4.4; inter-method τ with 95% CI; Table 4.5; Claim C3
H2 / RQ2	How closely does each score agree with the allowance checks and with the transfer checks?	Tables 4.3 and 4.5; Claim C2
H3 / RQ3	What does each score cost to compute at equal n ?	Tables 4.1–4.2; Claim C1
RQ4	Do t_1 scores predict t_2 new approvals and new senders, and does any PageRank do better than the raw degree it smooths?	§4.6; Tables 4.10, 4.11, 4.12, 4.13 and 4.16; Claim C4
RQ5	Does the coupled operator C-PR improve on a walk over the transfers alone, and does it rank wallets better than a walk over either layer alone?	§4.3.1, §4.6 and §4.6.2; Tables 4.3, 4.4, 4.11, 4.14 and 4.15
C1	Runtime and graph size of each score	Table 4.1
C2	Same-window agreement of each score with the checks of its own edge type, partly built in and, for EndorseRank, set by its ties	Tables 4.3 and 4.5
C3	EndorseRank and AWP order the cohort differently, mainly through which wallets are in the allowance graph	Table 4.5; Figure 4.3
C4	Out of window no PageRank ranks a label as well as the raw degree of the same edge type, in either window	Tables 4.10, 4.11, 4.12 and 4.16
C5	Coupling the two edge types does not beat its two layers walked alone, and the registered replication does not confirm its exploratory gain over the transfer walk: the gain on new approvals replicates, but non-inferiority on new senders fails, and against AWP it fails clearly	Tables 4.3, 4.4, 4.11 and 4.15; §4.6.2
C6	Post hoc: allowance count leads transfer count on later liquidation avoidance; PageRank level unresolved; mostly contract accounts; lower on the profitable share	Tables 4.18 and 4.19

to the number of sampled wallets (RQ3). Measurement: solve time (mean and standard deviation of five timed runs after one warm-up run), memory, iterations, $|V|$ and $|E|$, all under one protocol. Criterion: the time ratio is reported only together with the ratio of $|E|$, so that any difference can be traced to the size of the graph.

- O4. Test whether scores frozen at t_1 predict new approvals and new transfer senders at t_2 among the cohort’s spenders, and whether any PageRank does better than the raw degree it smooths (RQ4). Measurement: Kendall τ with 95% intervals between every score computed on 28 February 2026, the two raw degrees included, and both labels counted from March to May 2026 on the $n = 1,335$ spenders, with paired contrasts of each PageRank against its own degree. Criterion: a score predicts a label when its interval excludes 0, and a PageRank improves on its degree only when the interval of the paired difference lies above 0.
- O5. Test whether one walk across both edge types improves on a walk over the transfers alone, and whether it beats a walk over either layer alone (RQ5). Measurement: C-PR at $\lambda = 0.5$ (with 0.25 and 0.75 as sensitivity values) and S-PR, compared through paired intervals on $\Delta\tau$ with the two layers walked alone and with EndorseRank and AWP, on the holdout and same-window labels. Criteria, each fixed before the data it is judged on: (a) the rule of 14 September 2026, under which C-PR does no worse than the allowance walk on new approvals and better than the transfer walk on new senders on the spring holdout, and in the same window lies within the interval of the leading layer on the transfer and allowance families and above both layers on Sybil stability; (b) the rule registered on 1 October 2026, under which C-PR beats the transfer walk on new approval pairs in the June–August window and is not worse by more than 0.02 on new senders or on the traders’ liquidation-free close rate. A registered sensitivity analysis repeats (b) with AWP in place of the transfer walk; it does not change the rule.

1.6 Research questions

Three hypotheses go with the first three research questions. H1: on the same wallet set, EndorseRank and AWP produce different wallet orderings. H2: each score agrees with the checks built from the edges it walks on. H3: at the same value of n , a score's computation time follows the number of edges in its graph, so EndorseRank, on the sparsest graph, is the fastest to compute. H1–H3 hold largely by construction and act as checks of the instruments. The falsifiable tests are RQ4, whose degree contrasts were first computed after the spring labels were known and then repeated in the registered window, and RQ5, answered by rules fixed before the relevant data (Section 3.7). Table 1.2 summarizes the questions.

- RQ1. On the same Arbitrum wallet set, do EndorseRank and AWP produce different wallet orderings?
- RQ2. How closely does each score agree with current allowance checks and with current transfer checks (Spearman's ρ and Kendall's τ)?
- RQ3. What does each score cost to compute on the same wallet set, as measured by time, memory, and iterations?
- RQ4. Given scores computed at time t_1 , do the scores predict t_2 new owner–spender approvals to inbound spenders and new transfer senders, and does any PageRank predict them better than the raw degree it smooths at t_1 ?
- RQ5. Does a coupled PageRank running on both the allowance and the transfer layer improve on a walk over the transfers alone, such as AWP, and does it rank wallets better than a walk over either layer alone, out of sample as well as in sample?

RQ4 replaces the trading-success validation of the research proposal, whose outcome-based reference was circular (Section 2.8). RQ5 moves from comparing the two edge types to combining them; its rule for improving on the transfer walk came after the spring labels had been seen, so it is judged only on the June–August window registered on 1 October 2026 (Sections 3.7 and 3.7.2).

1.7 Summary of the literature review

Chapter 2 reviews four areas. In graph ranking, PageRank^[15] ranks nodes by recursively propagated endorsement at a per-iteration cost linear in the number of edges^[23], and EigenTrust^[16] and TrustRank^[17] adapt it to peer-to-peer trust and web spam with normalization and seed sets. In on-chain reputation, Do and Do^[4] apply PageRank and HodgeRank to Ethereum transactions, and AWP^[3] weights transaction edges by value and age and validates its ranking against inbound transfer count and value, statistics drawn from the same data as its graph. Lin et al.^[12] propagate account risk through the transaction network, and surveys of blockchain reputation systems^{[25],[26]} find explicit ratings and transfer volume to be the usual trust signals. Deployed DeFi tooling such as Human Passport instead aggregates on-chain and off-chain credentials to decide whether an address belongs to a distinct person^[64]. None of this work takes the ERC-20 allowance^[18] as a ranking edge; address clustering uses approvals only to link addresses of one entity^[40]. In DeFi risk and measurement, surveys^{[8],[27]} place behavioral counterparty risk beside contract, oracle, market and governance risk, and wash trading^{[28],[29]}, extractable value^[30], scam tokens^[31] and airdrop farming^[32] show that much transfer activity is artificial. In measurement theory, construct validity^{[22],[33]} separates what a score measures from external criteria, and the bootstrap^{[34],[35]} gives distribution-free intervals for rank statistics. The gap (Section 2.21) is an allowance-edge ranking evaluated on one cohort beside the transfer method and the raw degrees, with confidence intervals, paired differences, a fixed timing protocol and a temporal holdout.

1.8 Summary of the research methodology

Chapter 3 applies a quantitative, comparative design to public ledger data. ERC-20 Approval and Transfer logs on Arbitrum One from 1 December 2025 to 31 May 2026 are extracted from Google BigQuery for a matched cohort of $n = 5,521$ GMX V2 wallets, fixed from GMX data alone. One power-iteration PageRank implementation scores the latest-allowance graph and the transfer graph, and the coupled operator runs on the same solver. Each score is compared with

Spearman’s ρ and Kendall’s τ against same-window transfer, allowance and Sybil-stability proxies and, out of window, against new approvals and new transfer senders on the spender cohort, with the raw degrees at the freeze as baselines. A registered second window repeats the holdout from June to August 2026 and adds a trader label. Every coefficient has a 95% interval from 400 paired wallet resamples (seed 42), and method comparisons are paired. A pre-specified post hoc analysis, planned after the registered labels and their single-score coefficients were known but before its contrasts were computed, compares allowance-side and transfer-side scores on the traders’ later liquidation avoidance in both windows (2,000 paired resamples; Bonferroni 97.5% intervals for its two decision contrasts). Runtime, memory, iterations and graph size are recorded under one timing protocol. The code, configuration and machine-readable summaries behind every table are in a public repository (Section 3.13), and the study uses only public data (Section 3.16).

1.9 Contributions

The thesis makes five contributions to the study of on-chain reputation and DeFi risk. EndorseRank and AWP both apply weighted PageRank ^[15] to a single edge type, each with its own restart distribution; only the coupled operator changes the transition rule. What is new is therefore an edge type, an evaluation design, an analytic account of what the allowance walk computes on this sample, and a registered test of whether the two edge types combine.

- Latest-allowance edge construction (EndorseRank): an endorsement graph in which an edge $u \rightarrow v$ stands for the latest in-window ERC-20 allowance, set by approve or permit, that the spender v holds from the owner u , weighted as AWP weights a transfer. The edge records a delegation of spending rights, not a history of payments. Approvals have been used to cluster the addresses of one entity ^[40], but no reputation method reviewed ranks on them (Section 2.21).
- Same-cohort evaluation with statistical inference: every score is computed on one set of wallets and checked against identical transfer, allowance and Sybil-stability proxies under the

AWP validation protocol ^[3], with 95% bootstrap intervals for every coefficient and paired intervals for the contrasts ^[34].

- Out-of-window tests with degree baselines: scores frozen at t_1 are tested against the new approvals and new transfer senders that arrive after t_1 , beside the raw degrees that PageRank smooths, which separates in-sample agreement from prediction and the propagated score from its local count (Claims C4 and C5). A pre-specified post hoc analysis adds an outcome built from neither edge type: on later liquidation avoidance the approval count ranked traders above the transfer count in both windows, while the like-for-like PageRank pair is not resolved. This count-level lead, with account type an unresolved confound, concerns forced-liquidation avoidance, not trading success (Claim C6).
- Closed form of EndorseRank on a depth-one graph: with uniform restarts and uniformly redistributed dangling mass, a spender whose in-neighbors have no in-edges scores $c(1 + d\delta(v))$, where c is the common score of nodes without in-edges, d the damping factor and $\delta(v)$ the owner-normalized in-strength, which divides each owner's endorsement over all the spenders it approves. This holds to within 1% for 96.4% and 97.4% of spenders at the two freezes (Section 5.3.2). Besides the time and token weights, that normalization is what separates EndorseRank from the raw approval count, and the finding that propagation adds nothing is limited to this one-hop sample.
- Coupled operator and its registered evaluation: C-PR blends a raw-amount allowance layer and a raw-amount transfer layer node by node with a fixed weight λ , and S-PR is a seeded ablation. A first rule, fixed with λ before any coupled score was computed, asked C-PR to beat both layers walked alone, and a second, registered before the logs of its window were extracted, asked it to improve on the transfer walk; both fail and are reported as failed (Claim C5). The ablation shows that the allowance signal survives when allowance edges carry the walk and is lost when they only decide where it restarts.

1.10 Scope and limitations

The study covers Arbitrum One from 2025-12-01 to 2026-05-31. Its main sample is the $n = 5,521$ GMX V2 wallets with at least three closes in that period, and its holdout sample the $n = 1,335$ spenders that held a positive allowance on 28 February 2026. Because EndorseRank separates spenders and not the owners who grant approvals, and most matched traders are owners, the holdout is run on spenders; the registered June–August window adds a trader label (Section 3.7.2). Only events with a cohort wallet at one end are extracted, so each graph is a one-hop ego network around the cohort, the allowance walk has a depth of about one, and multi-hop propagation is not tested (Section 5.3.2). Most approval-receiving traders are contract accounts, so the post hoc liquidation result cannot separate the allowance signal from account type (Section 4.6.3). Runtime is also measured on subgraphs of the cohort graph drawn through a larger address pool ($N = 99,080$) that adds no edges. Default labels in unsecured lending are outside the scope of this study, extended baselines are archived (Appendix 8.4), and no claim is made that the results generalize beyond Arbitrum One or the months studied. Amounts are raw token base units. EndorseRank and AWP pass them through a bounded function, so every approval or transfer of ten base units or more counts about the same, while in the raw layers of C-PR unlimited approvals decide the rows of many owners (Chapter 3). Adversarial manipulation of allowances or transfers is a real threat: Chapter 5 shows that a closed farm of Sybil addresses can inflate EndorseRank cheaply, and the Sybil-adjusted proxies describe some manipulation patterns without protecting the score against them ^[11].

1.11 Research roadmap

Chapter 2 reviews graph-based on-chain reputation, ERC-20 allowance semantics and the reputation tooling deployed in DeFi, and develops the theoretical framework. Chapter 3 defines the cohorts, the graphs and scores, the construct checks, the temporal holdouts, the timing protocol, and the standards for reproducibility and ethics. Chapter 4 reports the benchmark (Claim C1), the

same-window alignment of every score (C2), rank divergence (C3), robustness checks, the temporal holdouts with the registered replication (C4 and C5) and the post hoc liquidation-avoidance analysis (C6); Appendix 8.5 collects the direct comparisons between EndorseRank and AWP. Chapter 5 interprets the findings by research question and objective, works out their implications, models the cost of Sybil attacks on allowance graphs, and weighs threats to validity and limitations. Chapter 6 answers RQ1–RQ5, states the contribution to knowledge and sets out future work.

2 Literature Review and Theoretical Framework

The research problem set out in Chapter 1 is that transfer-graph reputation scores give no role to spending authorization, an explicit act of trust that every ERC-20 token records, and that their graphs grow with every payment. The objectives O1–O5 and research questions RQ1–RQ5 follow from it. To compare EndorseRank with Adaptive Weighted PageRank (AWP) on this problem, we run each method’s graph through PageRank and check the resulting rankings with rank correlations. The framework combines hyperlink–citation propagation, domain alignment and computational complexity. The gap that remains is taken up in Chapters 3 and 4.

2.1 Blockchain and DeFi foundations

A public blockchain records transactions in hashed blocks, and each block is secured by the ones added after it ^[5]. Once a majority of the participating nodes has accepted a block, rewriting it is costly. That cost allows parties to settle on the ledger without each of them needing a known legal identity (*trust-minimized* settlement).

Bitcoin built a peer-to-peer electronic cash network on this kind of ledger ^[5]. Ethereum keeps the append-only ledger but lets an account hold a program alongside its balance. A transaction can then call one of the program’s functions and change the contract’s state ^[7], and these functions run on the Ethereum Virtual Machine (EVM) ^[36]. Programs stored on the ledger in this way are called smart contracts; any honest node that replays a function call gets the same output ^[6]. As a result, collateralization ratios, liquidation conditions and fee schedules can be enforced by the contract instead of by a trusted third party ^[1].

Collectively, these contracts make up Decentralized Finance (DeFi). Between them they do the work of exchanges, lending platforms, derivatives markets and asset managers; Werner et al.^[8] define

the protocol classes, and Schär^[1] describes how protocols are layered on one another. Because one protocol can interact with another, new financial instruments can be composed from existing ones^[37]. The same property lets a failure spread across several protocols. Successful attacks have included flash-loan exploits^[38] and price-oracle manipulation^[39]. Transaction histories on these chains are public and machine-readable, yet they record actions, not legal identities^[2].

2.1.1 Pseudonymity and the credit problem

Conventional credit scoring ties a credit history to a legal entity. On a public blockchain the basic unit of analysis is instead an address (a wallet), which may stand for a legal entity, a trust, an algorithmic agent or a smart contract. One entity can own several addresses^[11], and statistical clustering can recover ownership only in part^[40]. On-chain lending therefore requires borrowers to post collateral worth more than the amount they borrow; if the collateral value falls below a set threshold, a liquidator may repay part of the debt and receive the corresponding collateral at a discount^[41]. The arrangement is secure, but it ties up capital. An on-chain credit score tries to quantify from on-chain behavior alone how likely an actor is to repay. Recent proposals draw on prospect theory^[13], on machine learning applied to lending data^[14], and on models of credit risk spreading through account networks^[12].

2.2 Ethereum, Layer-2 rollups, and Arbitrum One

Most DeFi applications have been deployed on the Ethereum mainnet, historically the main venue for DeFi innovation and liquidity, but growing adoption there has raised gas costs and constrained throughput^[42]. Layer-2 (L2) protocols try to keep Ethereum's security guarantees while making transactions cheaper. Some execute transactions off-chain, while others batch them and post a succinct summary of the execution to the Ethereum mainnet^[43]. Thibault et al.^[44] organize L2 scaling approaches into a taxonomy. Optimistic L2s such as Arbitrum treat each submitted batch of transactions as valid unless evidence against it is presented within a dispute period^[45].

Arbitrum One is an optimistic L2 of this kind and can execute EVM-based smart contracts. It has become a popular platform for DeFi applications, among them GMX, an exchange for perpetual swaps ^[46]. We use Arbitrum One because the public BigQuery dataset includes both its ERC-20 Approval and Transfer events, and because its fees are low enough that a six-month period contains dense activity. We make no claim that our results generalize to other L2s or to Ethereum mainnet without further validation. We do believe, however, that the proposed framework (approval and transfer graphs, construct checks, and rank correlation) applies to any chain that emits the same logs.

2.3 ERC-20 tokens and the allowance lifecycle

The ERC-20 standard defines a common interface for fungible tokens, made up of transfer, approve, transferFrom, allowance and the associated events ^[18]. After the owner calls approve(spender, amount), the token contract records that the spender may move up to amount out of the owner's balance. The spender can then call transferFrom to carry out a token transfer. With EIP-2612 permit, an allowance can be updated through an off-chain signature that settles in a single transaction ^[19]. The user experience improves, and the underlying semantics of authorization are unchanged.

Figure 2.1 traces an ERC-20 allowance through its lifecycle. An allowance is an authorization and involves no actual transfer. Token owners can grant allowances to routers, vaults or trading contracts that they trust to act within set limits. Revoking (approving zero) and updating (setting a new non-zero allowance) both change the allowance currently in force. EndorseRank accounts for this by taking the most recent allowance value for each owner-spender pair as the edge weight, so a newer allowance replaces the previous one outright; old values do not decay gradually over time. In the available data, allowances are sparse compared with transfers. A single approval can cover many transfers, and many wallets have never granted an approval to any external account. The sparsity is a computational advantage, and it also marks a semantic difference, since EndorseRank

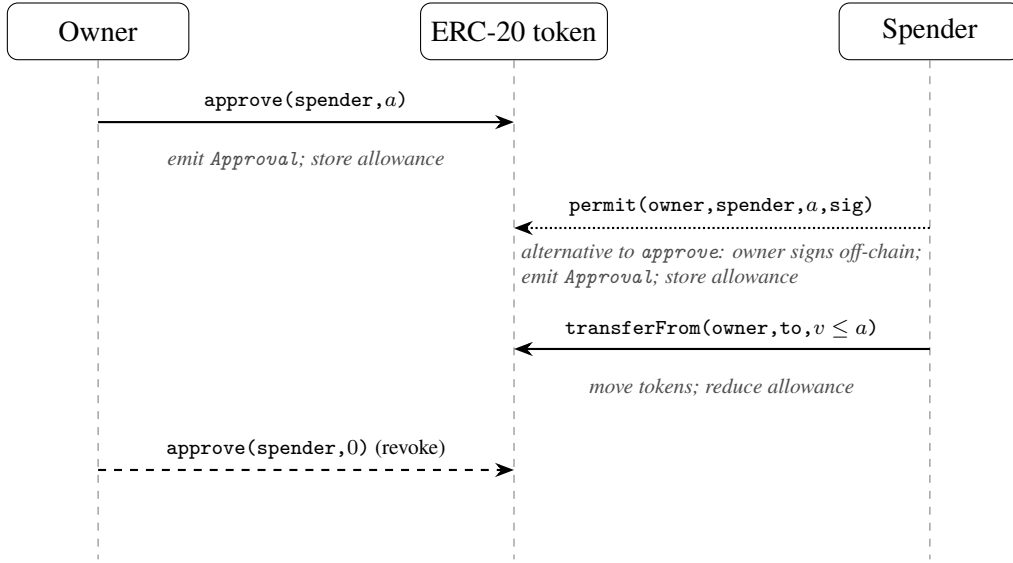


Figure 2.1: ERC-20 allowance lifecycle: approve, or its signature-based alternative permit (dotted), an optional transferFrom, and revoke. An EndorseRank edge sums, over tokens, one term for the most recent Approval event of its owner–spender pair in the observation window, and the term weights that approval by its age and amount as AWP weights a transfer. A later approval replaces an earlier one.

quantifies authorization to spend whereas a transfer graph records payment flows.

2.4 Graph-based on-chain reputation

PageRank and its variants rank nodes by the influence that flows to them along directed edges^[15]. Two recent studies apply this approach to blockchain transaction graphs, and the two should be distinguished. Do and Do^[4] run PageRank and HodgeRank on Ethereum transaction graphs to measure social credit; their main contribution is a definition of connectivity between accounts for credit scoring. Do, Do and Nguyen^[3] propose Adaptive Weighted PageRank (AWP), the transfer-graph method that this thesis evaluates beside EndorseRank (Section 2.18). Lin et al.^[12] argue that transaction connectivity can miss risk-relevant structure unless risk is propagated through the network, and Wu et al.^[24] reach a similar conclusion for phishing detection, where network embeddings of transaction graphs outperform per-account features. Surveys of blockchain-based reputation systems^[25] and of smart-contract reputation designs^[26] find that most deployments still take explicit

ratings or transfer volume as their trust signal, and a survey of knowledge discovery in cryptocurrency transaction data ^[47] names degree- and value-based centralities as the main descriptors of wallets. Packin and Lev-Aretz^[2] stress that such systems must be auditable even though their users are pseudonymous.

Outside the research literature, trust in deployed DeFi and web3 systems is usually established another way. An address builds standing through activity that anyone can inspect on the ledger, and tooling bound to the address summarizes it. The most widely used tool, Human Passport (formerly Bitcoin Passport), collects verifiable credentials, called stamps, for checks such as government identity, biometrics, vouching and earlier web2 and web3 activity. Each stamp carries a weight that reflects how hard it is to forge, and an address passes when the weighted sum reaches a threshold ^[64]. Passport also classifies EVM addresses on several networks, Arbitrum among them, as human or Sybil from their on-chain activity ^[64]. Its purpose is Sybil resistance, so it combines on-chain with off-chain evidence and works as a gate and not as a graded ordering. A review of proof-of-personhood protocols finds that they trade Sybil resistance against self-sovereignty and privacy ^[65], and a recent security review names credential farming, the bulk acquisition of cheap stamps, as the main residual weakness of Passport ^[66]. Such tools answer whether an address is controlled by a distinct person. EndorseRank answers a different question, which addresses other addresses authorize to spend their tokens, and it orders them from on-chain records alone (Table 2.1).

Graph-based methods are attractive because they (i) use only public information, (ii) produce continuous scores that suit ranking, and (iii) admit complexity analysis by number of edges. They are, however, vulnerable to Sybil attacks ^[11] and wash trading, and they suffer from the mismatch between graph centrality and actual default risk; PageRank scores are not credit scores in the regulatory sense (Section 2.8).

A propagated score must also beat a simpler one. Where degree correlations are weak, as on the Web graph, PageRank is on average proportional to in-degree and can be approximated from it ^[70]. New links concentrate on nodes that already have many, as preferential attachment assumes ^[67] and measured network growth shows ^[68], so link prediction, on Ethereum transaction networks too ^[74],

uses degree-based predictors as standard baselines ^[69].

2.5 PageRank formalism

Consider a directed graph $G = (V, E)$ with $|V|$ nodes, and write $w_{ij} > 0$ for the weight of the edge from i to j . That weight stands for influence from i to j , in either the endorsement or the transfer sense introduced above. Let W be the corresponding weighted adjacency matrix, and let $o_i = \sum_j w_{ij}$ be the out-strength of node i . The row-stochastic transition matrix P is then

$$P_{ij} = \begin{cases} w_{ij}/o_i & \text{if } o_i > 0, \\ 1/|V| & \text{if } o_i = 0 \end{cases} \quad (2.1)$$

(dangling nodes spread their mass evenly). Given a damping parameter $d \in (0, 1)$, the Google matrix is

$$G_{\text{PR}} = d P^\top + \frac{1-d}{|V|} \mathbf{1}\mathbf{1}^\top, \quad (2.2)$$

and the PageRank vector $\mathbf{r}$ satisfies $\mathbf{r} = G_{\text{PR}}\mathbf{r}$ with $\mathbf{r} \geq 0$ and $\sum_i r_i = 1$ ^{[15],[23]}. Element by element, with the dangling rows of P written as a separate middle term, this becomes

$$r_j = d \sum_{i:(i \rightarrow j) \in E} r_i \frac{w_{ij}}{o_i} + \frac{d}{|V|} \sum_{i:o_i=0} r_i + \frac{1-d}{|V|}. \quad (2.3)$$

2.5.1 Random-surfer interpretation

Equation (2.3) can be read probabilistically. At each step a surfer at a node with out-edges follows one of them with probability d , choosing it in proportion to its weight, or teleports to a uniformly chosen node with probability $1 - d$ (Figure 2.2). At a node without out-edges the surfer always jumps to a uniformly chosen node; this is the dangling rule of Equation (2.1). The stationary distribution then ranks nodes by how often they are visited in the long run. Damping keeps rank sinks (groups of nodes with no edge leaving them) from absorbing all the mass. With $d < 1$ and uniform

teleportation, every entry of G_{PR} is positive, so G_{PR} is primitive and its Perron vector is unique ^[23]. Appendix 8.3.1 gives the argument.

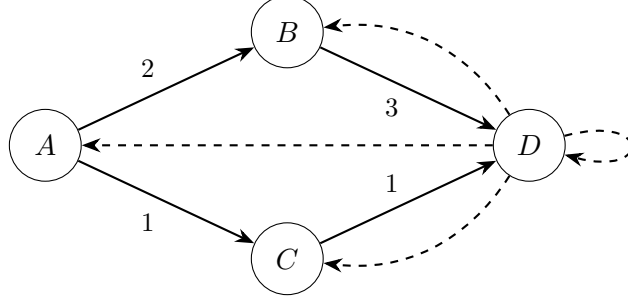


Figure 2.2: Random-surfer intuition for PageRank on the four-node graph of Section 2.13; edge labels are the allowance weights. Solid arrows are graph transitions: with total probability d the surfer follows one of them, chosen in proportion to its weight. Dashed arrows show uniform teleportation, which reaches every node (the current one included) with probability $(1 - d)/|V|$; the dashed fan is drawn only from D , although every node has one. D has no outgoing edge, so under the dangling rule of Equation (2.1) a surfer at D always jumps to a uniformly chosen node, each with probability $1/|V| = 0.25$. That rule, and not damping, keeps D from absorbing the chain.

2.5.2 Power iteration and complexity

In practice, $\mathbf{r}$ is usually obtained by power iteration from the uniform vector $\mathbf{r}^{(0)} = \mathbf{1}/|V|$,

$$\mathbf{r}^{(t+1)} = G_{\text{PR}} \mathbf{r}^{(t)}, \quad (2.4)$$

until $\|\mathbf{r}^{(t+1)} - \mathbf{r}^{(t)}\|_1 < \varepsilon$ or until the iteration cap I_{max} is reached. On a sparse graph, each iteration costs $O(|E|)$ arithmetic operations ^[48]. A method that generates fewer edges, for instance by using latest-allowance snapshots instead of full transfer histories, therefore lowers the cost per iteration and sometimes the memory required. Hypothesis H3 is motivated by this complexity fact.

By default this work sets damping to the conventional $d = 0.85$ ^[15] and uses two study-specific solver settings, a convergence tolerance $\varepsilon = 10^{-8}$ and a maximum of $I_{\text{max}} = 300$ iterations.¹

Damping is varied over $d \in \{0.75, 0.85, 0.95\}$ in the robustness section.

¹The tolerance and the iteration cap were chosen to keep run-times reproducible; they are not put forward as part of the classical PageRank formulation.

2.6 Reputation systems lineage

PageRank belongs to a wider family of spectral and propagation-based reputation mechanisms. Kleinberg’s HITS algorithm assigns each page two scores, one as a hub and one as an authority [49]. EigenTrust extends the idea to peer-to-peer networks and normalizes local trust, so a malicious node cannot raise its own score at will [16]. To suppress web spam, TrustRank starts from a seed set of trusted nodes and propagates their trust outward [17]. Reputation is relational in all of these methods. Where trust flows only from a vetted seed set, as in TrustRank, a newcomer cannot raise its score without endorsements from nodes that already hold reputation. With uniform teleportation, which EndorseRank uses (Section 3.6.2), that protection is weaker. Gyöngyi and Garcia-Molina derive the link farms and farm alliances that most raise a target’s PageRank [71], and Cheng and Friedman show that symmetric reputation functions such as PageRank can be manipulated with Sybils, whereas some asymmetric functions based on flows or paths from a trusted node resist them [72]. The Sybil cost model of Section 5.6 prices such a farm on an allowance graph.

On-chain, the Sybil attack is still the main concern [11]. New wallets cost little to create, whereas wallets that receive allowances or transfers from distinct counterparties cost more. The Sybil-adjusted stability proxies used in this work (inbound counterparty ratio, tenure, active months) partly capture that diversity of interaction, but they do not prevent a malicious actor from manipulating them.

Machine-learning credit models that use DeFi features [14] are trained to predict labeled defaults, and they give up interpretability to do so. EndorseRank and AWP keep to interpretable, audit-friendly propagation rules [2].

2.7 ERC-20 allowances as endorsement

Through the ERC-20 approve function and EIP-2612 permit, users state explicitly how much a spender may move on their behalf [18],[19]. EndorseRank takes the latest allowance from an owner to a spender on each token as one endorsement, weights it as AWP weights a transfer, and runs PageRank on the resulting directed graph. Newer approvals supersede older ones, so each pair and

token contributes a single event.

Concretely, for each owner–spender pair the latest in-window approval on each token contributes its time weight times its bounded value weight, as AWP weights a transfer; a pair whose latest value is zero (a revocation) is dropped, and the terms are summed over tokens (Equation (3.7) in Section 3.6.2). The resulting graph G_{approve} is usually much sparser than the transfer graph over the same wallets and window, and PageRank on it yields the EndorseRank scores s_{ER} .

Treating allowances as endorsements is a deliberate choice, and an imprecise one. Users approve routers because it is convenient, not necessarily to make a social statement about whether the router is trustworthy. Contract addresses and EOAs coexist as well. Even so, an allowance signals approval more explicitly than a received transfer, which could come from market making, wash trading ^[29], airdrop farming ^[32] or routing without any intent to endorse.

Approvals have been read as a relation between addresses before: Victor’s self-authorization heuristic uses ERC-20 approvals to cluster addresses that it attributes to one entity ^[40]. That reading links addresses and ranks nothing. It is also a caution, since an approval between two addresses of one entity is a self-endorsement, the edge that the Sybil cost model of Section 5.6 prices at gas alone.

2.8 What this research does not claim

The research proposal planned to validate both methods against GMX V2 trading success and to bound that validation with an outcome-based reference, GF-PR: PageRank on a star graph built from realized profit and loss. GF-PR is circular. It is computed from the same profit-and-loss stream that defines the trading-success proxies, so its near-perfect agreement with the realized-gain proxy is the correlation of a variable with a monotone transformation of itself, and the research question and claim that depended on it were withdrawn with it. The same-window trading proxies remain only as archived families (Table 8.2). Trading outcomes return out of window, in an exploratory run on the matched traders (Section 4.6.1), as the liquidation-free close rate of the registered June–August window (Section 3.7.2), and in a pre-specified post hoc comparison of the two edge types

(Section 4.6.3). That comparison needs an outcome built from neither edge type. Construct validity expects a score to agree with checks of the construct it was built to measure ^{[22],[33]}, so each edge type's lead on its own checks is partly built in; the GMX V2 trader outcomes give neither score that head start, and for counterparty-risk reputation the relevant one is forced liquidation. No score in this thesis is offered as a predictor of trading profit, loss avoidance or liquidation, or as a measure of creditworthiness or default risk; the post hoc liquidation comparison would need a registered test before any such use.

2.9 Rank correlation theory

Reputation scores and validation proxies are both continuous variables, and neither has a natural boundary for classification. Following the AWP validation design^[3] and classical psychometrics^[22], this research measures two kinds of rank agreement, monotonic and pairwise. Bootstrap confidence intervals quantify the sampling uncertainty of every reported coefficient ^[34].

Spearman's rank correlation ^[20] is the Pearson correlation of the ranks. For paired ranks (R_i, S_i) of n items without ties, it reduces to

$$\rho = 1 - \frac{6 \sum_{i=1}^n (R_i - S_i)^2}{n(n^2 - 1)}. \quad (2.5)$$

This no-ties form does not hold when ties occur. Spearman's ρ is then the Pearson correlation of average ranks (mid-ranks), in which tied items share the mean of the positions they occupy, and every reported ρ is computed this way. In both forms ρ measures monotonic association.

Kendall's τ ^[21] rests on counts of concordant and discordant pairs. Of the $n_0 = n(n - 1)/2$ pairs (i, j) with $i < j$, n_c are concordant (both rankings order them the same way) and n_d are discordant (the rankings order them oppositely); a pair tied in either ranking is neither. Let $n_1 = \sum_k t_k(t_k - 1)/2$, where t_k is the size of the k -th group of tied values in the first ranking, and define

n_2 in the same way for the second ranking. The τ -b variant is then

$$\tau_b = \frac{n_c - n_d}{\sqrt{(n_0 - n_1)(n_0 - n_2)}}, \quad (2.6)$$

and without the tie corrections it becomes $\tau_a = (n_c - n_d)/n_0$. Every τ reported in this thesis is τ_b , computed on the raw scores and proxies. The difference matters for heavily tied scores, and EndorseRank on the matched cohort is one (Section 4.4). For agreement about pairwise orderings, Kendall’s τ is usually the easier coefficient to interpret, and Chapter 4 uses it as the main family-level summary. Its values are compared with those of other methods on the same cohort; no uncited absolute benchmark is used.

2.10 Theoretical framework

This study compares EndorseRank and AWP on reputation structure (H1), construct alignment (H2) and computational efficiency (H3), and then on prediction out of window (RQ4) and on coupling the two layers (RQ5). Three bodies of theory frame the first part: hyperlink–citation propagation, construct (domain) alignment, and computational complexity. Two further arguments bear on the second part.

- Hyperlink–citation semantics: In the original PageRank algorithm ^[15], an inbound hyperlink counts as a citation. Do and Do^[4] apply this reasoning to Ethereum transfers, and AWP^[3] weights each transfer edge by the age and the value of its transfers, through a logistic decay in age and a bounded transform of value (Section 3.6.1). EndorseRank extends citation-based propagation to ERC-20 approve and permit edges. An allowance is an explicit, on-chain delegation of spending authority that remains in force until it is revoked or replaced ^{[18],[19]}, whereas a transfer may reflect trading or routing without any intent to trust. Graph-based risk models ^[12] show that network structure can carry information beyond raw activity counts, which argues for a directed endorsement graph alongside transfer volume.

- Domain alignment framework: Measurement theory uses construct alignment to ask how closely an indicator follows the construct it is meant to track ^[22]. Messick^[33] widens the idea into one unified judgment about what a score means and what using it implies. With rank-based reputation scores, Spearman's ρ and Kendall's τ measure monotonic and pairwise agreement in ordering and need no classification threshold, as in the AWP validation protocol^[3]. Graph coherence is checked with contemporaneous transfer and allowance statistics. The out-of-window checks are temporal holdouts: new approvals and new transfer senders for spenders, and, in the registered June–August window, the liquidation-free close rate for traders.
- Computational complexity: Each PageRank iteration costs $O(|E|)$ ^[15]. Allowance graphs are sparser than transfer graphs, with fewer edges $|E|$, so an iteration needs less time and memory, and in practice convergence often takes fewer iterations too. This argument underlies H3. Both methods use the same solver, so any difference in runtime is an expected effect of edge sparsity and does not count as an algorithmic contribution.
- Local counts as a baseline: Where wallets that already have many counterparties gain the most new ones (Section 2.4), the raw degree is a strong predictor of future arrivals, and a propagated score can beat it only when who the endorsers are matters more than how many. RQ4 therefore sets each PageRank against its own raw degree at the freeze date.
- Layer coupling: Multilayer PageRank treats several relations as layers over one node set ^{[62],[63]}. Mixing the layers node by node, as C-PR does, leaves a question open for layers of very different density: a node with no edge in the sparse layer follows the dense one, so the signal of the sparse layer may or may not survive. RQ5 therefore asks two things: whether the mixture improves on the transfer walk, and whether it beats both layers.

The conceptual model in Figure 2.3 links the scores of the thesis to H1–H3 and RQ1–RQ4. RQ5 is the dashed box on the right. It takes input from H2/RQ2 and from RQ4 and asks whether the two constructs combine.

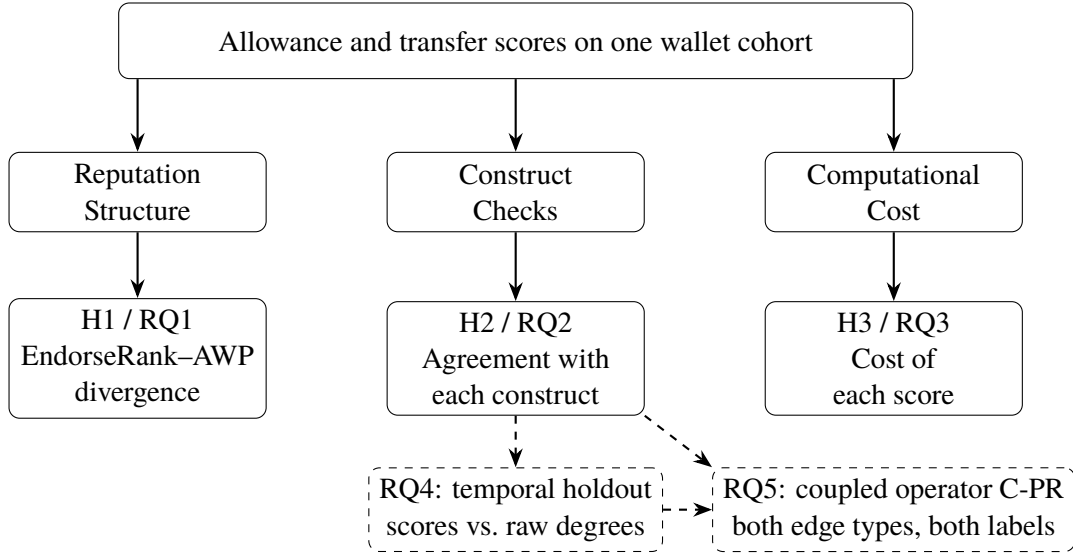


Figure 2.3: Conceptual model linking the allowance and transfer scores of one wallet cohort to reputation structure (H1/RQ1), construct checks (H2/RQ2), computational cost (H3/RQ3) and the temporal holdout, in which each PageRank is set against the raw degree it smooths (RQ4). The dashed box on the right is RQ5, the coupled operator C-PR.

Only EndorseRank and AWP enter H1. H2/RQ2 and H3/RQ3 concern every score, and RQ4 is a time-split test on the spender cohort, with the raw degrees at the freeze as its baselines; it does not rank trading outcomes.

2.11 Terminology

The formal constructs are defined in Chapter 1. The terms below are used when results are interpreted:

- Single-layer methods: EndorseRank (allowance graph) and AWP (transfer graph), the two PageRank scores that walk on one edge type. The coupled walks of Section 3.6.3 and the raw degrees are evaluated beside them.
- Primary graph construct: the graph that a method is *designed* to rank, for example G_{approve} for EndorseRank, as distinct from the checks applied after scoring.
- Contemporaneous check: a local transfer or allowance statistic taken from the same window

as the score ^[22].

- Temporal holdout: scores are frozen at t_1 (28 February 2026), and new owner–spender approval pairs and new transfer senders are counted up to t_2 (31 May 2026); a registered second window freezes scores on 31 May 2026 and counts labels from June to August 2026 (Section 3.7.2).
- Matched wallet cohort ($n = 5,521$), spender holdout cohort ($n = 1,335$) and expanded wallet pool ($N = 99,080$): the three address sets of Section 4.1.1.

2.12 Domain alignment metrics

Construct alignment is measured by Spearman’s ρ and Kendall’s τ between scores and check rankings (Section 2.10), on contemporaneous transfer and allowance checks and on the holdout labels; Chapter 3 gives the operational definitions. The literature reviewed so far thus supports three design choices: PageRank as an interpretable propagation backbone, allowances as endorsement edges separate from transfers, and a check design based on rank correlation plus a time split.

2.13 Worked example: PageRank on a four-node endorsement graph

Take four wallets $\{A, B, C, D\}$ joined by the latest-allowance edges

$$A \rightarrow B (2), \quad A \rightarrow C (1), \quad B \rightarrow D (3), \quad C \rightarrow D (1),$$

with damping $d = 0.85$. The out-strengths are $o_A = 3$, $o_B = 3$, $o_C = 1$ and $o_D = 0$ (the last is dangling). The edges $B \rightarrow D$ and $C \rightarrow D$ are the only out-edges of their nodes, so each is followed with probability one and its weight has no effect on the scores. Teleportation adds $(1 - d)/4 = 0.0375$ to every node at every step, the first one included. One power-iteration step from the uniform vector $\mathbf{r}^{(0)} = \mathbf{1}/4$ gives $\mathbf{r}^{(1)} = (0.0906, 0.2323, 0.1615, 0.5156)$ for (A, B, C, D) . The mass already tilts toward D , since both B and C endorse D , while A splits its mass between

B and C . The iteration converges to the stationary vector $\mathbf{r} = (0.1375, 0.2154, 0.1765, 0.4706)$, so D comes first, B and C follow, and A is last. No wallet endorses A . Apart from teleportation, its whole score comes from the dangling rule, which spreads D 's mass evenly over the four nodes ($dr_D/4 = 0.1000$). The example shows that a node gains score by *receiving* endorsements from nodes that are endorsed in turn, which is precisely the citation logic of PageRank^{[15],[23]}.

Suppose the same wallets were linked by transfers instead of allowances. The transfers would in general join other pairs, or the same pairs with other per-pair totals. That gives another transition matrix P , and possibly a different ranking of exactly the same nodes. Hypothesis H1 is this situation in miniature: EndorseRank and AWP need not agree.

2.14 DeFi risk taxonomy and where reputation fits

Werner et al.^[8] classify DeFi risk by protocol layer; Zhou et al.^[27] group it by the kinds of attack seen in practice. Bugs in contract code and misconfigured upgrade keys are smart-contract risks^[50]. A stale price, or a feed under someone's control, is an oracle risk^[39]. Market risks^[51] cover volatility and liquidation cascades, including stress in stablecoin and funding markets^[52], and the market crash of March 2020 showed how these two layers can interact^[9]. Token-weighted voting, which can be modeled, falls under governance risk^[53]. We study a narrower layer, *counterparty* or *behavioral* risk: whether an address's past behavior makes it a safe counterparty, as a trader that handles leveraged exposure responsibly or as a contract that many owners trust to spend their tokens.

On-chain reputation scores cover only part of the behavioral risk layer and are no substitute for collateral, auditing, or oracle design. Their purpose is to compress a system's relational structure (who authorizes whom, who pays whom) into an ordinal measure that can be audited and reproduced with little effort^[2]. Within behavioral risk, EndorseRank looks at authorization and AWP at payment flows. Neither tries to predict exploits or governance attacks.

Finally, fake activity makes behavioral risk harder to measure. Wash trading inflates volume on

decentralized exchanges ^[28] and, on a larger scale, on unregulated centralized exchanges ^[29]. Front-running and related forms of extractable value create transfers that follow ordering strategies instead of relationships ^[54]; Qin et al. ^[30] estimate how much of that value is captured on Ethereum. Scam tokens on Uniswap ^[31] and cross-exchange arbitrage trades ^[55] likewise add transactions without forming sustained relationships. The transfer graph is particularly sensitive to fake volume, while the allowance graph is vulnerable to approval farming. Time-based splits and Sybil detection solve the problem only in part.

2.15 Related measurement traditions

Outside blockchains, reputation systems have been studied in e-commerce ^[56], in peer-to-peer networks ^[16], and in work on web spam ^[17]. Four insights from this work apply here: (i) reputation is a property of relations; (ii) when identities are easy to create, Sybil attacks become possible ^[11]; (iii) seeding and normalization limit the effect of inflation; and (iv) evaluation needs external benchmarks, because internal coherence alone is not enough ^[22]. EndorseRank takes up (i) directly and treats (ii) as a threat. It does not take up (iii), since it restarts uniformly (Section 2.6). On (iv) it goes part of the way: the contemporaneous checks, measured by Spearman’s ρ ^[20] and Kendall’s τ ^[21], test coherence, while the temporal holdouts test prediction out of window, on the same constructs for spenders and on liquidation for traders.

Personalized and topic-sensitive PageRank ^{[57],[58]} bias the teleportation vector toward a chosen group of nodes, the mechanism that TrustRank uses to turn a small seed set into a global ranking ^[17]; AWP’s restarts in proportion to sending activity are another instance ^[3]. Multiplex PageRank lets a node’s centrality on one layer shape the walk on another ^[62], and the multilayer framework of De Domenico et al. lets one walker move within and between the layers of an interconnected network ^[63]. Both treat layers as different relations over one node set, which matches allowances and transfers among the same addresses. C-PR takes the simplest form of this idea, a per-node weighted sum of two transition matrices with one weight, and S-PR borrows the personalized-teleportation

idea, with the allowance score as the teleportation vector (Section 3.6.3).

2.16 Accounts, gas, and why logs are the measurement surface

Ethereum has two kinds of account: externally owned accounts (EOAs) and contract accounts. A private key controls an EOA, whereas a contract account is governed by the code it stores^[59]. Every state-changing transaction consumes gas. Logs come from a contract's LOG instructions, which run when its code emits an event (the Solidity `emit` statement); writing to storage produces no log. A reputation system that means to be auditable must prefer logs to off-chain databases, because logs are a permanent part of the blockchain history that any full node or index provider can recompute independently^[60].

Gas prices also shape the graphs. Cheaper gas on L2 networks densifies G_{transfer} more than G_{approve} , because transfers recur routinely while an approval is granted once so that later transfers can go through; Chapter 4 reports the resulting graph sizes.

2.17 Approvals as security-relevant events

Unlimited approvals (`type(uint256).max`) save the owner from approving again, at a cost in safety. If the approved account is compromised, every one of the owner's tokens can be stolen until the owner resets the approval. An allowance is therefore a capability and not a record: whoever holds it can move the owner's tokens until it is spent or revoked^[18]. EndorseRank interprets a token approval as endorsing one spender's capability to move those tokens. This is why an allowance does not mean the same thing as a transfer, even though both involve the movement of tokens. An account could receive many transfers and never be granted permission to move funds on anyone else's behalf.

Wang et al. find that unlimited approvals make up the majority of ERC-20 approvals on Ethereum and quantify the risk they carry^[73]. They are the majority here too (Section 3.6.3 gives the counts) and appear as amounts of $2^{256} - 1 \approx 1.16 \times 10^{77}$ base units. EndorseRank's value weight counts

them like any other approval, while in the raw allowance layer of C-PR they take almost all of their owners' rows, and most owners hold at least one.

2.18 AWP in depth: what the transfer-graph baseline measures

Adaptive Weighted PageRank (AWP), proposed by Do, Do, and Nguyen^[3], ranks the addresses of a transaction graph. Each transaction adds to its edge a logistic function of its time multiplied by a bounded function of its value, so that no transaction adds more than one to the edge weight. The walk restarts at each address in proportion to its activeness, the sum over the addresses it has sent to of the time weight of its latest transaction to each. Ranking accuracy is then validated against inbound degree and inbound value, taken as proxies for the economic attention a wallet attracts. The form of the decay (Equation (3.3)) and the in-degree/in-value proxies in Chapters 3 and 4 are taken from the AWP paper. The paper gives no parameter values; Section 3.6.1 gives the values used here. The approach has the advantage of being entirely on-chain and rank-based, and it needs no off-chain identity of any kind.

EndorseRank keeps the same template, AWP's edge weights under PageRank plus a rank-correlation check, and changes what the edges mean. In Chapter 4 we test whether allowance edges yield a score that is construct-distinct, allowance-aligned, and cheaper, and whether scores taken at t_1 can predict new approvals by t_2 .

2.19 Comparison of graph reputation methods

Table 2.1 compares EndorseRank with the other approaches covered in this chapter by edge semantics, temporal model and validation design.

As Table 2.1 indicates, EndorseRank's contribution is the use of the latest allowance as the edge of a reputation ranking, evaluated out of window; it is neither the ranking algorithm nor the reading of approvals as a link between addresses (Section 2.7). On the operator side, our contribution is to couple the two relations into one walk, with a seeded ablation to measure how much this coupling

Table 2.1: Comparison of selected graph reputation methods (conceptual).

Method		Edge semantics	Temporal model	Typical validation
PageRank [15]	(web)	Hyperlinks	None / static snapshot	Retrieval quality
EigenTrust [16]		Local trust ratings	Iterative aggregation	P2P file authenticity
TrustRank [17]		Links from seed goods	Static / seeded	Spam demotion
PageRank/ HodgeRank	on	Token transfers	Static window	Not reproduced in this thesis
Ethereum [4]				
AWP [3]		Transactions, bounded value weights	Logistic decay of edges and restarts	In-degree / in-value
RiskProp [12]		Risk-bearing links	Propagation of risk	Account risk labels
Multiplex / multilayer PageRank [62],[63]		Several relations over one node set	Static layers	Versatility of top-ranked nodes
Human [64]	Passport	No edges; weighted credentials about the address holder, on-chain and off-chain	Current credential set	Admission threshold; known human and Sybil addresses
EndorseRank (this study)	(this study)	Latest allowances, AWP's weights	Logistic decay of each latest approval	Allowance checks; t_2 new approvals
C-PR / S-PR (this study)	(this study)	Allowances mixed with, or seeding, transfers; raw amounts	Allowance snapshot and logistic decay of transfers	Holdout labels; rules fixed in advance

contributes. The coupling is tested out of window, against both constructs, and under a pre-defined rule.

2.20 Summary of the reviewed literature

Table 2.2 collects in one place the sources underlying this study. Each row records a source’s methodology, the strengths this thesis draws on, the limitations it tries to address, and how the source relates to this work. The research gap stated in Section 2.21 is grounded in the evidence collected in this table.

Table 2.2: Summary of the reviewed literature: methodology, strengths, limitations, and relation to this study.

Source (year)	Methodology	Strengths	Limitations	Relation to this study
Page et al. (1998) ^[15] ; Langville and Meyer (2006) ^[23]	Eigenvector rank-ing of a hyperlink graph, solved by power iteration	Interpretable propaga-tion; $O(E)$ per itera-tion; damping yields a unique Perron vector	Edges can only mean hyperlinks; no notion of time	Same solver and com-plexity argument, used unchanged for both methods compared
Kamvar et al. (2003) ^[16] ; Gyöngyi et al. (2004) ^[17]	Aggregation of trust in P2P networks; propaga-tion from seeds to de-mote spam	Normalization and seed sets curb infla-tion; the adversary is modeled explicitly	Depends on explicit ratings or curated seeds; not applied on-chain	Informs the Sybil-cost discussion and the uniform-teleportation base-line
Do and Do (2023) ^[4]	PageRank and HodgeR-ank applied to Ethereum transaction graphs as a measure of social credit	Treats transfer connec-tivity as relevant to credit; data are fully public	Transfers conflate pay-ment with trust; al-lowance semantics ab-sent	Earlier transfer-graph work, kept separate from AWP in this thesis

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Table 2.2 (continued)

Source (year)	Methodology	Strengths	Limitations	Relation to this study
Do, Do, and Nguyen (2023), AWP ^[3]	Transaction PageRank with bounded value weights, logistic recency and restarts weighted by sending activity; rank correlation with in-degree/in-value as validation	Validates on ranks, so no default labels are needed; recency modeled explicitly	Proxies are computed from the same transfer data that forms the graph (mechanical alignment); dense graph; no parameter values reported	Baseline, used as published (Section 3.6.1); the decay parameters and b are set here, and EndorseRank takes the same edge weights
Lin et al. (2025), RiskProp ^[12]	Network propagation of account risk, paired with a de-anonymization score	Structure shown to add information beyond activity counts; evaluated with labels	Needs risk labels; aimed at fraud, not trust or authorization	Argues for finer edge semantics; a simplified liquidation-weighted transfer walk is archived (Appendix 8.4)
Wu et al. (2022) ^[24]	Phishing detection through network embedding of Ethereum transaction graphs	Graph features outperform per-account features; large labeled dataset	Supervised; its labels mark phishing, not creditworthiness	Backs the premise that graph position is informative; not a competing method
Hassija et al. (2020) ^[13] ; Kotb (2023) ^[14]	Blockchain-data credit scoring using prospect theory and machine learning	Aimed directly at lending decisions	Need labels or off-chain features; only partly interpretable	A contrasting class; EndorseRank remains within auditable graph propagation
Bellini et al. (2020) ^[25] ; Almasoud et al. (2020) ^[26]	Surveys of blockchain-based trust and reputation systems	Chart the design space and find explicit ratings and volume-based signals dominant	Authorization edges left unevaluated; nothing measured on L2	Show that none of the surveyed designs uses latest-allowance edges

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Table 2.2 (continued)

Source (year)	Methodology	Strengths	Limitations	Relation to this study
Victor (2020) ^[40] ; Victor and Wein- traud (2021) ^[28] ; Cong et al. (2023) ^[29]	Heuristics for cluster- ing addresses, one of which links addresses through ERC-20 ap- provals; detection of wash trading on DEXs and exchanges	Measure the share of on-chain activity that is artificial or multi- address	No reputation score proposed	Closest prior use of approvals, as a clus- tering link and not a ranking edge; show why transfer-based signals are open to inflation
Cronbach and Meehl (1955) ^[22] ; Messick (1995) ^[33]	Psychometric construct validity	Distinguish what a score is designed to measure from external criteria	Not specific to blockchains	Basis for treating the contemporaneous checks as construct checks and the temporal holdouts as out-of-window evidence
Human Passport (n.d.) ^[64] ; Siddarth et al. (2020) ^[65] ; Bordeianu and Popescu (2026) ^[66]	Weighted credentials about an address holder, on-chain and off-chain, against a passing threshold; reviews of proof-of-personhood defenses	Deployed at scale; uses off-chain evi- dence against Sybil attacks	A gate and not a graded ordering; relations between ad- dresses play no part; credential farming remains	Shows how trust is es- tablished in practice, and what EndorseR- ank does not offer: no personhood guar- antee
Efron (1979) ^[34] ; DiCiccio and Efron (1996) ^[35]	Bootstrap resampling and bootstrap-based confidence intervals	Uncertainty for rank statistics with no distributional as- sumption; paired contrasts	Units must be ex- changeable	Source of the 95% intervals and $\Delta\tau$ contrasts reported in Chapter 4

2.21 Research gap

Five gaps emerge from Table 2.2 and Section 2.15. First, every on-chain reputation method discussed here (PageRank/HodgeRank ^[4], AWP ^[3], RiskProp ^[12], and the designs in the surveys ^[25]) builds its graph from transfers or explicit ratings, and the attestation tooling deployed in DeFi scores the holder of an address without any graph of relations between addresses ^[64]. None uses the ERC-20 allowance, the on-chain record of a deliberate delegation of spending power that every ERC-20 token carries. Approvals have served address clustering and security analysis (Sections 2.7 and 2.17), but in the work reviewed here they have not served as the edges of a reputation ranking. It remains unknown whether a latest-allowance endorsement graph yields a distinct and meaningful ordering (O1 and RQ1). A second gap lies in validation. The transfer-graph baseline checks its scores against in-degree and in-value computed from the same transfer data used to build the graph, so some agreement is unavoidable; no report applies this procedure to an allowance construct and its transfer counterpart with confidence intervals and paired difference tests on the same set of addresses (O2 and RQ2). Third, runtimes for blockchain PageRank are often published without edge counts, memory use or iteration counts under a fixed protocol, which makes it hard to attribute any efficiency claim to sparsity as opposed to implementation (O3 and RQ3). The fourth gap is methodological and has to do with time. Graph statistics from the same window are often presented as validation, although such scores cannot be shown to predict future behavior; the out-of-window check used here is instead a temporal holdout with a future-tense label for each construct and raw degree at the time of the freeze as the baseline (O4 and RQ4). Finally, multilayer ranking has been developed for social, transport and citation networks ^{[62],[63]} but has not been used on two on-chain relations over one set of addresses. No study reports whether combining an authorization layer with a payment layer improves on the payment layer alone, or beats both layers, under rules fixed before the outcome is seen (O5 and RQ5).

2.22 Open problems motivating this research

The literature leaves several problems in on-chain reputation unresolved, and these motivate the present study:

1. Edge semantics: When identities are pseudonymous, which on-chain relations capture trust or authorization best ^[2]?
2. Validation without defaults: On what basis can methods be compared when loan-default labels are scarce ^[4]?
3. Computational cost: As graphs grow, can reputation scores still be refreshed at interactive latencies ^[23]?
4. Time-split checks: What keeps an evaluation from mistaking same-window graph statistics for out-of-window evidence ^[22]?
5. Adversarial robustness: How much does it cost to forge reputation when identities are cheap ^[11]?

EndorseRank takes on (1)–(4) through allowance edges, contemporaneous checks, runtime benchmarks, and a temporal holdout. Problem (5) is analyzed at the model level in Chapter 5; a full empirical adversarial evaluation is left for future work.

Chapter 3 turns this gap statement into an operational design: the data window and cohorts, the construction of both graphs and their solution with one shared PageRank implementation, the proxy families, the temporal holdout, and the timing and bootstrap protocols behind the results of Chapter 4.

3 Research Methodology

Chapter 2 ended with five gaps. This chapter gives each of them a measurable procedure and sets out how EndorseRank, Adaptive Weighted PageRank (AWP) and the coupled walks built from their two layers are evaluated as reputation scores on Arbitrum One. EndorseRank runs on the ERC-20 allowance graph and AWP on the transfer graph ^[3]. Both weight their edges as AWP's authors specify and are solved with weighted PageRank ^[15]; AWP restarts the walk in proportion to sending activity, and EndorseRank restarts it uniformly.

Methodologically, the study follows construct (domain) alignment. Reputation scores are continuous rankings, and the checks rest on observable wallet-level statistics instead of binary default labels ^[22]; each ranking method is one indicator of a reputation construct that cannot be observed directly. The claim is comparative, every score set against the checks of each construct and against the raw degrees it smooths; it is not a prediction of trading profit or credit default.

Every procedure runs on public blockchain logs inside a fixed observation window. Seeds are deterministic and the evaluation pipeline is open, so anyone who reruns the extraction, or holds the cached parquet files, can regenerate the reported tables.

Four principles guide each operational choice.

- Same-seed comparability: every score is computed on an identical wallet set, so no difference between scores can come from sampling.
- Fixing choices in advance: the observation window, topic hashes, damping, decay parameters, bootstrap resamples and seed, and benchmark seeds are set in a versioned configuration. The coupled operator, its weights and the first rule for RQ5 were written into a plan dated 14 September 2026, before any coupled score was computed. That plan and the configuration encoding the rule entered version control only on 28 September, in the commit that also holds the first coupled spring results, so their order rests on the dated plan and not on the commit

history. The second rule for RQ5 and its June–August window were registered on 1 October 2026, before any log from that window was extracted (Section 3.7.2). EndorseRank and AWP were scored after the same-window checks and the spring labels were known, so apart from one registered sensitivity analysis that sets C-PR against AWP, no comparison that involves them was fixed before its labels were known; those of Section 3.7.3 were fixed after their labels were known but before they were computed. All are reported with intervals.

- Construction and timing: each cohort’s edge lists are built once, before any timer starts, and the benchmarks time the solver call.
- Scope: H1 and C3 compare the orderings of EndorseRank and AWP; the other hypotheses and Claims C1, C2 and C4 concern every score, C5 the coupled operator and its ablation, and C6 (post hoc) the approval and transfer counts on trader outcomes. RQ4 and RQ5 draw on the spender holdout cohort, and the registered replication adds a trader cohort.

3.1 Notation

The notation used in this chapter and in Chapters 4 and 5 is collected in Table 3.1 below, and a condensed list appears in Table II. Vectors are set in bold and sets in calligraphic type. Wallet addresses are treated as opaque identifiers; we do not map them to any external identity.

Except where stated otherwise, PageRank means the weighted variant of Page et al.^[15], whose update at iteration t is

$$\mathbf{r}^{(t+1)} = (1 - d) \boldsymbol{\pi} + d P^\top \mathbf{r}^{(t)} + d (\mathbf{r}^{(t)\top} \mathbf{1}_{\text{dang}}) \boldsymbol{\pi}, \quad (3.1)$$

where P is the transition matrix of Equation (3.2) below, $\boldsymbol{\pi}$ is the restart distribution over the nodes, uniform ($\pi_u = 1/|V|$) unless a method specifies another (AWP in Section 3.6.1, S-PR in Section 3.6.3), and $\mathbf{1}_{\text{dang}}$ is the indicator vector of the dangling nodes, with entries $\mathbf{1}[o(u) = 0]$. The third term therefore collects the mass held by nodes without out-edges. With $w(u, v)$ denoting the weight of the directed edge from u to v and $o(u) = \sum_v w(u, v)$ the out-strength, the transition

Table 3.1: Notation for graphs, scores, proxies, and parameters.

Symbol	Meaning
$\mathcal{W}$	Matched wallet cohort ($ \mathcal{W} = n = 5,521$)
$\mathcal{W}_{\text{pool}}$	Expanded wallet pool ($ \mathcal{W}_{\text{pool}} = N = 99,080$)
$G = (V, E)$	Graph passed to the PageRank solver: V holds the end points of the retained edges E
$G_{\text{approve}} = (V_A, E_A, w_A)$	EndorseRank allowance graph
$G_{\text{transfer}} = (V_T, E_T, w_T)$	AWP transfer graph
$a^*(u, v, \text{token})$	Latest in-window allowance of owner u to spender v on one token (raw base units)
$\Delta t^*(u, v, \text{token})$	Age of the approval that set a^*
$w_A(u, v)$	EndorseRank edge weight, owner $u \rightarrow$ spender v : $\sigma(\Delta t^*)V(a^*)$ summed over tokens
$w_T(u, v)$	AWP edge weight, sender $u \rightarrow$ receiver v : $\sigma(\Delta t_e)V(x_e)$ summed over transfers
x_e	Token amount of transfer e (raw base units)
$V(z), b$	Bounded value transform of AWP, $V(z) = 2/(1 + e^{-bz}) - 1$, with $b = 1$ on raw base units
X_u	Activeness of address u , the restart weight of AWP
$\tilde{w}_A(u, v), \tilde{w}_T(u, v)$	Weights of the two layers of C-PR, in raw amounts
$o(u); o_A(u), o_T(u)$	Out-strength $\sum_v w(u, v)$ of node u ; the same in the allowance and transfer layers
P	Transition matrix, $P_{u,v} = w(u, v)/o(u)$; rows of dangling nodes are zero
$\mathbf{1}[\cdot]$	Indicator: 1 if the condition holds, 0 otherwise
$\mathbf{1}_{\text{dang}}$	Indicator vector of the dangling nodes, entries $\mathbf{1}[o(u) = 0]$
$\mathbf{r}, \mathbf{r}^{(t)}$	PageRank vector and its iterate after t steps
π	Restart (teleportation) distribution, also used for dangling mass: uniform for EndorseRank and C-PR, proportional to X_u for AWP, and for S-PR the allowance layer’s score restricted to V_T and renormalized
d	PageRank damping factor (default 0.85)
ε	PageRank convergence tolerance (10^{-8})
I_{max}	Maximum PageRank iterations (300)
λ	Allowance-layer weight of the coupled operator C-PR ($\lambda = 0.5$ primary; 0.25, 0.75 sensitivity; at $\lambda = 1$ and $\lambda = 0$ one layer is walked alone)
$\alpha_A(u), \alpha_T(u)$	Layer weights in node u ’s C-PR row; $D_\lambda(u)$ is their common denominator
$P^{(\lambda)}$	Coupled transition matrix of C-PR
$\mathbf{s}_M$	Reputation score vector for method M
$\mathbf{p}_q$	Proxy value vector for proxy q
$\sigma(\Delta t)$	Logistic time-decay on the age of an approval or transfer
k, t_0	Decay steepness (0.01) and midpoint (180 days)
Δt	Age of an event in days before the decay anchor (T_{obs} ; t_1 in the holdout)
T_{obs}	Observation end timestamp (2026-05-31 23:59:59 UTC)
t_1, t_2	Holdout score freeze (2026-02-28 23:59:59 UTC) and end of the label window (2026-05-31 23:59:59 UTC)
ρ	Spearman’s rank correlation
R_i, S_i	Mid-ranks of wallet i ’s score and proxy value (tied values share their average position)
τ, τ_b	Kendall’s rank correlation; every reported value is the tie-corrected τ_b
n_c, n_d	Numbers of concordant and discordant wallet pairs
$n_0; n_1, n_2$	All pairs, $n(n-1)/2$; pairs tied in the first and in the second ranking

matrix is

$$P_{u,v} = \begin{cases} w(u,v)/o(u) & \text{if } o(u) > 0, \\ 0 & \text{otherwise.} \end{cases} \quad (3.2)$$

The rows of dangling nodes are zero, so P is substochastic; the mass lost at those nodes returns through the third term of Equation (3.1) and is spread according to π . Filling each zero row with $1/|V|$ gives the row-stochastic matrix of Equation (2.1), and with uniform π Equation (3.1) is then the Google-matrix form of Equation (2.2). The iteration starts from $\mathbf{r}^{(0)} = \mathbf{1}/|V|$ and stops once $\|\mathbf{r}^{(t+1)} - \mathbf{r}^{(t)}\|_1 < \varepsilon$ or after $I_{\max}$ iterations. V is the set of end points of the retained edges, and the final vector is normalized to sum to one over V . Wallets are also listed in descending order of score, ties broken by address, but these ordinal ranks serve only for listing; the correlations of Section 3.10 are computed on the raw scores, where tied scores stay tied.

The damping factor d interpolates between pure propagation ($d \rightarrow 1$) and uniform scores ($d \rightarrow 0$). Its default of 0.85 matches the classical web setting and AWP^[3], and the robustness sweep varies it with edge weights held fixed (Section 3.12). The strict $\varepsilon = 10^{-8}$ and $I_{\max} = 300$ keep the measured runtimes tied to the structure of the graph and not to early termination.

3.2 Data sources and sampling

This subsection fixes the chain, time interval, public tables, events and wallets of the main sample; no later step brings in events from outside it. Figure 3.1 places every kind of on-chain record used in the thesis on one calendar, with the months each analysis scores, the months it holds out as labels, and what each record is used for.

- **Primary chain:** All events come from Arbitrum One (chain ID 42161), an optimistic rollup on Ethereum^[45] whose logs are in the public BigQuery export. One chain avoids cross-chain aliases, where one user's addresses on several chains would appear as distinct nodes, and every wallet in the sample faces the same fees and latencies.
- **Observation window:** Six months, from 2025-12-01 00:00:00 UTC to 2026-05-31 23:59:59

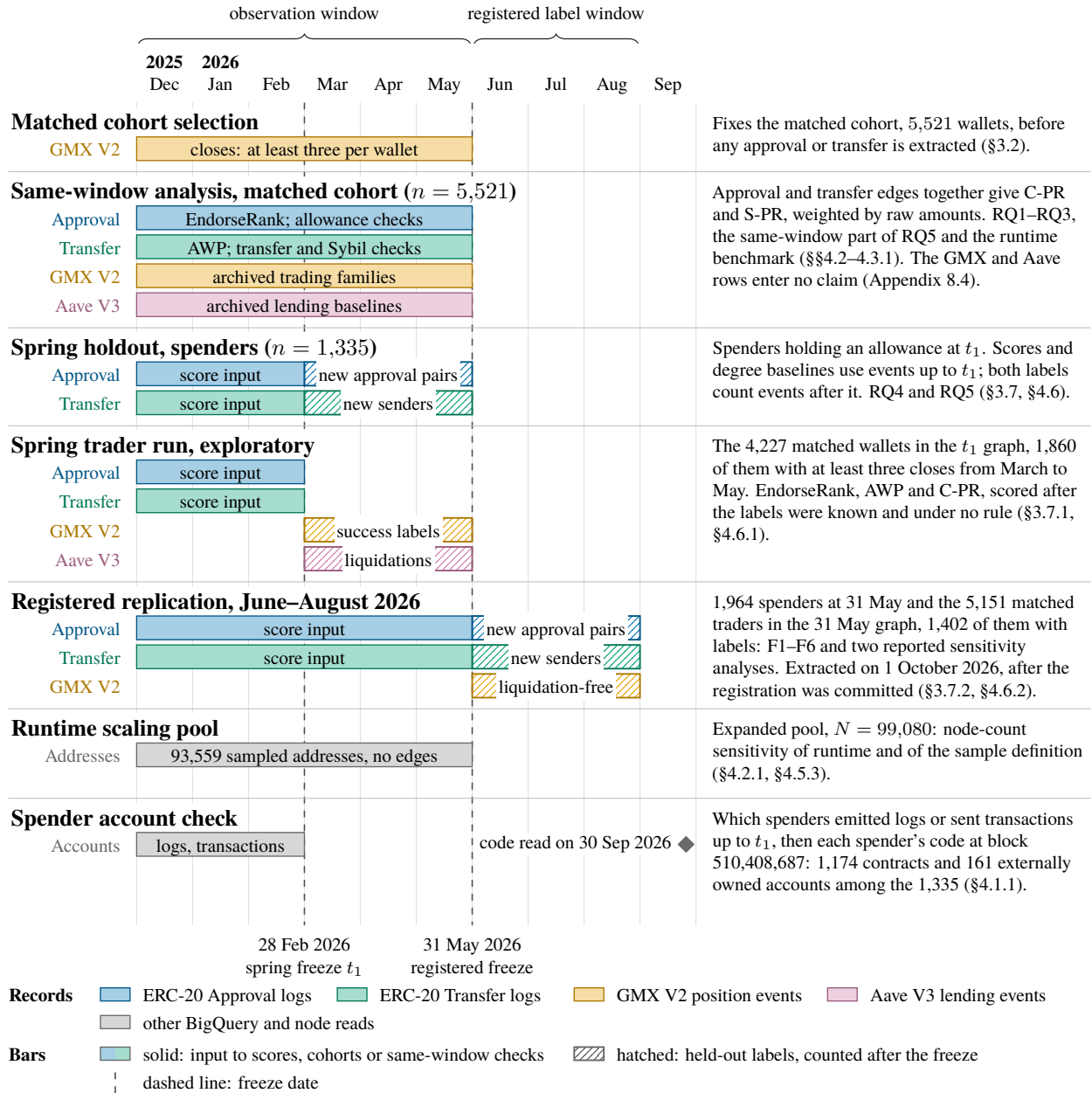


Figure 3.1: Blockchain data used in this thesis, all from Arbitrum One. Each lane is one analysis, and each bar is one kind of on-chain record placed on the months it covers. Solid bars are inputs to scores, to cohorts or to same-window checks. Hatched bars are labels counted after a freeze date and kept out of the scores; the June–August logs were extracted on 1 October 2026, after the replication was registered. The text in a bar says what the records compute, and the right-hand column names the cohort, the research questions and the sections that report the results. The GMX records select the matched cohort, and the approval and transfer queries are filtered to it.

UTC; the queries take 2026-06-01 as an exclusive end. Six months is long enough for the tenure and active-month proxies to vary, and both graphs use exactly the same dates. Extraction runs month by month to stay within the per-query byte budget (Section 3.3). The time decay is anchored at $T_{\text{obs}} = 2026-05-31\ 23:59:59\ \text{UTC}$, so an event on the final day has $0 \leq \Delta t < 1$ and one on the first day is about 182 days old.

- BigQuery source: Logs are read from the public BigQuery table

`bigquery-public-data.goog_blockchain_arbitrum_one_us.logs.`

One record is one event emitted by a contract, with its block timestamp, address, topics and data. Queries filter by block timestamp, by contract address where relevant, and by event topic hashes. Anyone with a Google Cloud account can rerun the same query against the same table; Appendix 8.2 shows the query patterns.

- Reputation events: EndorseRank uses ERC-20 Approval logs^[18], which also cover approvals made through EIP-2612 permit^[19], and AWP uses Transfer logs. Only events in which a cohort wallet is owner or spender, or sender or receiver, are kept. This bounds the extracted volume and makes each graph a one-hop ego network around the cohort, since an edge between two wallets outside it is never extracted. Most allowance-graph owners therefore have no in-edges (4,479 of the 4,590 at the end of the window), and the allowance walk, with a depth of about one, does not test multi-hop propagation (Section 5.3.2).
- Matched cohort: Alignment is applied to the $n = 5,521$ GMX V2 wallets on Arbitrum One with at least three closes between 2025-12-01 and 2026-05-31, a close being one GMX PositionDecrease event, so that partial closes and liquidations count. The list comes from GMX event data alone and is fixed before any approval or transfer is extracted, so the cohort does not depend on the reputation graphs and differences between scores cannot come from sampling. A cohort wallet need not appear in either graph: 966 are touched by no allowance edge, 381 by no transfer edge and 370 by neither, and the method concerned scores such a wallet 0 (Section 3.6).

- Exclusions: Mempool transactions, off-chain order books, centralized exchange accounts and bridges lie outside the data, and no attempt is made to merge addresses controlled by one operator into one entity.

3.3 Two-phase collection strategy

Data collection and evaluation run as two separate phases, so the tables, benchmarks and robustness runs can be rebuilt as often as needed without cloud costs, by us or by anyone who has run the extraction.

- Phase 1 (reputation streams): With the cohort fixed from GMX data, we extract every `Approval` event in which a cohort wallet is owner or spender and every `Transfer` event in which it is sender or recipient, one query per month, each with its own dry-run cost estimate. Approvals are reduced to the most recent allowance per owner–spender–token triple in (block number, log index) order, and an observed allowance of 0 drops the edge. Transfer amounts stay in raw base units, and zero-value transfers are dropped. These tables are the only inputs to graph construction.
- Phase 2 (local evaluation): From the Phase 1 parquet files a local machine computes the scores, the checks, the alignment statistics, the benchmarks and the robustness runs. The expanded pool is formed from addresses in the reputation subgraphs and in supplemental parquet files, so no second chain scan is needed.
- Cost guardrails: A query whose dry-run estimate exceeds a per-query budget of 150 GiB¹ is aborted unless the operator confirms it with an extra flag. For each query the wallets scanned, rows returned, bytes processed and query time are logged with the pipeline (Appendix 8.1), so the cost of acquisition can be audited.

¹This per-query byte limit is enforced by the study’s own pipeline; BigQuery does not impose it.

3.4 Pipeline architecture

Figure 3.2 traces the whole pipeline, from public logs to the research tables. All remote steps write immutable parquet files. Given its inputs and the versioned configuration file, every local step is deterministic.

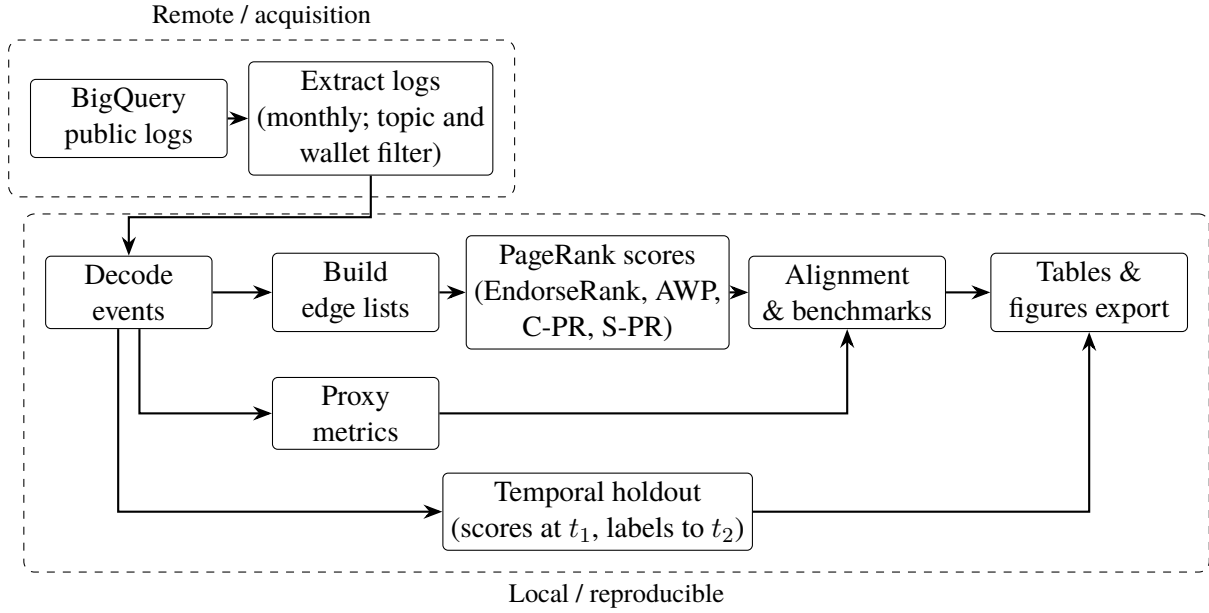


Figure 3.2: Pipeline overview. Approval and transfer logs are filtered and extracted remotely in monthly chunks. Locally, the events are decoded and turned into edge lists, from which EndorseRank, AWP, C-PR and S-PR are computed; the proxy metrics come from the same events on a parallel branch. The alignment and benchmark calculations and the temporal holdout follow, and the result tables and figures are exported last.

Phase 1 scripts write monthly shards and the aggregated latest-allowance and transfer tables, which the Phase 2 scripts read to build the graphs, compute the scores and produce the tables of Chapter 4. The whole pipeline can also run offline on a synthetic fixture dataset.

3.5 Event decoding

A log emitted by a smart contract has topics and a data payload but no field names. Decoding turns each row into a typed record for graph construction and the checks, and an error here would corrupt

both graphs silently. The topic hashes and field mappings are listed below.

- Log anatomy: Each log records the emitting contract, a sequence of topics and an ABI-encoded data blob. The first topic is the keccak-256 hash of the event signature; indexed parameters follow as 32-byte words (addresses left-padded), and non-indexed parameters go into the data field in ABI order ^[18]. ERC-721 tokens emit `Transfer` and `Approval` under the same signatures but put the token ID in a fourth topic and leave the data empty, so their logs decode to zero and are dropped with the other zero-value records. SQL filters by topic, and Python decodes the numeric fields.
- ERC-20 `Approval`: Its topic hash is

`0x8c5be1e5ebec7d5bd14f71427d1e84f3dd0314c0f7b2291e5b200ac8c7c3b925,`

the keccak-256 hash of `Approval(address,address,uint256)`. Owner and spender occupy Topics 1 and 2, and the allowance is decoded from the data field in the token's base units, without scaling by decimals. Unlimited allowances (`type(uint256).max`) keep their raw value. A `permit` emits the same `Approval` event, so an allowance counts whether it came from an on-chain `approve()` call or from an off-chain signature ^[19].

- ERC-20 `Transfer`: Its topic hash is

`0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef,`

the keccak-256 hash of `Transfer(address,address,uint256)`. Sender and recipient are indexed, and the value stays in base units. Mint and burn logs, which use the zero address, are kept whenever a cohort wallet is involved. Self-transfers stay in the events table but are left out of the inbound-counterparty computations (Section 3.9), since they reveal no counterparty.

- Address normalization and units: Addresses are lower-cased before the join. Amounts are the `uint256` value as a double-precision float in raw base units, not divided by decimals, converted to USD, capped or winsorized, so an unlimited allowance enters as $2^{256} - 1 \approx 1.16 \times 10^{77}$ (Sections 3.6.2 and 3.6.3 give its effect).

3.6 Graph construction

EndorseRank and AWP weight their edges the same way and run the same PageRank solver, Equation (3.1); they differ in the events that make the edges and in where the walk restarts. Each method keeps the edges with at least one endpoint in the seed wallet set $\mathcal{W}$, which brings in their one-hop neighbors as nodes, and reports a score for every wallet in $\mathcal{W}$; a wallet missing from the subgraph scores 0.

Edges in EndorseRank stand for endorsements and edges in AWP for transfers (Section 2.7). Both methods weight an event by its age, so that recent events count for more, and pass its amount through a bounded function, so that one event counts at most once. AWP is described first, since EndorseRank takes its weights from it.

3.6.1 AWP transfer graph

AWP is the method of Do, Do and Nguyen, presented at IEEE RIVF 2023^[3]. A transfer e of token amount x_e , made Δt_e days before the observation window closes at T_{obs} , enters the edge from its sender to its receiver with the weight $\sigma(\Delta t_e) V(x_e)$. The time weight is the logistic decay

$$\sigma(\Delta t) = \frac{1}{1 + \exp(k(\Delta t - t_0))}, \quad k = 0.01, \quad t_0 = 180 \text{ days}, \quad (3.3)$$

and the value weight is

$$V(z) = \frac{2}{1 + e^{-bz}} - 1, \quad b = 1, \quad (3.4)$$

which is 0 at $z = 0$ and rises towards 1 as the amount grows. The edge weights are sums over transfers:

$$w_T(u, v) = \sum_{e: u \rightarrow v} \sigma(\Delta t_e) V(x_e). \quad (3.5)$$

The sum runs over every transfer from u to v in the window, whatever the token. A self-transfer ($u = v$) is kept and becomes a self-loop on u . The walk restarts at an address in proportion to its

activeness,

$$X_u = \sum_{v \neq u} \max_{e: u \rightarrow v} \sigma(\Delta t_e), \quad (3.6)$$

which adds up, over the addresses that u has sent to, the time weight of its latest transfer to each. The restart distribution of Equation (3.1) is $\pi_u = X_u / \sum_{v \in V_T} X_v$, and it also takes the mass of the dangling nodes. An address that never sends to another has $X_u = 0$ and receives no restart mass, so its score comes from its in-edges alone. The damping factor is $d = 0.85$.

The paper scores Ethereum transactions and gives no parameter values; its time weight is the logistic function of Equation (3.3) written in calendar time instead of age, and k , t_0 and b are this study's choices. With amounts in raw base units (Section 3.5), $V(x_e)$ is within 10^{-4} of 1 for every transfer of 10 base units or more, which covers 99.5% of the transfers in the window. An edge weight is therefore, in effect, the number of transfers on it, each counted by its time weight, and the activeness of an address is the number of addresses it has sent to, counted the same way. Over the ages in the window the decay is close to linear (Figure 3.3): the weight falls from $\sigma(0) = 0.858$ through $\sigma(90) = 0.711$ to $\sigma(182) = 0.495$, and only events from the first two days of the window are older than t_0 .

Algorithm 1 states the construction and scoring steps formally.

3.6.2 EndorseRank allowance graph

EndorseRank puts AWP's edge weights on the allowance graph. Let $\mathcal{A}$ denote the set of latest allowances after preprocessing. For each (token, owner–spender) triple only the most recent Approval event inside the observation window is kept, in (block number, log index) order, and a zero allowance deletes the edge. The surviving allowance $a^*(u, v, \text{token})$ was set $\Delta t^*(u, v, \text{token})$ days before T_{obs} , and it enters the edge as one event of that age and amount:

$$w_A(u, v) = \sum_{\text{token}} \sigma(\Delta t^*(u, v, \text{token})) V(a^*(u, v, \text{token})), \quad (3.7)$$

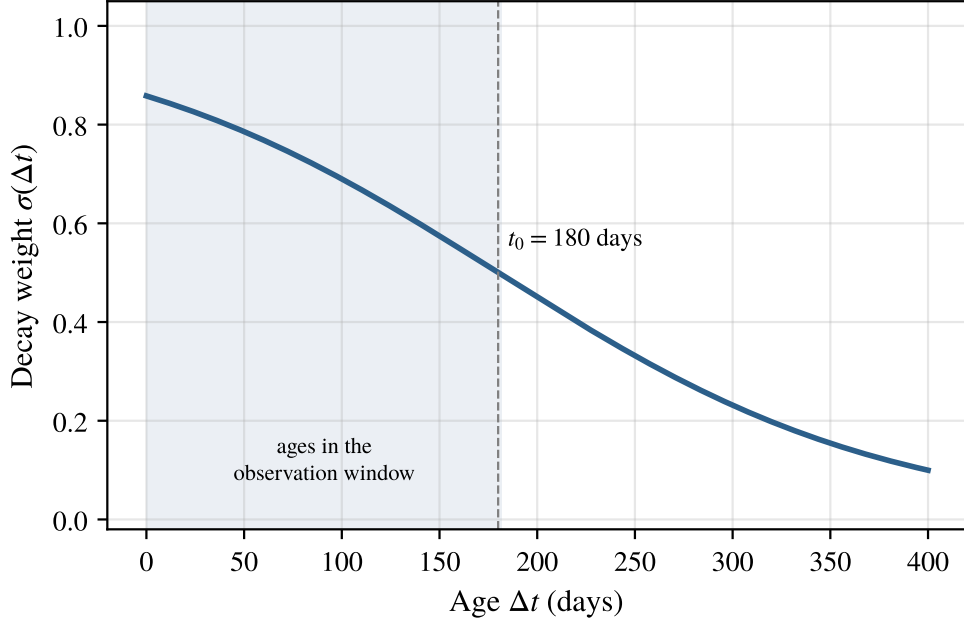


Figure 3.3: Logistic time-decay $\sigma(\Delta t)$ of AWP and EndorseRank ($k = 0.01$, $t_0 = 180$ days). The horizontal axis shows event age in days before 2026-05-31 UTC, and the vertical axis shows the time weight of a transfer or an approval. The shaded band marks the ages that occur in the observation window (0 to 182 days), over which the weight falls only from 0.86 to 0.50.

Algorithm 1 AWP graph construction and scoring

Require: Transfers $\mathcal{T}$, seed wallets $\mathcal{W}$, observation end T_{obs} , parameters $k, t_0, b, d, \varepsilon, I_{\text{max}}$

Ensure: Scores s_{AWP} on $\mathcal{W}$

- 1: $E \leftarrow \emptyset$; $X_u \leftarrow 0$ for every address u
 - 2: **for** each transfer $e = (u, v, x_e, t_e)$ in $\mathcal{T}$ **do**
 - 3: $\Delta t \leftarrow (T_{\text{obs}} - t_e)$ in days
 - 4: accumulate $\sigma(\Delta t) V(x_e)$ on edge $(u \rightarrow v)$ ▷ Equations (3.3)–(3.5); $u = v$ gives a self-loop
 - 5: **end for**
 - 6: **for** each sender u **do**
 - 7: $X_u \leftarrow \sum_{v \neq u} \max_{e: u \rightarrow v} \sigma(\Delta t_e)$ ▷ Equation (3.6)
 - 8: **end for**
 - 9: remove non-positive edges; retain edges incident to $\mathcal{W}$
 - 10: $\pi_u \leftarrow X_u / \sum_v X_v$ for the nodes u of the retained edges
 - 11: $\mathbf{s} \leftarrow \text{WeightedPageRank}(E, \boldsymbol{\pi}, d, \varepsilon, I_{\text{max}})$ ▷ Equations (3.1)–(3.2)
 - 12: **return** $s_{\text{AWP}}[i] \leftarrow \mathbf{s}[i]$ for $i \in \mathcal{W}$ (else 0)
-

with σ , V , k , t_0 and b as for AWP. A directed edge $u \rightarrow v$ means that owner u endorses spender v . Almost every allowance is far above 10 base units (99.9% of them), so $w_A(u, v)$ is in effect the number of tokens on which u has approved v , each counted by the age of its approval. An unlimited allowance counts no more than a small one. The damping factor is $d = 0.85$.

The walk restarts uniformly. AWP’s restart rule, carried over to allowances, would restart the walk at owners in proportion to the number of spenders they approved recently. That rewards granting endorsements, while EndorseRank is meant to rank the wallets that receive them. We made this choice after scoring both restart rules on the data of this study, so the variant with AWP’s restarts is reported next to EndorseRank (Table 4.9).

The weight keeps the amount and the time of the latest approval; it is not the allowance in force at the end of the window. Within a pair, on each token, the newest approval replaces the older ones^[18]. Approvals granted before 1 December 2025 are not observed, a finite allowance that transferFrom has since reduced keeps its approved value, and an owner who approves an allowance-manager contract such as Uniswap’s Permit2 appears only with an edge to that contract, not to the application that later spends through it.

The steps for building the graph and running PageRank are listed in Algorithm 2.

Algorithm 2 EndorseRank graph construction and scoring

Require: Latest allowances $\mathcal{A}$, seed wallets $\mathcal{W}$, observation end T_{obs} , parameters $k, t_0, b, d, \varepsilon, I_{\text{max}}$

Ensure: Scores s_{ER} on $\mathcal{W}$

- 1: $E \leftarrow \emptyset$
 - 2: **for** each latest allowance $(u, v, \text{token}, a^*, t^*)$ in $\mathcal{A}$ **do**
 - 3: $\Delta t^* \leftarrow (T_{\text{obs}} - t^*)$ in days
 - 4: accumulate $\sigma(\Delta t^*) V(a^*)$ on edge $(u \rightarrow v)$ ▷ Equation (3.7)
 - 5: **end for**
 - 6: remove non-positive edges; retain edges incident to $\mathcal{W}$
 - 7: $\mathbf{s} \leftarrow \text{WeightedPageRank}(E, \text{uniform } \boldsymbol{\pi}, d, \varepsilon, I_{\text{max}})$ ▷ Equations (3.1)–(3.2)
 - 8: **return** $\text{s}_{\text{ER}}[i] \leftarrow \mathbf{s}[i]$ for $i \in \mathcal{W}$ (else 0)
-

Approvals are rare and transfers common, so on the matched cohort the approval subgraph is much sparser (Chapter 4); since one PageRank iteration costs $O(|E|)$, this sparsity is the structural basis

of H3.

For both methods, edge lists are stored as (from, to, weight) triples; duplicate edges are merged by summing their weights, and non-positive weights are dropped before the solve. The solver builds a sparse compressed-row transition matrix and the restart vector, iterates Equation (3.1) and returns a score for each node, and the cohort scores are aligned to $\mathcal{W}$ with zeros for missing entries.

3.6.3 Coupled operator over both layers (C-PR) and the seeded ablation (S-PR)

EndorseRank and AWP each walk on a single edge type, and RQ5 asks whether one walk over both edge types ranks wallets better than a walk over one. The coupled operator was fixed on 14 September 2026 on two layers that weight each event by its raw amount:

$$\tilde{w}_A(u, v) = \sum_{\text{token}} a^*(u, v, \text{token}), \quad \tilde{w}_T(u, v) = \sum_{e: u \rightarrow v} x_e \sigma(\Delta t_e). \quad (3.8)$$

The allowance layer has the edges of EndorseRank and the transfer layer those of AWP; only the weights differ. Amounts enter in raw base units, with no bounded transform, decimal adjustment or cap, and the allowance layer carries no time decay. Both decision rules for RQ5 were fixed on these layers, so the operator is used as it was specified. The two layers form one graph on the union node set $V = V_A \cup V_T$, with transition matrices P_A and P_T from Equation (3.2). For wallet u the coupled row is a convex mixture of the two layer rows, renormalized over the layers in which u has out-edges:

$$P_{u,\cdot}^{(\lambda)} = \alpha_A(u) P_{A,u,\cdot} + \alpha_T(u) P_{T,u,\cdot}, \quad (3.9)$$

$$\alpha_A(u) = \frac{\lambda \mathbf{1}[o_A(u) > 0]}{D_\lambda(u)}, \quad \alpha_T(u) = \frac{(1 - \lambda) \mathbf{1}[o_T(u) > 0]}{D_\lambda(u)},$$

where $D_\lambda(u) = \lambda \mathbf{1}[o_A(u) > 0] + (1 - \lambda) \mathbf{1}[o_T(u) > 0]$, and $o_A(u)$ and $o_T(u)$ are the out-strengths of u in the two layers. A wallet with out-edges in only one layer follows that layer with weight one. A layer of weight 0 is dropped with its edges and nodes, and a wallet with $D_\lambda(u) = 0$ is dangling,

its mass handled by the third term of Equation (3.1). Teleportation is uniform, $d = 0.85$, and $\mathbf{s}_{\text{CPR}}$ is the PageRank vector of Equation (3.1) with $P = P^{(\lambda)}$, under the solver settings of the single-layer methods.

At $\lambda = 1$ the operator walks the allowance layer alone and at $\lambda = 0$ the transfer layer alone; we write these walks C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$). They are the comparators of both rules for RQ5 (Sections 3.7 and 3.7.2), and the registration documents call them EndorseRank and AWP. Unit tests check on a fixture that each end point gives the scores of its layer walked alone with uniform restarts. The family is not continuous at the end points. For $0 < \lambda < 1$ the node set is $V_A \cup V_T$ and a node with out-edges in only one layer follows it whatever λ is, whereas at $\lambda = 1$ a node whose out-edges are all transfers is dangling or absent; as λ approaches 1, C-PR therefore tends to a walk that still follows the transfer edges of such nodes, and the same holds at the other end. The values $\lambda = 0.5$ (primary) and $\lambda \in \{0.25, 0.75\}$ (sensitivity) were fixed in the configuration on 14 September 2026, before any coupled score had been computed.

Raw amounts make the allowance layer very uneven. Of the 21,585 latest allowances, 13,055 (60.5%) are unlimited, and in the allowance subgraph 51% of the edges carry one; 3,986 of the 4,590 owners hold at least one. In double precision a finite allowance next to an unlimited one gets a negligible share of the owner’s row, so such an owner’s row is split among its unlimited approvals, and the finite allowances of the 1,987 owners who hold both kinds have almost no effect. In the transfer layer, amounts of different tokens are added on their raw scales, so a token with many decimals outweighs one with few. EndorseRank and AWP avoid both effects through V .

In the S-PR ablation the transfer layer is walked, but the uniform teleportation vector π of Equation (3.1) is replaced by the score of C-PR ($\lambda = 1$), restricted to V_T and renormalized over it. Let $\mathbf{r}_A$ be the PageRank vector of the allowance layer walked alone on V_A , extended by $r_{A,u} = 0$ for $u \notin V_A$. Then

$$\pi_u = \frac{r_{A,u}}{\sum_{v \in V_T} r_{A,v}} \quad (u \in V_T),$$

so a transfer-graph node without an allowance score receives zero restart mass. If no node of V_T

lies in V_A , the denominator is zero and π falls back to the uniform distribution on V_T . The same vector redistributes dangling mass. The result is the TrustRank^[17] construction, but with allowance-derived seeds instead of hand-labeled ones. Allowances decide where the walk restarts; transfers decide how it spreads. A comparison of S-PR with C-PR separates what the allowance edge contributes as a propagation medium from what it contributes as a seed.

The edge set determines what the coupled operator costs. Because C-PR walks on $E_A \cup E_T$, one of its iterations costs as much as an iteration on each layer, and its runtime is expected to be at or above AWP's. The speed result of Claim C1 is therefore EndorseRank's alone. S-PR needs one solve on each layer. Both operators are timed under the benchmark protocol of Section 3.11.

3.7 Temporal holdout protocol

In the spring holdout, scores use every extracted event from the start of the data, December 1, 2025, up to $t_1 = \text{February 28, 2026, 23:59:59 UTC}$. Labels are counted on events from March 1, 2026, 00:00:00 UTC, up to $t_2 = \text{May 31, 2026, 23:59:59 UTC}$. The cohort consists of the $n = 1,335$ spenders that hold at least one positive latest allowance at t_1 in the extracted approval data. Every score in the holdout table (EndorseRank, AWP, C-PR at the three values of λ and at its two end points, S-PR and two raw-degree baselines) is computed from events up to t_1 alone. The time decay is anchored at t_1 as well: Δt in Equation (3.3) is measured back from t_1 , so ages stay below 90 days and the weights lie between about 0.71 and 0.86.

We count two labels per wallet, one for each construct, so that in the future window each single-layer method is tested on its own construct and the hybrids on both:

1. New approval pairs: the count of owner–spender pairs, attributed to the spender, that receive a positive approval in the label window but were absent from the latest-positive snapshot at t_1 . This label measures the endorsement construct.
2. New transfer senders: the count of distinct addresses that send the wallet a transfer in the label window without having sent it one between the data start (December 1, 2025) and t_1 ,

self-transfers excluded. It measures the flow construct and is the transfer-side counterpart of the first label.

For baselines we take each wallet’s in-approve degree (the number of distinct owners with a positive allowance) and its transfer in-degree (the number of distinct senders other than the wallet itself), both at t_1 . Each single-layer PageRank smooths one of these local statistics. Because their rows appear in the table, the increment of each PageRank over its own raw degree can be read off directly. That increment answers RQ4; the PageRank coefficient on its own does not.

Uncertainty comes from the paired wallet bootstrap of Section 3.10 ($B = 400$, one shared index matrix), so every difference between two coefficients on the same label is a paired difference. The contrasts fixed before the coupled scores were computed concern the coupled operator and its layers: C-PR ($\lambda = 1$) minus in-approve degree on new approval pairs; C-PR minus C-PR ($\lambda = 1$) on new approval pairs, with C-PR minus C-PR ($\lambda = 0$) on new transfer senders; and the same two contrasts for S-PR. The contrasts that involve EndorseRank and AWP, each against its own degree and C-PR against each of them, were computed after the labels were known; the differences between EndorseRank and AWP are in Appendix 8.5.

The first rule for RQ5, fixed on 14 September 2026, judges C-PR against its own layers walked alone and has two parts. The primary part uses the holdout: C-PR at $\lambda = 0.5$ should do no worse than C-PR ($\lambda = 1$) on new approval pairs (the interval of $\Delta\tau = \tau_{\text{CPR}} - \tau_{\lambda=1}$ contains zero or lies above it) and should be better than C-PR ($\lambda = 0$) on new transfer senders (the interval of $\tau_{\text{CPR}} - \tau_{\lambda=0}$ lies above zero). The secondary part uses the same window and the matched cohort. Its condition is that C-PR’s point estimate lies within the 95% interval of the better of the two layers on the transfer family and on the allowance family, and that C-PR exceeds both layers on the Sybil-stability family. The paired contrasts of Table 4.4 are reported beside this rule as a stricter check; they do not decide it. If its primary part fails, the second half of RQ5, whether C-PR beats a walk on either layer alone, receives a negative answer, and the result is reported with its intervals. The first half, whether C-PR improves on a walk over the transfers alone, is not covered by this rule; Section 3.7.2 describes how it is tested. Once labels have been observed, λ is not tuned. When this rule was fixed, the spring

labels and the holdout results of the two layers walked alone were already known; only the coupled scores had not been computed.

The pipeline also computed a revoke rate, an approval-then-outbound drain heuristic and Aave borrower liquidations over the label window. None of them is offered as supporting evidence. Chapter 4 gives the revoke and drain coefficients, which rise with the number of owners a spender has and so cannot separate trust from use; on this spender set the Aave events were either empty or measured the wrong construct.

3.7.1 Trader outcomes in the spring window

The spender cohort says little about the traders who make up the matched cohort, because most of them receive no approvals. On 29 August 2026, before the coupled operator existed, an exploratory run applied the spring freeze to the matched cohort and chose GMX success labels counted from March to May: the number of profitable closes, the realized gain and the share of closes that were profitable. The cohort is the 4,227 matched wallets in the freeze-date graph, and the labels are defined for the 1,860 of them with at least three closes in that window. EndorseRank, AWP and C-PR were scored on this cohort after the labels were known, with the 400 paired resamples of the spender holdout, and Chapter 4 reports the run as exploratory. It has no GMX liquidation label, and its Aave liquidation label covers too few wallets to read.

3.7.2 Registered replication: June to August 2026

The spring holdout also showed something that no rule had asked about. Against its transfer layer walked alone, C-PR ranked the spenders that later gained new approvals far better, and it was level with that walk on new senders. We noticed this only after the labels were in, so it counts as a hypothesis and not as a finding, and it is tested again on a window that had not been extracted when the test was fixed. The design was written on 28 September 2026 in a configuration file and a written plan, both committed to the public repository before the extraction (Appendix 8.1). On 1 October 2026 the status in the configuration was set to registered in a commit to the same

repository, and the June–August logs were extracted later that day. The extraction script refuses to run until both files are committed with the status set to registered; a testing override exists, and any use of it is written into the extraction manifest, which records none.

The method is unchanged. C-PR is used at $\lambda = 0.5$, as fixed on 14 September, together with the two sensitivity values and S-PR. The solver settings, the SQL, the extraction filter of 5,521 wallets, the GMX decoder and the paired bootstrap (400 resamples, seed 42) are those used for the spring. The score window is longer: scores are frozen on 31 May 2026 at 23:59:59 UTC and use every extracted event from 1 December 2025 onwards, so the ages of transfers and approvals run up to 182 days instead of 90. Labels are counted from 1 June to 31 August 2026.

Two cohorts are evaluated. The spender cohort is defined as before: every spender holding at least one positive latest allowance at the freeze. The trader cohort is the matched cohort restricted to the wallets in the freeze-date graph, and its membership rests on closes made before the freeze. The spenders keep their two labels. The traders get the liquidation-free close rate, one minus the share of a wallet’s June–August closes that ended in liquidation, defined for wallets with at least three closes in the window. It is the nearest thing to a default that these data contain, it is built from neither graph, and because it is a rate it does not reward a wallet merely for trading more. Two things about it were known when it was chosen. One is its prevalence. From March to May, 1,434 of the 2,962 matched wallets with at least three closes had at least one liquidation; among the 1,860 of them that were in the freeze-date graph, which is how the trader cohort is defined, 923 did. The other is its same-window counterpart. The liquidation-free rate over December to May is one of the three proxies behind the archived liquidation family of Table 8.2, and on its own it gave $\tau = 0.075$ for the allowance layer walked alone and 0.083 for the transfer layer walked alone. It had never been computed for C-PR, and never out of window.

The rule compares C-PR with its transfer layer walked alone, C-PR ($\lambda = 0$), through the paired difference $\Delta\tau = \tau_{\text{CPR}} - \tau_{\lambda=0}$:

- F1, spenders, new approval pairs: the 95% interval of $\Delta\tau$ lies above zero;

- F2, spenders, new transfer senders: the lower end of the interval lies above -0.02 ;
- F3, traders, liquidation-free close rate: the lower end of the interval lies above -0.02 .

C-PR counts as improving on a walk over the transfers alone only if all three hold, so no correction for multiple testing is needed. The margin of 0.02 is the half-width of the paired interval for the F2 contrast in the spring; a smaller difference could not be resolved at these sample sizes. One weakness of F3 was written down with the rule. If neither score predicts liquidation, F3 can pass with both coefficients near zero, so the two coefficients are reported next to the verdict. Three further results are reported in a fixed order and do not enter the decision: C-PR against C-PR ($\lambda = 0$) on the liquidation-free close rate as a superiority test (F4), C-PR against C-PR ($\lambda = 1$) on the same label (F5), and the primary part of the 14 September rule, rerun on the new window (F6).

Two sensitivity analyses were added on 1 October 2026, after AWP had been scored on the spring data and before any June–August log was extracted. Both are reported next to the decision, and neither enters it. The first repeats F1–F4 with AWP (Section 3.6.1) in place of C-PR ($\lambda = 0$). The second repeats F1–F6 with the cohort wallets that a graph leaves out kept as isolated nodes instead of scored zero (Section 4.4). Only the two layers walked alone change under the second rule, since every spender is in the allowance graph and every trader in the cohort is in the union graph that C-PR walks. EndorseRank was not among the registered scores. It was scored on the same cohorts, labels and wallet resamples after the registered evaluation, and those results are reported apart from the decision.

3.7.3 Outcomes built from neither edge type

A comparison of the two edge types on the GMX trader labels, which are built from neither, was fixed in a plan committed on 9 October 2026 (commit 20e250f), after the registered labels and the single-score coefficients of Tables 4.17 and 4.16 were known but before the analysis ran; it is a pre-specified post hoc analysis of existing data ^[75]. It reuses the trader cohorts and label code of Sections 3.7.1 and 3.7.2 in two windows: W1, frozen on 31 May 2026 with labels from June to August (1,402 traders), and W0, frozen on 28 February 2026 with labels from March to May (1,860

traders).

Four pairs set allowance against transfer: in-approve degree and transfer in-degree; EndorseRank with AWP's activity restarts and AWP, the like-for-like PageRank pair; EndorseRank and AWP; and the raw-amount layers C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$). C-PR ($\lambda = 0$) minus AWP isolates AWP's weights and restarts. All six trader labels are reported: the liquidation-free close rate (primary), a no-liquidation flag, profitable closes, realized gain, and the profitable and non-loss shares of closes. Two contrasts on the liquidation-free close rate in W1 decide: P, in-approve degree minus transfer in-degree, and Q, EndorseRank with activity restarts minus AWP, each with a Bonferroni 97.5% interval from $B = 2,000$ paired resamples (seed 42). Because the chance of at least one liquidation rises with the number of closes, τ_b is also computed within strata of 3–4, 5–9, 10–24 and 25 or more closes in the label window and weighted by each stratum's wallet pairs. Recipients (traders with an in-approve degree above zero at the freeze) are compared with the other traders and matched to the account types read for the spring spender cohort (Section 4.1.1); an influence check drops the 10 recipients of highest in-approve degree.

R1, a count-level lead on liquidation avoidance, needs the 97.5% interval of P and the 95% interval of the stratified P above zero in W1, and a positive P in W0 (and on externally owned recipients alone, if there are at least 30). R2, a PageRank-level lead, needs the 97.5% interval of Q above zero in W1 and a positive Q in W0. R3 applies if P on the profitable or the non-loss share in W1 is below zero with a 95% interval excluding zero; the allowance signal is then read as specific to liquidation avoidance and as running against loss avoidance. The wording of the conclusion for each outcome was fixed in the same commit.

3.8 Construct-check design

Every score enters the same-window evaluation: EndorseRank, AWP, C-PR at its five values of λ and S-PR. For each score M and check q the pipeline reports Spearman's ρ and Kendall's τ between $\mathbf{s}_M$ and $\mathbf{p}_q$ with 95% bootstrap intervals (Section 3.10), each check oriented so that a higher value

means higher expected reputation, as in the AWP validation protocol^[3].

The same-window evaluation is organized around two contemporaneous families:

1. Transfer (the proxies used to validate AWP^[3]): the wallet's inbound transfer activity.
2. Allowance: inbound approval activity, where the wallet is the spender.

Both families are graph-adjacent: their inputs are the event classes that make up G_{transfer} and G_{approve} , although each check is a local statistic and not a global score. Strong alignment of AWP with the transfer checks, or of EndorseRank with the allowance checks, shows coherence within a construct and is not evidence of independent predictive power. A third family, the Sybil-adjusted activity summaries of Section 3.9.3, enters the secondary rule for C-PR, but no hypothesis rests on it.

The temporal holdout of Section 3.7 provides the out-of-window check. Packin and Lev-Aretz^[2] warn that decentralized credit scores can become opaque; publishing explicit formulas (Section 3.9) and keeping same-window checks apart from the time split reduce that opacity.

3.9 Operational definitions of proxies

In what follows, $\mathcal{W}$ denotes the matched cohort. The notation is that of the check-computation module in the open evaluation pipeline.

3.9.1 Transfer family

Write $\mathcal{T}_{\text{in}}(w)$ for the set of transfers received by wallet w in the window and u_e for the sender of transfer e . Self-transfers belong to $\mathcal{T}_{\text{in}}(w)$, so a wallet that sends tokens to itself counts itself as one sender, and amounts x_e are summed in raw base units across tokens. Define

$$\text{in_degree}(w) = |\{u_e : e \in \mathcal{T}_{\text{in}}(w)\}|, \quad (3.10)$$

$$\text{in_value}(w) = \sum_{e \in \mathcal{T}_{\text{in}}(w)} x_e. \quad (3.11)$$

Both are baselines defined in the AWP paper^[3], on the premise that economically important addresses receive transfers from more distinct counterparties and in greater total value. Unlike PageRank they are local and ignore the senders' reputations, so comparing AWP with them shows how far propagation, recency weighting and restarts at active senders move its ordering away from the raw inflow.

3.9.2 Allowance family

On the allowance side, w is the spender (owner $\rightarrow w$). With a^* the latest allowances of Equation (3.7), define

$$\text{in_approve_degree}(w) = |\{u : a^*(u, w, \cdot) > 0\}|, \quad (3.12)$$

$$\text{in_approve_value}(w) = \sum_{u, \text{token}} a^*(u, w, \text{token}). \quad (3.13)$$

Like the transfer baselines, they count and sum incoming approvals without following multi-hop endorsement chains.

3.9.3 Sybil-adjusted stability family

The inbound counters leave out self-transfers ($u = w$). Let $N_{\text{tx}}(w)$ be the number of transfers that w receives from other addresses and $N_{\text{uniq}}(w)$ the number of distinct inbound counterparties. Tenure and active months use a second set of rows. Every transfer that involves a cohort wallet is attributed to its sender if the sender is in the cohort, and otherwise to its recipient; a transfer between two cohort wallets thus counts only for the sender. We take $t_{\min}(w)$ and $t_{\max}(w)$ to be the earliest and latest timestamps among the rows attributed to w . $N_{\text{mon}}(w)$ is the larger of two counts of distinct calendar months: the months in which w received a transfer from another address, and

the months of the rows attributed to w . Define

$$\text{inbound_counterparty_ratio}(w) = \frac{N_{\text{uniq}}(w)}{\max(N_{\text{tx}}(w), 1)}, \quad (3.14)$$

$$\text{transfer_tenure_days}(w) = \frac{t_{\max}(w) - t_{\min}(w)}{86400} \quad (\text{seconds to days}), \quad (3.15)$$

$$\text{active_months}(w) = N_{\text{mon}}(w). \quad (3.16)$$

The counterparty ratio penalizes wallets that receive many transfers from few addresses, a simple wash pattern. It does not penalize the usual Sybil pattern, in which many fresh addresses send one transfer each; a wallet fed that way scores close to the maximum of one. Tenure and active months track how persistent on-chain activity is over the observation window. These proxies are therefore weak Sybil checks, and colluding adversaries can satisfy all three. They do, however, operationalize the intuition that sustained, multi-counterparty activity says more about reputation than bursty self-dealing ^[2].

3.10 Statistical tests

Neither the reputation scores nor the validation proxies are binary default indicators; both are continuous and are handled as ranks. Following the AWP validation protocol^[3] and Cronbach and Meehl’s approach to construct validity^[22], concordance is measured with rank correlations, which are invariant under any monotonic re-scaling of either variable. Each coefficient is reported with a bootstrap confidence interval.

- Spearman’s ρ : the Pearson correlation of the mid-ranks R_i and S_i of the raw score and the raw proxy value of wallet i , tied values sharing the average of their positions (Section 2.9). It measures monotonic agreement: if higher reputation consistently goes with higher proxy values, ρ is positive whatever the nonlinearity in the raw scores.
- Kendall’s τ : every reported value is the tie-corrected τ_b of Equation (2.6), computed on the raw scores and proxies. Without ties, $(1 + \tau)/2$ is the probability that a randomly chosen pair

is ordered the same way by both rankings. With ties that reading belongs to $\tau_a = (n_c - n_d)/n_0$ and not to τ_b , and ties are common here: on the matched cohort, 67.3% of all wallet pairs are tied on EndorseRank.

- Why both: the AWP protocol ^[3] reports both, and reporting both keeps the conclusions from hinging on one coefficient. The score distributions are heavy-tailed, with most wallets near zero and a few hubs holding much of the mass, which is why rank coefficients are used instead of a Pearson correlation on raw scores, which the hubs would dominate.
- Family aggregation: A family is summarized by the plain average over its proxies. The transfer-family mean τ , for example, is $\frac{1}{2}(\tau_{\text{in_degree}} + \tau_{\text{in_value}})$. On a fixed cohort and fixed proxies, a comparison between scores is only a relative statement about those scores and implies no universal property of reputation. Our conclusions are drawn from the confidence intervals, and no absolute τ cutoff is used as a decision criterion.
- Uncertainty: Sampling uncertainty is estimated with a paired wallet bootstrap ^[34]. The matched cohort is resampled with replacement $B = 400$ times (seed 42, fixed in the configuration), and one index matrix is reused for every method and proxy, so the resampled statistics are paired across methods. For each proxy τ , family mean and inter-method difference, the 2.5th and 97.5th percentiles of the bootstrap distribution form a 95% percentile interval ^[35]. Because τ_b is a smooth function of U-statistics, the percentile interval is first-order accurate at $n = 5,521$. The small number of resamples keeps the $O(B n \log n)$ cost of the tie-corrected τ tractable across all methods and proxies, at the price of endpoints precise to about ± 0.005 . Only the analysis of Section 3.7.3 departs from this, with $B = 2,000$.
- Difference tests: Every comparison is a paired contrast on the same resamples. Resample by resample we compute $\Delta\tau = \tau_A - \tau_B$ and report the 95% percentile interval of its bootstrap distribution; a contrast counts as a difference when its interval excludes zero. Because resampling is paired, the contrast keeps the covariance of the two estimates, $\text{Var}(\hat{\tau}_A - \hat{\tau}_B) = \text{Var} \hat{\tau}_A + \text{Var} \hat{\tau}_B - 2 \text{Cov}(\hat{\tau}_A, \hat{\tau}_B)$, and with positive covariance its interval is narrower than

the two separate intervals suggest. The contrasts that decide a rule were fixed before the data they are judged on: those of the 14 September rule (Section 3.7) and those of the registered replication (Section 3.7.2). C-PR against its transfer layer walked alone on new approval pairs in the spring was not among them; it is marked as exploratory wherever it appears. In the same window, eight contrasts set C-PR and S-PR against whichever layer, walked alone, leads each family (the transfer walk on transfer, the allowance walk on allowance, both on Sybil stability), and four set C-PR against AWP on transfer, against EndorseRank on allowance and against both on Sybil stability. Five same-window contrasts that involve EndorseRank and AWP, none fixed in advance, are reported in Appendix 8.5. The post hoc contrasts of Section 3.7.3 were fixed after their labels were known.

- Directionality and imputation: Each proxy is oriented so that a higher value means a better reputation. After re-alignment to $\mathcal{W}$, a wallet absent from a given proxy source is imputed as zero. With sparse transactions this is a conservative choice, and both scoring approaches apply it in the same way. After this step every cohort wallet has a value in every score and proxy vector, so each correlation uses the whole of $\mathcal{W}$.
- Non-inferiority: Two of the three conditions of the registered replication ask only that C-PR not fall behind its transfer layer walked alone by more than a margin fixed in advance, $\delta = 0.02$. Such a contrast passes when the lower end of its 95% interval lies above $-\delta$. The test is weaker than a difference test and says nothing about superiority on that label. It is used where the claim under test is that C-PR gives up nothing that matters.
- What is excluded: The study attempts no causal inference, trains no supervised models on separate training and testing sets, and reports no p -values; it asks whether the interval of a pre-specified contrast excludes the null value.

3.11 Computational benchmark (Claim C1)

Claim C1 concerns the computational cost of each score. The benchmark times the solver call that turns an edge list into scores. It leaves out the one-time cost of building the edge lists, mostly loading and preprocessing, which a scoring service that refreshes its scores periodically would amortize.

- **Timed component:** The edge lists and AWP’s restart weights are built before the timer starts. The timed call indexes the nodes, assembles the sparse weighted adjacency matrix, row-normalizes it, forms the restart vector and runs power iteration with $d = 0.85$, $\varepsilon = 10^{-8}$ and $I_{\max} = 300$. In a separate profiling run of the same call, matrix assembly took about 94% of the wall time and the power iterations about 1%. Both steps are $O(|E|)$, and a deployment would repeat both at every update. The same graphs give the alignment scores.
- **Measurement protocol:** Every configuration is measured in one session. An untimed warm-up run, on which alone peak memory is tracked, is followed by five timed runs, from which we report the mean and standard deviation of the run times and the mean iteration count, with the nodes and edges of the input graph. Results are memoized under a key formed from the edge multiset, damping, tolerance, iteration cap and repeat number. The same graph under the same settings occurs in the main benchmark, in the $d = 0.85$ row of the damping sweep, in the in-cohort scaling stage with $n = 5,521$ and in the full-pool stage, whose induced subgraph is the cohort graph (Section 3.12.1); those tables report the same values by design, and they are not independent measurements of one quantity. Every runtime table lists edge counts, since edges drive the cost of an algorithm with $O(|E|)$ work per iteration^[15], and the runtimes are written to the machine-readable output (Section 3.13).
- **Cohorts:** The main benchmark runs on the matched wallet cohort ($n = 5,521$). For node-count sensitivity we draw deterministic subsamples from the expanded wallet pool (Section 3.12).
- **Fairness:** Every algorithm runs with the same solver, tolerance, iteration cap, damping (outside the damping sweep) and seed wallet set, so differences in cost come from the graphs

and from AWP’s restart vector, which can change the number of iterations, and not from the implementation. Hardware and software are recorded in the evaluation manifest, and the cost of two scores is compared through time ratios, each reported beside the corresponding edge ratio.

3.12 Scaling and robustness extensions

We also examine how runtime responds to node count on a larger address set, and whether the family-level pattern survives changes to the damping, the token set and the sample. These analyses qualify Claim C1 and test the reach of the family-level conclusions; neither supplies an alternative primary sample. Unless stated otherwise, alignment claims in Chapter 4 refer to the matched cohort, and node-count sensitivity refers to the expanded pool.

3.12.1 Expanded wallet pool and node-count sensitivity of runtime

We define the expanded wallet pool $\mathcal{W}_{\text{pool}}$ as the union of the matched cohort (5,521 addresses) and 93,559 additional addresses sampled from monthly Arbitrum One ERC-20 activity in the observation window, taken from the supplemental active-wallet parquet that our sampling script produces. This gives $N = |\mathcal{W}_{\text{pool}}| = 99,080$ addresses for the study. Intermediate-sized benchmarks are drawn from the pool by deterministic SHA256 subsampling with seed string `benchmark-tier2-v1`: wallets are sorted by

SHA256(*seed:address*) with *seed* = `benchmark-tier2-v1`,

that is, by the hash of one string made of the seed, a colon and the lower-cased address, and the subsample is the first n wallets in that order. The seed string is a constant fixed in our configuration. We report three stages, $n = 10,000$, $n = 50,000$ and the full pool $n = N = 99,080$, all on the same approval and transfer parquet; in-cohort runs with $n \leq 5,521$ are in Appendix 8.1. Since the hashing is deterministic, the subsamples cannot drift between library versions the way pseudorandom draws do.

Two features of this design limit what the experiment can show. At every stage the graph keeps the edges of the cohort-window approval and transfer parquet that have at least one end point among the sampled wallets, and the solver’s node set consists of the end points of those edges. The edge count $|E|$ at a given stage thus depends on how many cohort wallets the hash order selects, so it does not follow n : for both methods the pool stage with $n = 10,000$ has fewer edges than the in-cohort stage with $n = 2,000$. The other feature concerns the supplemental pool addresses, whose own approval and transfer logs were meant to be extracted in a secondary step. That extraction did not finish (Appendix 8.1). As a result, the full-pool stage adds no edges beyond the cohort parquet, and it reproduces the cohort graph exactly. It adds no nodes either, because a sampled address that touches no retained edge never enters the solver graph; the full-pool allowance graph has the cohort’s 6,442 nodes. The pool stages therefore do not measure how runtime reacts to added isolated nodes, and they do not scale the edge count that drives per-iteration cost. Every scaling table reports per-stage $|V|$ and $|E|$ for both methods, and the scaling statement in Chapter 4 goes no further than those columns allow.

3.12.2 Damping sensitivity

On the matched cohort we vary the damping over $d \in \{0.75, 0.85, 0.95\}$, with the graphs and proxies held fixed. A smaller d raises teleportation; a larger one gives multi-hop paths more weight and may slow convergence. The sweep asks whether alignment on the allowance and transfer families depends on the teleportation probability.

3.12.3 Top-token subgraph

For this check we keep only 20 ERC-20 tokens and recompute EndorseRank and AWP on that subgraph. The first selection takes the tokens with the largest summed amounts over the window, adding raw transfer amounts and raw allowance values. Amounts are raw base units and an unlimited approval counts as $2^{256} - 1$, so this ranking favors tokens with many unlimited approvals and large supplies. A second selection, added after the results were known, takes the tokens with the

most rows (transfer events plus latest allowances). Neither is a ranking by economic value. The check asks whether the family-level results change when every other token is dropped.

3.12.4 Sample-definition sensitivity

At each stage the alignment statistics are recomputed on the subsamples of the expanded pool, drawn with the same deterministic seed. The pool subsamples differ from the cohort in size and in composition: many of their wallets appear in one graph only, so their score and proxy in the other graph are zero, and ties at zero inflate rank agreement mechanically. The per-stage coefficients are therefore a sensitivity check on the sample definition and not evidence of stability; the question is how far each coefficient moves when the sample changes.

3.13 Reproducibility

No table in Chapter 4 was transcribed by hand. Once the Phase 1 extracts are on disk, one evaluation driver recomputes every check, every alignment statistic with its interval and every benchmark, and writes them to a machine-readable summary, from which a separate stage exports the tables of Chapter 4 and Appendix 8.1. The driver takes cached extracts or synthetic fixtures as input and has switches for the damping, top-token and sample-definition sweeps. One versioned configuration file holds the observation window, damping, decay parameters, event-topic IDs, bootstrap count and seed, and benchmark seeds. In fixture mode the pipeline needs no cloud credentials and runs offline.

The reproducible elements are these: deterministic wallet subsampling by SHA256 hashes; a fixed PageRank tolerance and iteration cap; one bootstrap seed and one resample-index matrix; the proxy formulas of Section 3.9; the benchmarking procedure of Section 3.11, with per-run timings kept in the logs; the coupling weight λ and the holdout logic of Section 3.7, both fixed in the configuration file before the run began; and the extracted manifest files. From those same extracts the alignment and benchmark tables can be rebuilt without any new BigQuery queries. The machine-readable

summaries behind all the tables, together with the active configuration, are committed to the Git repository under `data/3-published-results-for-thesis/`, in the same folder as the source code. The numbers in this thesis can thus be checked against those summaries without rerunning the pipeline. Appendix 8.1 lists the GitHub repository URL, the exact commit hash that generated the tables in this thesis, the configuration file and the summary files.

3.14 Sampling considerations and missingness

The matched cohort is a constructed sample, not a random draw from all Arbitrum addresses: eligibility needs at least three GMX V2 closes in the window, and approval and transfer activity plays no part in it. Same-window findings therefore apply to active GMX V2 traders on Arbitrum One, and holdout findings to the spenders in these traders' approval data. Wallets that did not trade on GMX V2 in the window, dormant wallets and long-term holders among them, are excluded by design, and cohort wallets outside a graph score 0 under that method (Section 3.2).

Data can be missing in a few ways. An amount that cannot be parsed is read as zero and dropped with the zero-value records. EndorseRank does not see approvals that emit no standard ERC-20 event, and AWP does not see transfers made off-chain or inside custodial accounts. Both are thus biased toward on-chain, standard-token behavior, which is acceptable for an on-chain reputation study but matters once results are extrapolated to all users of a protocol.

The pool ($N = 99,080$) widens the set of sampled addresses for the runtime experiments, but the solver graph never grows beyond the cohort graph (Section 3.12.1), so alignment is always computed on the matched cohort.

3.15 Implementation notes

The PageRank code is written in Python, with a SciPy sparse transition operator and NumPy power iteration. Each configuration builds its edge lists once. The timing benchmarks leave out I/O and preprocessing and cover operator assembly and iteration only (Section 3.11). Optional fixture gen-

eration is the one place where randomness enters; given the parquet inputs and configuration seeds, evaluation on real data is deterministic. Small floating-point differences between platforms are to be expected, but they would be negligible against the three-decimal-place precision of the τ metric. Figures are generated from the same evaluation artifacts as the tables, so the plotted values agree with the tabulated numbers.

3.16 Ethics

Only publicly available blockchain data are used in this research, and it involves no human subjects. The institutional ethics review (Unisa SoC/CAES) was completed before any empirical data collection began. Appendix 8.7.1 contains the ethics clearance certificate (reference [Unisa SoC/CAES ethics clearance ref: TBD], dated [TBD]).

- Privacy and anonymity: An address on Arbitrum One is a pseudonym and does not by itself reveal the real-world identity behind it ^[2]. Pseudonymity is not the same as anonymity, however, and in theory a determined adversary could tie an address to a person through exchange deposits, online profiles or heuristics. In this work we deliberately make no such links. Our data collection is limited to on-chain `Approval` and `Transfer` events, and we scraped no off-chain personally identifiable information (such as IP addresses, exchange KYC records, social media handles or deanonymization graphs). We did not cluster addresses into real-world entities, publish leaderboards of addresses likely to be personally identifiable, or try to join a wallet’s transaction history with other databases to infer identity. The published results are aggregate measures only (correlation coefficients, runtimes, family means) and contain no individualized rankings of named participants. Where the main text quotes examples, they name broad categories of user (for example, ‘spenders with high allowances’) and never specific addresses.
- Data minimization: The processed tables contain only the fields that graph construction and the checks require, namely the relevant addresses, token amounts and timestamps. Raw logs

are kept so that the research pipeline can be audited. They will not be released for the purpose of deanonymizing users. Wherever possible, query filters were scoped to wallets, so as to avoid indiscriminate full-chain dumps outside the scope of the study design.

- **Data usage and compliance:** Every query ran against public data hosted in the `bigquery-public-data.goog_blockchain_arbitrum_one_us` dataset and complied with the Google Cloud terms of service and rate limits. Dry-run byte estimates and per-query maximums prevented unnecessary scans. We sent no transactions and accessed no privileged or non-public indexing endpoints, and no smart contract was invoked in any way that might interfere with the normal operation of the network.
- **Fairness and transparency:** New participants, low-frequency actors and wallets that grant no public approvals may all be undervalued by graph-based reputation systems ^[2]. Both `EndorseRank` and `AWP` carry structural biases of this kind. In the allowance graph, for instance, an address that has received no incoming spend authority is at a disadvantage relative to addresses that have; transfer graphs, in turn, favor addresses embedded in value-flow networks. Chapter 5 discusses the limitations, among them cold-start effects and the Sybil susceptibility of approval graphs. Every code path, parameter choice and evaluation metric is documented in the open pipeline (Appendix 8.1), so that others can review the work and replicate it independently.
- **Responsible disclosure and impact:** `EndorseRank` may inform risk assessment in DeFi, but it is not financial advice, credit advice or a consumer reporting product. Lending, trading and compliance decisions still need human oversight. Chapter 5 acknowledges and discusses the potential for adversarial manipulation, for example Sybil attacks on G_{approve} or wash transfers on G_{transfer} . The study reports aggregated, cohort-level findings and no wallet-level rankings intended for targeting individual users. Any future operational deployment would require additional governance, appeal mechanisms and fairness monitoring, all of which lie beyond the scope of this research.

- Summary of methodological commitments: Before results are interpreted, the design has already fixed the chain, window, cohort rules, graph formulas, proxy definitions, bootstrap and timing protocols, and statistical tests. The primary claims concern all the scores together, on the matched wallet cohort and the spender holdout, with each PageRank set against the raw degree it smooths. The expanded pool and the sensitivity sweeps test efficiency and how far results depend on parameters and sample definition.

Chapter 4 reports what this protocol yields for Arbitrum One: the dataset and cohorts, the runtime benchmark, the same-window alignment of every score, the rank divergence, the sensitivity sweeps, the temporal holdouts, the registered replication and the outcomes built from neither edge type, and finally Claims C1–C6.

4 Implementation and Empirical Results

This chapter runs the design of Chapter 3 on Arbitrum One as a single case study: one chain and one observation window, one matched cohort for every same-window comparison, a spender cohort and a trader cohort for the temporal holdouts, and one expanded address pool for node-count sensitivity, with a registered replication on June to August 2026 and a pre-specified post hoc comparison on trader outcomes at the end of Section 4.6. Every table is generated from the evaluation summaries of Section 3.13, each Kendall τ comes with its confidence interval, and each PageRank is read against the raw degree it smooths.

4.1 Case study setting: Arbitrum One

Arbitrum One was chosen because it is a heavily used Ethereum rollup with mature DeFi protocols (Section 2.2), because its logs are in the public BigQuery export that the pipeline reads, and because one of its margin-trading protocols, GMX V2, supplies the wallet seed set. The observation window, 1 December 2025 to 31 May 2026, spans several market regimes, and within it the time weight of an event never falls below about half of the weight of a recent one (Figure 3.3).

4.1.1 Cohorts

Three address sets are used.

- Matched cohort ($n = 5,521$). The GMX V2 wallets with at least three closes in the window, fixed from GMX data before any approval or transfer was extracted (Section 3.2). Membership does not depend on either graph: no allowance edge touches 966 cohort wallets, no transfer edge touches 381, and 370 are touched by neither; a wallet absent from a graph scores zero under that method. Every same-window alignment, difference test and benchmark is computed

on this cohort.

- Holdout spender cohort ($n = 1,335$). Every spender that holds at least one positive latest allowance at t_1 (28 February 2026, 23:59:59 UTC) in the extracted approval data. Most are contracts. Read from an Arbitrum node at the end of September 2026 (block 510,408,687), 1,174 of the 1,335 addresses held contract code and 161 were externally owned accounts, one of which delegates to contract code under EIP-7702. All 100 spenders that EndorseRank ranks highest are contracts, as are 221 of the 222 that gained a new approval pair from March to May. A BigQuery check of the scoring window agrees where it can decide, since every spender that emitted an event log holds code. Of the 686 spenders that neither emitted a log nor sent a transaction in the window, 530 are contracts, among them routers whose events other contracts emit, such as the GMX exchange router named in the pipeline configuration, and 156 are externally owned accounts. The scores of these wallets are tested against owner–spender approval pairs and transfer senders first seen from 1 March to 31 May 2026 (Section 3.7).
- Expanded wallet pool ($N = 99,080$). The matched cohort plus 93,559 addresses sampled from the chain’s monthly ERC-20 activity, used only for the node-count sensitivity of runtime and the sensitivity to sample definition (Section 3.12.1). The added addresses bring no edges, so the full pool’s solver graph is the cohort graph.

Table 4.1 lists the graph sizes. On the same wallets the endorsement graph has about one-ninth as many edges as the transfer graph, because EndorseRank keeps one latest allowance per token, owner and spender, whereas AWP adds a term for every transfer. The sampling is the same for both, so the differences between the scores trace back to their edges and restarts.

4.1.2 Implementation

Every algorithm runs on one Python implementation of PageRank ($d = 0.85$, $\varepsilon = 10^{-8}$, $I_{\max} = 300$). EndorseRank and AWP differ in their edge lists and restart vectors, and C-PR mixes its two edge lists node by node (Equation (3.9)). Each run writes its scores, per-proxy statistics, bootstrap intervals and timings to one file. The tables of this chapter come from runs made on 2 October 2026,

except those of the registered replication, evaluated on 1 October; Appendix 8.1 lists the runs, the repository revision, the configuration file and the summary files.

4.2 Computational benchmark (Claim C1)

Claim C1 concerns what each score costs to compute on the same input. Following the timing procedure of Section 3.11, Table 4.1 reports for each algorithm, on the matched cohort, the PageRank solve time and its standard deviation over the five timed repeats, the peak memory of the warm-up solve, the number of iterations to convergence and the size of the evaluation subgraph.

Table 4.1: Computational benchmark on the matched wallet cohort ($n = 5,521$; AWP/EndorseRank runtime ratio $9.2\times$). Runtime = mean wall-clock seconds over 5 timed PageRank solves after one untimed warm-up; SD over the same 5 runs. $|V|$, $|E|$ = nodes and edges of each method’s evaluation subgraph. C-PR walks on the union of the allowance layer and the transfer layer, so $|E|$ is the sum of the two edge sets; S-PR runs on the transfer layer and its runtime includes the solve of the allowance layer that supplies its restart vector.

Method	Runtime (s)	SD (s)	Peak mem. (MB)	Iters	$ V $	$ E $
EndorseRank	0.547	0.014	4.6	37.0	6,442	14,727
AWP	5.029	0.062	38.2	91.0	30,791	137,087
C-PR ($\lambda = 0.5$)	5.667	0.091	41.3	76.0	31,242	151,814
S-PR	5.600	0.020	39.1	145.0	30,791	137,087

Most of the timed call is spent before the first iteration, when the sparse matrix is assembled from the edge list in a per-edge Python loop. A separate profiling run of the same call on the same edge lists attributes about 94% of the wall time to matrix assembly (0.52 of 0.55 s for EndorseRank, 4.84 of 5.17 s for AWP). The power iterations take about 1%: about 6 ms for EndorseRank’s 37 iterations and 67 ms for AWP’s 91. Node indexing takes most of the rest, and forming AWP’s restart vector about 11 ms. Assembly and each iteration are both linear in $|E|$ (Appendix 8.3.3), so the measured ratio of $9.2\times$ tracks the edge ratio of $9.3\times$. EndorseRank also converges in fewer iterations. The allowance graph has no cycle apart from one self-loop, and 99.0% of its edges end at spenders without out-edges, so every path stops after a step and the mass returns through the restart

distribution; the transfer graph has a strongly connected core of 9,278 addresses, and 28.5% of its edges are reciprocated, so the walk circulates for longer. That difference is real, but it adds little to the wall time. Peak memory likewise grows with $|E|$, and run-to-run noise is about two orders of magnitude smaller than the gap between the methods (standard deviations of 0.014 s and 0.062 s). The two hybrid rows show what the speed result covers and where it stops. C-PR walks on the union of both edge sets (151,814 edges, 31,242 nodes) and needs 5.67 s, longer than AWP. It converges in fewer iterations (76 against 91), but it has more edges to assemble, and assembly dominates the call. S-PR runs on the transfer layer and takes 5.60 s. Both its time and its iteration count include the solve of the allowance layer that supplies its restart vector: the 145 iterations are 37 for that solve plus 108 for the seeded transfer walk. The low cost thus belongs to EndorseRank alone: any algorithm that queries the transfer layer, on its own or combined with allowance information, pays the cost of that query.

Two caveats attach to this result. The time covers only the solver call on edge lists built in advance; a full refresh would add parsing, decoding and edge-list construction, which all algorithms share wholly or in part. And a faster method is not necessarily a better one, so the runtime has to be read against the alignment and holdout results that follow.

4.2.1 Node-count sensitivity on the expanded pool

Deterministic SHA256 subsampling of the expanded pool yields three increasing node counts. Table 4.2 reports the runtime at each of them, together with the nodes and edges of every algorithm’s subgraph at that stage. The in-cohort stages ($n \leq 5,521$) are in Appendix 8.1 (Table 8.1).

Two features of Table 4.2 follow from Section 3.12.1. The full-pool row is identical to the matched-cohort row of Table 4.1 and to the $d = 0.85$ row of Table 4.6, by design: no logs were parsed for the extra addresses, so the subgraph induced at $N = 99,080$ is the cohort subgraph, and each distinct graph is timed once per session. And $|E|$ grows with n within each scaling table but not across the two samplings: the pool stage with $n = 10,000$ has fewer EndorseRank edges (2,055) than the in-cohort stage with $n = 1,000$ (2,584, Table 8.1), because only the cohort wallets, about 6% of the

Table 4.2: Runtime versus evaluation-set size on the expanded wallet pool ($N = 99,080$; SHA256 subsampling seed `benchmark-tier2-v1`). Each stage keeps the edges that touch at least one sampled wallet, so $|V|$ and $|E|$ are reported per stage; the pool adds addresses but no additional extracted edges beyond the matched-cohort subgraph, hence the full-pool row reproduces the matched-cohort graph. Speedup = AWP/ER runtime ratio within the same row. Runtime = mean wall-clock seconds over 5 timed PageRank solves after one untimed warm-up; SD over the same 5 runs.

n	ER (s)	AWP (s)	Speedup	ER $ V $	ER $ E $	AWP $ V $	AWP $ E $
10,000	0.075	1.078	$14.4\times$	1,296	2,055	8,489	26,732
50,000	0.342	3.414	$10.0\times$	4,440	9,190	19,990	92,887
99,080	0.547	5.029	$9.2\times$	6,442	14,727	30,791	137,087

pool, had their logs extracted. What the table supports is a bounded claim. Across stages whose edge counts differ by a factor of five to seven, AWP takes 8.6 to 14.4 times as long as EndorseRank, and EndorseRank stays at or below 0.55 s; the largest ratio belongs to the smallest pool stage, where AWP needs 97 iterations and EndorseRank 32. How either algorithm scales as edges grow with the number of addresses is not shown, because the second-stage extraction that was to supply the extra edges did not finish. Figure 4.1 plots the in-cohort and pool stages on one log axis.

4.3 Same-window alignment (Claim C2)

Claim C2 asks what each score agrees with in the window it is computed from. Table 4.3 reports the family-mean Kendall τ of every score on the transfer, allowance and Sybil-stability families, each with a 95% paired-bootstrap confidence interval. The raw degrees have no row because in this window they are proxies: in-degree belongs to the transfer family and in-approve degree to the allowance family. Apart from the secondary rule for C-PR fixed on 14 September (Section 3.7), none of the comparisons below was fixed in advance (Chapter 3).

The scores divide by the edges they walk on. Those that walk on transfers, AWP, C-PR at every λ below 1 and S-PR, reach 0.494 to 0.612 on the transfer family and stay near zero on the allowance family, where only S-PR, whose restarts come from the allowance walk, reaches 0.116. The two that walk on allowances alone, EndorseRank and C-PR ($\lambda = 1$), reach 0.375 and 0.355 on the

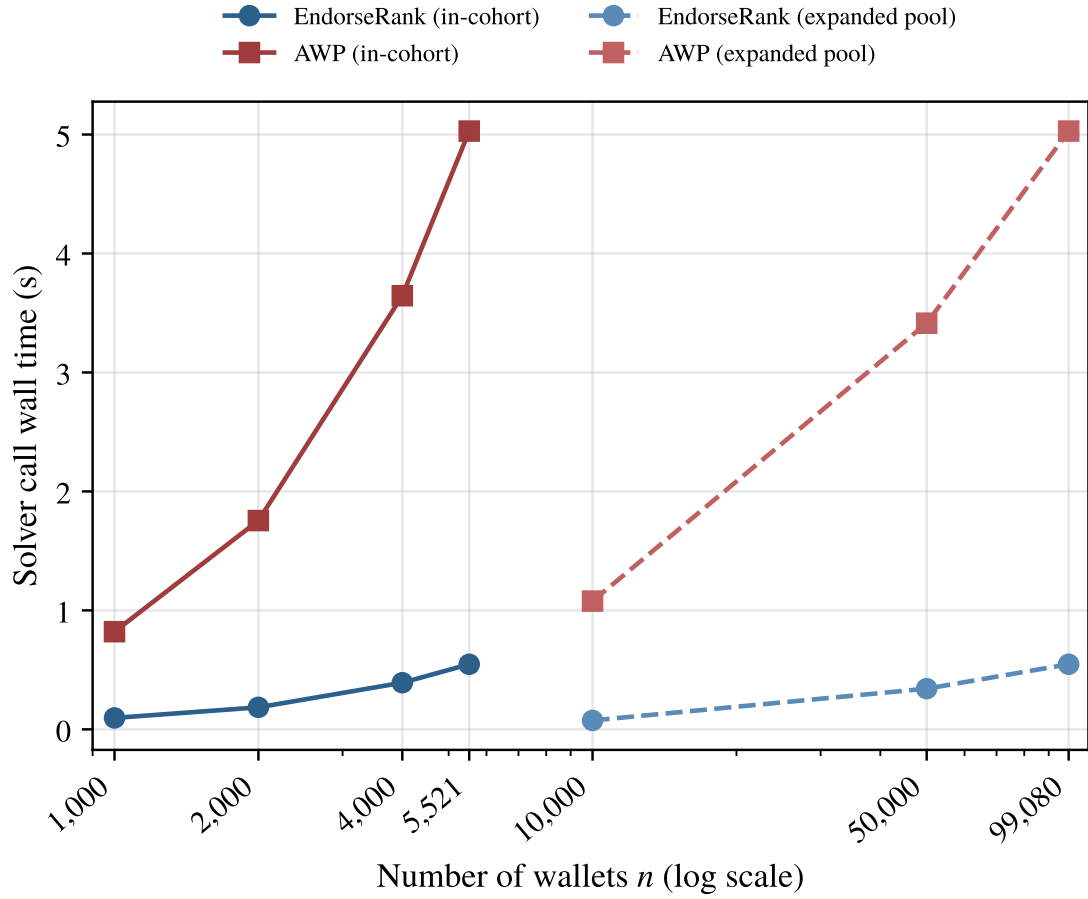


Figure 4.1: Wall-clock time of one solver call against the number of sampled wallets (log x -axis). Solid lines: in-cohort stages (Table 8.1). Dashed lines: expanded-pool stages (Table 4.2), which add no extracted edges, so the rightmost pool point is the full matched cohort. Runtime follows the edge count of each stage and not n .

Table 4.3: Same-window family-mean Kendall τ with bootstrap intervals for every score of this thesis (matched cohort, $n = 5,521$). The scores are EndorseRank and AWP, the coupled operator at five values of λ and the endorsement-seeded walk; $\lambda = 1$ is the allowance layer walked alone (latest allowances in raw amounts) and $\lambda = 0$ the transfer layer walked alone (raw amounts discounted by the decay of AWP); both restart uniformly. The raw degrees have no row because they are proxies of the transfer and allowance families. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Method	Transfer	Allowance	Sybil
EndorseRank	0.330 [0.311, 0.350]	0.375 [0.342, 0.401]	0.231 [0.212, 0.250]
AWP	0.612 [0.602, 0.622]	-0.024 [-0.043, -0.002]	0.278 [0.264, 0.292]
C-PR ($\lambda = 1$)	0.333 [0.314, 0.353]	0.355 [0.322, 0.384]	0.233 [0.214, 0.253]
C-PR ($\lambda = 0$)	0.534 [0.522, 0.547]	-0.030 [-0.045, -0.017]	0.286 [0.274, 0.300]
C-PR ($\lambda = 0.25$)	0.531 [0.520, 0.545]	-0.009 [-0.020, 0.004]	0.288 [0.276, 0.302]
C-PR ($\lambda = 0.5$)	0.527 [0.515, 0.540]	-0.001 [-0.013, 0.013]	0.290 [0.279, 0.304]
C-PR ($\lambda = 0.75$)	0.519 [0.507, 0.533]	0.006 [-0.007, 0.020]	0.292 [0.281, 0.306]
S-PR	0.494 [0.481, 0.508]	0.116 [0.097, 0.134]	0.220 [0.207, 0.235]

allowance family and about 0.33 on the transfer family. On the Sybil-stability family every score lies between 0.220 and 0.292. Part of this agreement is built in. The main proxies of each family come from the same events as one of the graphs, in-degree and in-value from the transfers and in-approve degree and value from the latest allowances, and a PageRank smooths the in-degree of its own graph. Agreement with the proxies of a score’s own edge type is therefore an intended-construct check and cannot count as an external test. The holdout of Section 4.6 asks the same question out of window.

Appendix 8.5.2 breaks the family means of EndorseRank and AWP into their proxies. AWP’s largest coefficient is with transfer in-degree ($\tau = 0.728$); with in-value it is much lower (0.495). That is what its weights lead one to expect: V counts every transfer of ten base units or more about once, whatever its amount, so AWP follows the number of incoming transfers more closely than their summed value. It does not collapse to the in-degree either. Propagation, time weights and restarts at active senders shift a measurable share of pairwise orderings, and out of window AWP ranks new senders below the in-degree (Section 4.6). On the Sybil-stability family AWP follows tenure and active months, and it is clearly negative on the inbound-counterparty ratio (-0.228). Its edge

weights count transfers, so a wallet that receives many transfers from a few senders ranks high, and that is the pattern the ratio penalizes.

EndorseRank’s coefficients are decided by its ties (Section 4.4). Only 131 cohort wallets lie above the common value of the wallets that receive no approval, both allowance proxies are positive for the same 131 wallets, and EndorseRank lifts all of them above its common value. Its two allowance coefficients (0.376 and 0.373) therefore mostly measure the same split and sit at the ceiling that the ties allow (0.377 and 0.378). Against a proxy without ties the ceiling is about 0.57, so EndorseRank’s coefficients on different families are not on one scale. Its transfer coefficient (0.330) mainly records that wallets in the allowance graph also receive more transfers than wallets outside it, and both coefficients change sharply when the wallets outside the graph are kept as isolated nodes (Section 4.4). C-PR ($\lambda = 1$) walks the same allowance graph with raw amounts, also restarts uniformly, and stays within 0.02 of EndorseRank on every family.

The same-window trading families of the research proposal, liquidation, inverse risk and GMX trading success (Section 2.8), are archived in Table 8.2. Every score aligns weakly with them. No family mean exceeds 0.24, and once two inverse-risk proxies are oriented correctly (Appendix 8.1.5), every score is negative on the inverse-risk family. The summed loss and the worst close get larger the more a wallet trades, and on the non-loss close rate, a share that does not, the coefficients of the scores lie between -0.031 and 0.024 .

Claim C2 is supported in a narrow sense. Every score that walks on one edge type agrees with the proxies of that edge type, with an interval well clear of zero, and the mixtures follow the denser of their two layers. The agreement is partly built in, and for EndorseRank it is set by the rule that scores a wallet outside the graph zero.

4.3.1 The coupled operator (RQ5)

Two hybrids were defined in Section 3.6.3. The coupled operator C-PR mixes the raw-amount allowance and transfer layers in each wallet’s transition row with weight λ , and the endorsement-seeded walk S-PR keeps the transfer layer but restarts from the allowance walk. Table 4.4 gives

their paired contrasts in this window: first the eight of Section 3.10 against whichever layer leads each family, then four against EndorseRank and AWP. RQ5 is decided by the out-of-window tests in Section 4.6. Here we only locate the hybrids on the checks that the single-layer scores were built to pass.

Table 4.4: Same-window paired contrasts for the coupled operator (C-PR, $\lambda = 0.5$) and the endorsement-seeded walk (S-PR) against the layer that leads each family walked alone, the allowance layer (C-PR ($\lambda = 1$)) or the transfer layer (C-PR ($\lambda = 0$)), and of C-PR against EndorseRank and AWP (matched cohort, $n = 5,521$). Family-mean τ ; the same wallet resamples are used for a and b . The contrasts with the endpoints are a stricter check beside the secondary criterion fixed on 14 September 2026 (Section 3.7), which compares C-PR’s point estimate with the leading layer’s interval in Table 4.3. An interval below 0 means a is significantly below b ; one above 0 means it exceeds it. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Contrast (a minus b)	τ_a	τ_b	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
<i>Against the two layers walked alone</i>					
C-PR transfer minus C-PR ($\lambda = 0$) transfer	0.527	0.534	-0.007	[-0.009, -0.005]	0.000
C-PR allowance minus C-PR ($\lambda = 1$) allowance	-0.001	0.355	-0.356	[-0.388, -0.323]	0.000
C-PR sybil-stability minus C-PR ($\lambda = 0$) sybil-stability	0.290	0.286	+0.004	[0.003, 0.006]	1.000
C-PR sybil-stability minus C-PR ($\lambda = 1$) sybil-stability	0.290	0.233	+0.057	[0.043, 0.069]	1.000
S-PR transfer minus C-PR ($\lambda = 0$) transfer	0.494	0.534	-0.040	[-0.049, -0.030]	0.000
S-PR allowance minus C-PR ($\lambda = 1$) allowance	0.116	0.355	-0.239	[-0.263, -0.215]	0.000
S-PR sybil-stability minus C-PR ($\lambda = 0$) sybil-stability	0.220	0.286	-0.065	[-0.072, -0.058]	0.000
S-PR sybil-stability minus C-PR ($\lambda = 1$) sybil-stability	0.220	0.233	-0.013	[-0.025, -0.003]	0.007
<i>Against EndorseRank and AWP</i>					
C-PR transfer minus AWP transfer	0.527	0.612	-0.085	[-0.095, -0.075]	0.000
C-PR allowance minus EndorseRank allowance	-0.001	0.375	-0.375	[-0.404, -0.344]	0.000
C-PR sybil-stability minus AWP sybil-stability	0.290	0.278	+0.012	[0.005, 0.020]	1.000
C-PR sybil-stability minus EndorseRank sybil-stability	0.290	0.231	+0.059	[0.045, 0.072]	1.000

C-PR behaves like its transfer layer with a small allowance-layer correction. Its transfer-family coefficient falls from 0.531 at $\lambda = 0.25$ to 0.519 at $\lambda = 0.75$, always below C-PR ($\lambda = 0$) at 0.534 and within 0.015 of it. At the primary setting the paired difference to that walk is -0.007 (interval $[-0.009, -0.005]$), small but clear of zero. AWP ranks the transfer family better than either: C-PR lies 0.085 below it ($[-0.095, -0.075]$). On the allowance family C-PR cannot be told apart from zero at any λ and lies far below its allowance layer ($\Delta\tau = -0.356$, interval $[-0.388, -0.323]$) and below EndorseRank (-0.375 , $[-0.404, -0.344]$). The cause is the sparsity of the allowance

layer. The transfer layer has about nine times the allowance layer's edges, and 30,791 nodes against 6,442 (Table 4.1), so most nodes of the union graph have no allowance out-edge. At those nodes the mixing rule of Equation (3.9) falls back on the transfer row whatever λ is; λ acts only at the minority of nodes with out-edges in both layers. Once the walk leaves that minority, it travels on transfer edges until it teleports. Raising λ to 0.75 shifts the allowance coefficient by less than one hundredth. The mixture recovers the allowance construct only at the end point $\lambda = 1$; at $\lambda = 0.75$ its allowance coefficient is still 0.006.

Only on the Sybil-adjusted stability family does C-PR exceed both layers walked alone, as the secondary criterion of Section 3.7 asked. It scores 0.290, against 0.286 for C-PR ($\lambda = 0$) and 0.233 for C-PR ($\lambda = 1$), and both paired intervals lie above zero ($[0.003, 0.006]$ and $[0.043, 0.069]$). It also exceeds AWP (0.278) and EndorseRank (0.231) on this family. The gain over the transfer walk is four thousandths, which makes it a difference in the statistical sense but not in any operational one.

On every family, S-PR does worse than the layer it borrows from. Restarting the transfer walk from the allowance walk drops the transfer coefficient to 0.494 ($\Delta\tau = -0.040$ against C-PR ($\lambda = 0$)), raises the allowance coefficient only as far as 0.116 ($\Delta\tau = -0.239$ against C-PR ($\lambda = 1$)), and pushes Sybil-adjusted stability below both layers. A restart vector changes where the walk lands after each teleport, but on a graph of this density it does not change where the walk spends its time. This is the same-window counterpart of the finding that Section 4.6 reaches out of window.

As fixed on 14 September, the secondary criterion asks C-PR's point estimate to lie within the 95% interval of the better layer on the transfer and allowance families, and C-PR to exceed both layers on Sybil-adjusted stability. At $\lambda = 0.5$, C-PR passes two of the three parts and fails the third. Its transfer coefficient (0.527) falls inside the interval of C-PR ($\lambda = 0$) ($[0.522, 0.547]$), and its Sybil-adjusted stability exceeds both layers. The allowance coefficient (-0.001), however, lies far outside the interval of C-PR ($\lambda = 1$) ($[0.322, 0.384]$), so the secondary rule fails. The paired contrasts of Table 4.4, reported alongside the rule as a stricter check, would fail the transfer part as well, since they place C-PR slightly but significantly below its transfer layer. In the same-window tables the

denser layer dominates the mixture. The holdout takes up the remaining question of whether the mixture has any predictive content that the single layers lack.

4.4 Rank divergence (H1 / RQ1)

Hypothesis H1 and RQ1 concern whether EndorseRank and AWP put the same wallets in different orders. The inter-method Kendall τ_b between the two score vectors is 0.360, with interval $[0.342, 0.379]$ (last row of Table 4.5), so the interval excludes both zero and one. The two rankings are related, yet far from interchangeable. Agreement runs above chance mainly because the wallets that EndorseRank scores zero also sit near the bottom of AWP (see below). Because 67.3% of wallet pairs tie on EndorseRank, only 32.7% of all pairs are ordered by both scores, and 18.6% of those pairs are discordant.

The marginal score distributions are in Figure 4.2, and the EndorseRank distribution is dominated by one tie block. EndorseRank takes only 99 distinct values on the cohort: 4,424 wallets share the score 9.62×10^{-5} that a wallet receiving no allowance gets from teleportation and redistributed dangling mass; 966 score zero because no allowance edge touches them; only 131 lie above the common value. Ties are rare under AWP, at 0.5% of wallet pairs. A marginal distribution cannot show how two scores order the same wallets, so Figure 4.3 plots the two rankings against each other, with tied wallets at their mid-rank as in τ_b . The EndorseRank tie blocks therefore appear as vertical columns. The 131 wallets that EndorseRank separates are spread over most of the AWP ranking (median AWP rank 3,096, interquartile range 1,520–4,850), so its leaders are not AWP’s leaders. The column of 4,424 tied wallets spans almost the whole AWP range (interquartile range 1,149–3,571). The 966 zero-score wallets sit mostly near the bottom of AWP (median AWP rank 4,966), and 370 of them score zero under AWP as well.

The tie blocks come from two rules, and one of them is a choice of this pipeline. The 4,424 wallets at the common value are in the allowance graph as owners that receive no approval, so PageRank gives them teleportation and dangling mass and nothing else. The 966 zero-score wallets are not in

the graph at all, and the pipeline scores a missing wallet zero (Section 3.6); kept as isolated nodes, they would have received the common value too. This rule decides most of EndorseRank’s ordering of the cohort. Of the 4.99 million wallet pairs that EndorseRank does not tie, 4.27 million (85.7%) set a wallet at the common value against a zero-score wallet. On the matched cohort, EndorseRank therefore mostly separates wallets in the allowance graph from wallets outside it. A wallet is in the graph when at least one latest allowance that it granted or received in the window is non-zero. Of the 966 outside it, 802 made no approval in the window; the other 164 did, but on every token and spender the latest approval they made set the allowance to zero.

Nothing in the graph favors the zero rule over the isolated-node alternative, so we reran the same-window alignment with the missing wallets kept as isolated nodes. This check was run after the results above were known. Under uniform teleportation an isolated node scores like any node without in-edges, so the 966 wallets join the block of 4,424 and every other score is multiplied by one common factor. AWP’s 381 missing wallets were handled the same way; under AWP’s restarts an isolated wallet that has sent nothing receives no restart mass, so it scores zero either way. Table 4.5 gives the result. No AWP coefficient moves. EndorseRank’s allowance coefficient rises to 0.990 (interval $[0.988, 0.992]$), since EndorseRank and the allowance proxies now split the cohort at the same place. Its transfer coefficient falls to -0.071 ($[-0.086, -0.056]$), and its Sybil-stability and trading-success coefficients to -0.036 and -0.017 . The agreement between the two scores falls from 0.360 to -0.025 ($[-0.044, -0.003]$). EndorseRank’s positive same-window coefficients on transfer, stability and trading, and its agreement with AWP, therefore measure which traders are in the allowance graph.

H1 is supported. The two scores order the cohort differently, and they are related mainly through the wallets that EndorseRank scores zero. What each score carries beyond its ties shows out of window, where every spender has received approvals and EndorseRank leaves almost no ties (Section 4.6). A protocol that relies on transfer-based scores cannot assume that its wallet ordering survives a switch to allowance-based scores or a blend with them.

Table 4.5: Same-window family-mean Kendall τ with wallets missing from a graph scored zero or kept as isolated nodes (matched cohort, $n = 5,521$). The main analysis scores a wallet that is missing from a method’s graph zero. The rule concerns 966 wallets for EndorseRank and 381 for AWP. Under EndorseRank’s uniform restarts an isolated node gets the score of a node without in-edges. AWP restarts only at addresses that sent a transfer, so an isolated wallet keeps the score zero and AWP’s coefficients are the same under both rules. The last row is the agreement between the two scores. The check was run after the results were known. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Family	Missing wallets scored zero		Missing wallets as isolated nodes	
	EndorseRank	AWP	EndorseRank	AWP
Transfer	0.330 [0.311, 0.350]	0.612 [0.602, 0.622]	-0.071 [-0.086, -0.056]	0.612 [0.602, 0.622]
Allowance	0.375 [0.342, 0.401]	-0.024 [-0.043, -0.002]	0.990 [0.988, 0.992]	-0.024 [-0.043, -0.002]
Sybil stability	0.231 [0.212, 0.250]	0.278 [0.264, 0.292]	-0.036 [-0.049, -0.022]	0.278 [0.264, 0.292]
Liquidation	0.093 [0.078, 0.110]	0.141 [0.129, 0.155]	0.082 [0.069, 0.096]	0.141 [0.129, 0.155]
Inverse risk	-0.055 [-0.071, -0.039]	-0.127 [-0.140, -0.115]	0.006 [-0.015, 0.023]	-0.127 [-0.140, -0.115]
GMX trading success	0.096 [0.080, 0.113]	0.236 [0.223, 0.250]	-0.017 [-0.033, -0.003]	0.236 [0.223, 0.250]
EndorseRank vs. AWP	0.360 [0.342, 0.379]		-0.025 [-0.044, -0.003]	

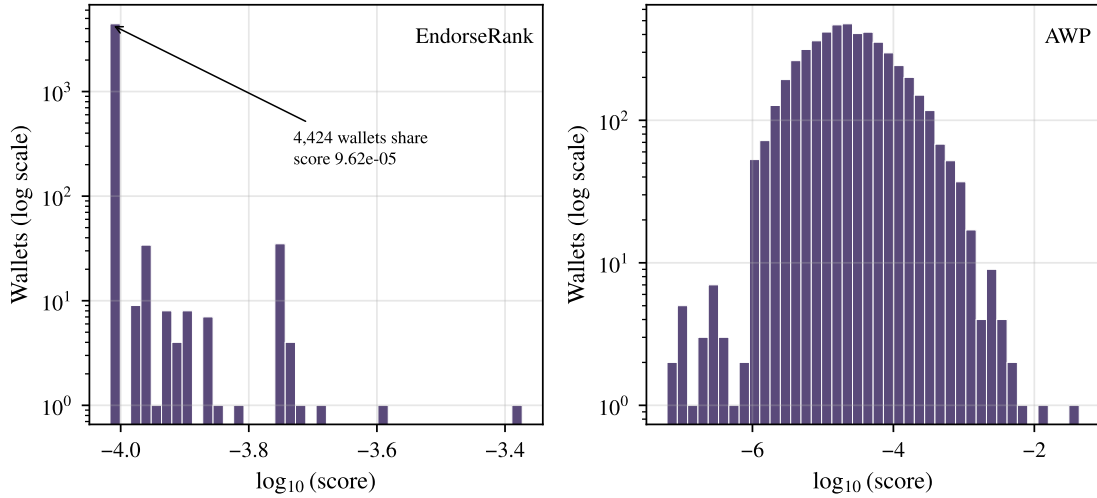


Figure 4.2: Marginal EndorseRank and AWP score distributions on the matched cohort ($n = 5,521$), drawn on a $\log_{10}$ scale for wallets with a positive score; counts are on a log scale. Zero-score wallets, which no edge of the respective graph touches, are left out: 966 for EndorseRank and 381 for AWP. For EndorseRank, 4,424 wallets share the score 9.62×10^{-5} of a wallet that receives no allowance, and only 131 lie above it.

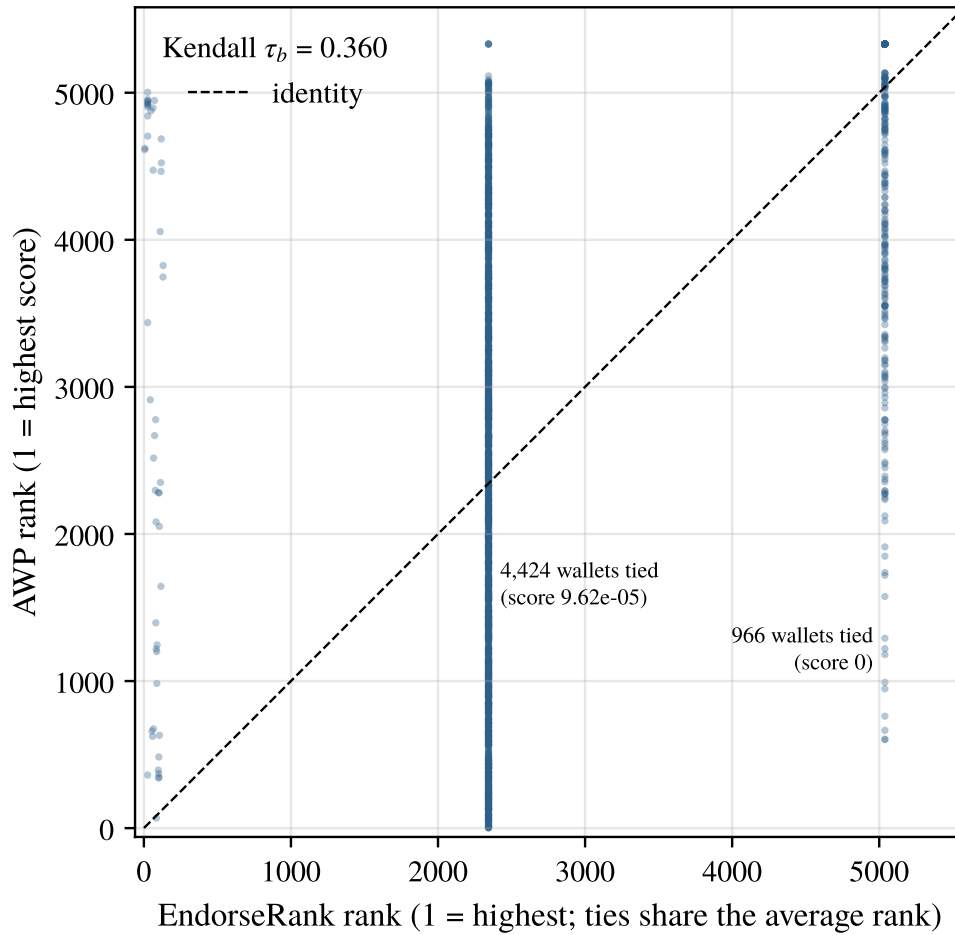


Figure 4.3: Rank–rank scatter of EndorseRank against AWP on the matched cohort (random sample of 2,000 wallets). Tied scores share their average rank, which is the tie treatment of Kendall’s τ_b . The two vertical columns are EndorseRank tie blocks: the 4,424 wallets with the common score 9.62×10^{-5} and the 966 wallets with score zero. A point on the dashed identity line has the same rank in both orderings.

4.5 Robustness and sensitivity checks

Three checks, run for EndorseRank and AWP, ask whether the results hinge on one damping value, one token mix or one sample definition. The first two hold the sample fixed; the third varies the sample definition itself, and there the coefficients are expected to move. A fourth check asks how much EndorseRank depends on its restart rule.

4.5.1 Damping

Table 4.6: Robustness: mean Kendall τ under PageRank damping $d \in \{0.75, 0.85, 0.95\}$ on the matched alignment cohort ($n = 5,521$). The $d = 0.85$ runtimes are the same measurements as Table 4.1 (one timing session; each graph–damping pair is timed once).

d	ER all.	AWP all.	ER tr.	AWP tr.	ER (s)	AWP (s)
0.75	0.375	-0.022	0.330	0.626	0.541	5.011
0.85	0.375	-0.024	0.330	0.612	0.547	5.029
0.95	0.375	-0.030	0.330	0.585	0.541	5.187

For EndorseRank the family means do not move at the third decimal place across $d \in \{0.75, 0.85, 0.95\}$. For AWP the transfer coefficient falls from 0.626 to 0.585 as d rises, and the allowance coefficient stays slightly negative. Neither score’s pattern is an artifact of the conventional $d = 0.85$. EndorseRank’s stability is largely structural. Only 131 cohort wallets lie above EndorseRank’s common value (Section 4.4); a change in d moves the common value but leaves the 4,424 wallets that share it tied, so the rank correlations can hardly move. Runtime barely moves either. EndorseRank takes 0.541, 0.547 and 0.541 s at $d = 0.75, 0.85$ and 0.95 , and AWP 5.011, 5.029 and 5.187 s, so the ratio stays between $9.2\times$ and $9.6\times$. Each graph–damping pair was timed once, in one session. Matrix assembly takes most of each call and does not depend on d , which is why the extra iterations at $d = 0.95$ add so little. Figure 4.4 plots the family means.

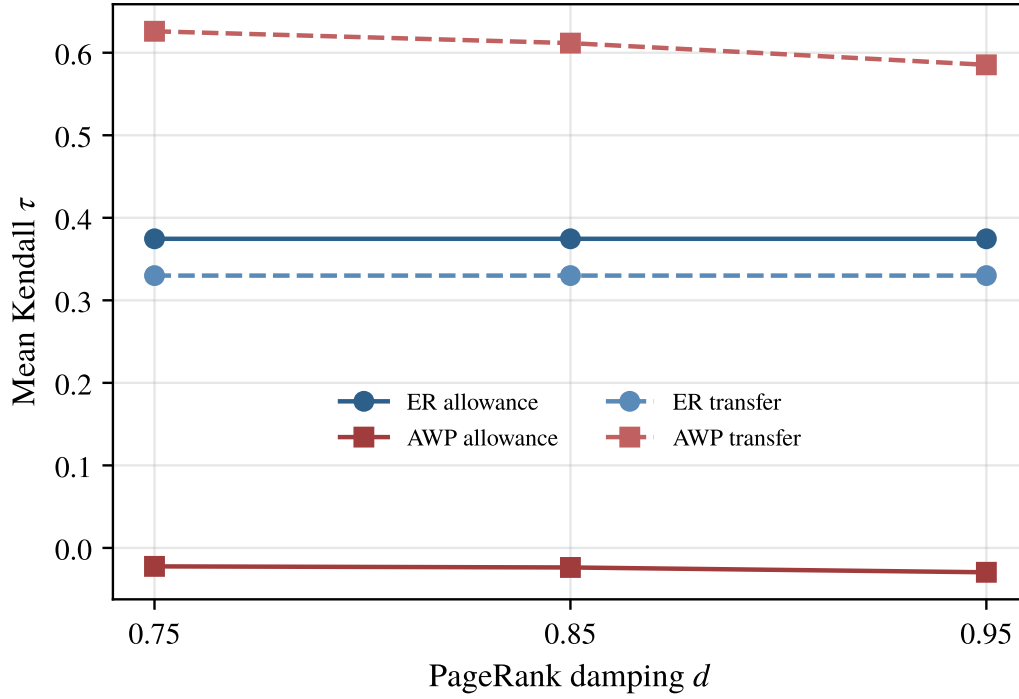


Figure 4.4: Damping sensitivity of family-mean Kendall τ for the allowance and transfer families across $d \in \{0.75, 0.85, 0.95\}$ (values from Table 4.6).

Table 4.7: Robustness: mean Kendall τ on the matched cohort when both graphs keep only 20 ERC-20 tokens, chosen by summed raw amount or by number of rows (transfer events and latest allowances), next to the value on all tokens from Table 4.3.

Method	Transfer			Allowance			Sybil		
	by amount	by count	all	by amount	by count	all	by amount	by count	all
EndorseRank	0.328	0.329	0.330	0.370	0.370	0.375	0.231	0.230	0.231
AWP	0.598	0.603	0.612	-0.021	-0.022	-0.024	0.273	0.275	0.278

4.5.2 Top-token subgraph

When both graphs are restricted to twenty tokens (Table 4.7), the largest change in a family mean is 0.014, for AWP on the transfer family when the tokens are the twenty with the largest summed raw amounts; with the twenty tokens that have the most rows it is 0.009. EndorseRank moves by 0.005 at most. The two lists share eleven tokens, so the pattern does not depend on the many tokens outside either set. Neither list ranks tokens by economic value: summed raw amounts favor tokens with many unlimited approvals, and row counts favor tokens that change hands often.

4.5.3 Sample-definition sensitivity

Table 4.8: Sensitivity of alignment to the evaluation set: mean Kendall τ and runtime at staged subsample sizes drawn from the expanded wallet pool ($N = 99,080$; seed `benchmark-tier2-v1`). Rows are evaluated on different wallet sets from the matched cohort, so τ values are not comparable with Table 4.3; they show how much the statistic moves with the evaluation population. Runtimes are the same measurements as Table 4.2.

n	ER all.	AWP all.	ER tr.	AWP tr.	ER (s)	AWP (s)	Speedup
10,000	0.343	0.239	0.821	0.900	0.075	1.078	14.37
50,000	0.333	0.230	0.827	0.909	0.342	3.414	9.98
99,080	0.332	0.230	0.826	0.909	0.547	5.029	9.19

In Table 4.8 the transfer and allowance family means are recomputed on the staged pool subsamples from the runtime scaling. The transfer-family means come out much larger than on the matched cohort (0.82–0.91), and AWP’s allowance coefficient turns positive (0.23–0.24), while EndorseRank’s allowance coefficient (0.332–0.343) ends below its cohort value of 0.375. None of these shifts is evidence about the scores. The pool subsamples let in wallets with no edges in one graph or in both; after alignment such a wallet has a zero score and a zero proxy, and tied zeros at the bottom of both rankings push Kendall τ up mechanically (Section 3.12.4). We therefore report the table as a sensitivity analysis of the sample definition, and its values must not be compared with those of Table 4.3. At every pool stage the EndorseRank graph has far fewer nodes than the sample has wallets (Table 4.2), so most pool wallets score zero on it, against 966 of the 5,521 cohort wallets.

4.5.4 Restarts of EndorseRank

EndorseRank restarts the walk uniformly, while AWP restarts it at addresses in proportion to their recent sending activity (Section 3.6.2). Carried over to allowances, AWP’s rule restarts the walk at owners in proportion to the number of spenders they approved recently. Table 4.9 scores both versions of EndorseRank, with AWP beside them. The restart rule was chosen after this comparison had been made.

Table 4.9: EndorseRank with uniform restarts and with restarts weighted by approving activity, AWP’s rule applied to the allowance graph, next to AWP: family-mean Kendall τ on the matched cohort ($n = 5,521$) and τ with the two spring labels on the spender cohort ($n = 1,335$). Both EndorseRank rows weight each latest allowance by $\sigma(\Delta t)V(z)$. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Method	Restarts	Same window, matched cohort			Spring holdout, spenders	
		Transfer	Allowance	Sybil	New approval pairs	New senders
EndorseRank	uniform	0.330 [0.311, 0.350]	0.375 [0.342, 0.401]	0.231 [0.212, 0.250]	0.394 [0.360, 0.431]	0.283 [0.247, 0.320]
EndorseRank	by approving activity	0.445 [0.433, 0.457]	0.044 [0.029, 0.062]	0.245 [0.233, 0.259]	0.372 [0.327, 0.417]	0.301 [0.254, 0.348]
AWP	by sending activity	0.612 [0.602, 0.622]	-0.024 [-0.043, -0.002]	0.278 [0.264, 0.292]	0.123 [0.068, 0.177]	0.344 [0.307, 0.383]

With AWP’s restarts, EndorseRank’s same-window allowance coefficient falls from 0.375 to 0.044, and its transfer coefficient rises from 0.330 to 0.445. The restart mass then goes to the owners that approve most. The cohort’s traders are such owners, so the score orders them by how much they approve, which is closer to their activity than to the endorsements they receive. Out of window the two versions are close. On the spring spenders they reach 0.394 and 0.372 on new approval pairs and 0.283 and 0.301 on new senders, with overlapping intervals, and in June–August 0.333 and 0.325, and 0.255 and 0.284 (Table 4.16). The uniform rule keeps the allowance construct in the same window and gives up nothing measurable out of window.

4.6 Temporal holdout (RQ4, RQ5)

A same-window check cannot separate a score from its own construct. The holdout instead tests whether centrality measured up to a cut-off predicts behavior after it. Following Section 3.7, scores on the spender cohort are frozen at 28 February 2026, and two labels are then counted for each wallet between 1 March and 31 May 2026: new owner–spender approval pairs (the endorsement construct) and new transfer senders (the flow construct). Table 4.10 sets every score computed at t_1 , the two raw-degree baselines included, against both labels. Table 4.11 gives the paired contrasts, first those fixed before the coupled scores were computed and then those computed after the labels were known.

Table 4.10: Temporal holdout on the spender cohort ($n = 1,335$): Kendall τ between scores frozen at 28 February 2026 and two labels counted on the wallet between 1 March 2026 and 31 May 2026: the number of new owner–spender approval pairs and the number of addresses that sent a transfer to the wallet for the first time. CI = 95% percentile bootstrap over wallets (400 paired resamples). 222 wallets received at least one new pair and 175 at least one new sender. Baseline rows are the raw degrees at t_1 that each single-layer PageRank smooths. C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$) are the allowance and transfer layers of the coupled operator walked alone.

Score at t_1	New approval pairs (τ [95% CI])	New transfer senders (τ [95% CI])
In-approve degree at t_1 (baseline)	0.711 [0.665, 0.759]	0.404 [0.348, 0.467]
Transfer in-degree at t_1 (baseline)	0.117 [0.049, 0.191]	0.472 [0.423, 0.520]
EndorseRank	0.394 [0.360, 0.431]	0.283 [0.247, 0.320]
AWP	0.123 [0.068, 0.177]	0.344 [0.307, 0.383]
C-PR ($\lambda = 1$)	0.479 [0.427, 0.529]	0.372 [0.325, 0.424]
C-PR ($\lambda = 0$)	0.044 [-0.011, 0.098]	0.270 [0.235, 0.310]
C-PR ($\lambda = 0.25$)	0.271 [0.231, 0.314]	0.266 [0.231, 0.306]
C-PR ($\lambda = 0.5$)	0.295 [0.252, 0.336]	0.277 [0.241, 0.318]
C-PR ($\lambda = 0.75$)	0.314 [0.271, 0.357]	0.287 [0.253, 0.330]
S-PR	0.114 [0.052, 0.164]	0.264 [0.229, 0.304]

The first finding answers the second part of RQ4 in the negative: no PageRank improves on the raw degree of its own edge type at t_1 . In-approve degree at t_1 predicts new approval pairs with $\tau = 0.711$. The walks over allowances come closest and still fall far short: C-PR ($\lambda = 1$) reaches 0.479, a paired difference of -0.233 (interval $[-0.291, -0.183]$, fixed before the run), and EndorseRank 0.394, a

Table 4.11: Paired bootstrap differences on the spender holdout ($n = 1,335,400$ resamples shared by a and b). The first block was fixed before the run: the increment of C-PR ($\lambda = 1$), the allowance layer walked alone, over its raw degree at t_1 , and the hybrid contrasts fixed on 14 September 2026, which compare C-PR and S-PR with the two layers they are built from, C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$). The second block was computed after the labels were known: the increments of EndorseRank and AWP over their raw degrees, C-PR against C-PR ($\lambda = 0$) on new approval pairs, and C-PR against EndorseRank and AWP. Primary criterion of 14 September (C-PR no worse than C-PR ($\lambda = 1$) on new approval pairs and better than C-PR ($\lambda = 0$) on new transfer senders): not met. The differences between EndorseRank and AWP are in Table 8.8.

Contrast (a minus b)	Label	τ_a	τ_b	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
<i>Fixed before the run</i>						
C-PR ($\lambda = 1$) minus t1 in-approve degree	new approval pairs	0.479	0.711	-0.233	[-0.291, -0.183]	0.000
C-PR minus C-PR ($\lambda = 1$)	new approval pairs	0.295	0.479	-0.184	[-0.213, -0.149]	0.000
C-PR minus C-PR ($\lambda = 0$)	new transfer senders	0.277	0.270	+0.007	[-0.012, 0.028]	0.735
S-PR minus C-PR ($\lambda = 1$)	new approval pairs	0.114	0.479	-0.365	[-0.431, -0.300]	0.000
S-PR minus C-PR ($\lambda = 0$)	new transfer senders	0.264	0.270	-0.006	[-0.017, 0.006]	0.163
<i>Computed after the labels were known</i>						
EndorseRank minus t1 in-approve degree	new approval pairs	0.394	0.711	-0.317	[-0.354, -0.283]	0.000
AWP minus t1 in-degree	new transfer senders	0.344	0.472	-0.127	[-0.150, -0.099]	0.000
C-PR minus C-PR ($\lambda = 0$)	new approval pairs	0.295	0.044	+0.251	[0.203, 0.306]	1.000
C-PR minus EndorseRank	new approval pairs	0.295	0.394	-0.099	[-0.133, -0.064]	0.000
C-PR minus AWP	new transfer senders	0.277	0.344	-0.067	[-0.095, -0.039]	0.000
C-PR minus AWP	new approval pairs	0.295	0.123	+0.172	[0.111, 0.226]	1.000

difference of -0.317 ($[-0.354, -0.283]$). Transfer in-degree at t_1 predicts new transfer senders with $\tau = 0.472$, against 0.344 for AWP ($\Delta\tau = -0.127$, interval $[-0.150, -0.099]$) and 0.270 for C-PR ($\lambda = 0$). Every interval lies well below zero. Within a three-month window, a wallet gains more new counterparties the more it already has, and the local count captures this directly. PageRank replaces the count with a weighted, propagated quantity. In EndorseRank an owner's endorsement is shared among all the spenders it approves, and in AWP a sender's among all the addresses it pays; on these labels that sharing acts as noise. If a PageRank adds anything beyond its degree, it does not show in predicting the future growth of its own construct.

Read across the table, the raw degree of each label's own edge type ranks that label best. On new approval pairs the two walks over allowances come next, then the interior mixtures (0.271 to 0.314), while the transfer in-degree, AWP, C-PR ($\lambda = 0$) and S-PR stay at or below 0.123 . On new transfer senders the order is less clean. The in-approve degree (0.404) and C-PR ($\lambda = 1$) (0.372) rank this label better than any walk over transfers, of which AWP reaches the most (0.344); on this cohort the spenders that many owners approve also gain many new senders. EndorseRank leaves almost no ties here: it gives $1,291$ distinct scores to the $1,335$ spenders, and its largest tie block holds 42 of them, so its coefficients rest on an ordering of nearly every spender. C-PR ($\lambda = 1$), the allowance layer walked with raw amounts, ranks both labels higher than EndorseRank. Its rows are decided by unlimited approvals (Section 3.6.3), so on this cohort the raw amounts kept part of the signal that counting each approval once gives up; the thesis does not test why.

Coverage does not explain why the walks fall short of their degrees. Of the $1,335$ spenders, 318 have no transfer edge up to the freeze and score zero under AWP. They gained new approval pairs more often than the rest (22% against 15%) and new senders less often (4% against 16%), so AWP's zeros lower its coefficient on the first label and raise it on the second. Table 4.12 restricts the holdout to the 977 spenders that had received a transfer from another address before the freeze. There AWP stays far below the transfer in-degree (0.388 against 0.701 ; -0.314 , $[-0.353, -0.276]$), and EndorseRank stays below the in-approve degree (0.340 against 0.679). The restriction was chosen after the results were known.

Table 4.12: Spender holdout restricted to the 977 of the 1,335 spenders that had received a transfer from another address before the freeze. The other 358 have no inbound transfer edge, and 318 spenders score zero under AWP. The restriction was tried after the results were known. The differences between EndorseRank and AWP on this subset are in Table 8.8. CI = 95% percentile bootstrap over wallets (400 resamples, seed 42).

Score at t_1	New approval pairs (τ [95% CI])	New transfer senders (τ [95% CI])
EndorseRank	0.340 [0.299, 0.386]	0.358 [0.316, 0.396]
AWP	0.298 [0.252, 0.343]	0.388 [0.348, 0.430]
In-approve degree at t_1	0.679 [0.616, 0.736]	0.570 [0.505, 0.635]
Transfer in-degree at t_1	0.569 [0.510, 0.631]	0.701 [0.651, 0.750]
AWP minus t1 in-degree (new transfer senders)	$\Delta\tau = -0.314$ [-0.353, -0.276]	

The second finding is that the coupled operator fails the rule of 14 September, although in an exploratory comparison it improves on its transfer layer walked alone. That rule asked C-PR to match C-PR ($\lambda = 1$) on new approval pairs and to beat C-PR ($\lambda = 0$) on new transfer senders. At $\lambda = 0.5$, C-PR does worse than the allowance walk on new approval pairs ($\Delta\tau = -0.184$, interval $[-0.213, -0.149]$) and no differently from the transfer walk on new transfer senders ($\Delta\tau = +0.007$, interval $[-0.012, 0.028]$), so both conditions of its primary part fail. Against the transfer walk alone, a comparison the rule did not include, C-PR does better. On new approval pairs C-PR reaches 0.295 against 0.044, a paired difference of $+0.251$ (interval $[0.203, 0.306]$, on the same resamples), and on new senders it shows no detectable loss. This contrast was computed after the labels were known, so it is treated as a hypothesis and tested again in the registered window below. Against EndorseRank and AWP, also computed after the labels were known, C-PR sits between the two on new approval pairs, 0.099 below EndorseRank ($[-0.133, -0.064]$) and 0.172 above AWP ($[0.111, 0.226]$), and it falls behind AWP on new senders (-0.067 , $[-0.095, -0.039]$). On the spring data the analogue of F2 would therefore fail against AWP. The registered replication reports that comparison next to its decision (Section 3.7.2). Across the three values of λ the order is monotone on both labels. On new approval pairs τ rises from 0.271 ($\lambda = 0.25$) through 0.295 to 0.314 ($\lambda = 0.75$), and on new transfer senders from 0.266 through 0.277 to 0.287. More allowance weight helps on both labels; for this cohort the transfer layer adds no out-of-window signal to what the allowance layer carries.

The seeded ablation shows where the allowance signal has to sit. S-PR matches the transfer walk on the flow label ($\Delta\tau = -0.006$, interval $[-0.017, 0.006]$) and falls far below the allowance walk on the endorsement label ($\Delta\tau = -0.365$). When transfer edges decide where the walk goes, a restart distribution derived from endorsements changes very little. On the spender holdout, the allowance signal survives when allowance edges carry the walk, fully in the allowance walk and diluted in C-PR, and it is lost when they only decide where the walk restarts.

Two caveats apply to the whole table. Each label is the same construction carried into the future (new approvals, new senders), so none of these spender results bears on credit or loss; the trader labels below are built from neither edge type but are not credit labels either. Both labels are also sparse. Of the 1,335 wallets, 222 gained a new approval pair and 175 a new sender; every coefficient rests entirely on how that small subset is ordered.

Other candidate labels were computed on the holdout, and they are reported here without being offered as support. Revocation counts and the drain heuristic both rise with EndorseRank ($\tau = 0.269$, interval $[0.226, 0.308]$, and 0.422 , interval $[0.389, 0.455]$), and more steeply still with the raw in-approve degree (0.483 and 0.575). For an endorsement reading the sign is wrong, since a trusted spender should lose approvals less often, not more. The likelier reading is volume. A spender with many owners sees more of everything owners do, revocations and drains included, so these labels cannot separate trust from use. Aave liquidations are empty in this spender set.

4.6.1 Trader outcomes (exploratory)

The trader run of Section 3.7.1 was scored after its labels were known and is reported as exploratory. Table 4.13 gives every score on the GMX labels of the 1,860 matched wallets with at least three closes from March to May.

The scores built on transfers rank the number of profitable closes and the realized gain. AWP reaches 0.210 and 0.191 , the transfer in-degree 0.198 and 0.169 , and the walks over the raw transfer layer and its mixtures 0.111 to 0.141 and 0.120 to 0.154 . EndorseRank and C-PR ($\lambda = 1$) reach about 0.07 and 0.05 , and the in-approve degree nothing. Both labels grow with how much a wallet trades.

Table 4.13: Spring trader run: Kendall τ between scores frozen at 28 February 2026 and GMX V2 labels counted from 1 March 2026 to 31 May 2026, on the 1,860 matched wallets with at least three closes in that window. The run was scored after its labels were known. Profitable closes and realized gain grow with the number of closes; the two shares do not. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Score at t_1	Profitable closes	Realized gain	Profitable share	Non-loss share
In-approve degree at t_1 (baseline)	-0.008 [-0.038, 0.026]	0.013 [-0.018, 0.049]	-0.057 [-0.095, -0.015]	-0.102 [-0.138, -0.063]
Transfer in-degree at t_1 (baseline)	0.198 [0.168, 0.228]	0.169 [0.135, 0.199]	0.016 [-0.017, 0.048]	0.011 [-0.017, 0.041]
EndorseRank	0.072 [0.041, 0.111]	0.048 [0.017, 0.082]	-0.016 [-0.053, 0.024]	-0.016 [-0.054, 0.023]
AWP	0.210 [0.181, 0.238]	0.191 [0.159, 0.219]	0.016 [-0.015, 0.044]	-0.020 [-0.051, 0.006]
C-PR ($\lambda = 1$)	0.073 [0.043, 0.111]	0.048 [0.018, 0.082]	-0.015 [-0.051, 0.027]	-0.016 [-0.054, 0.023]
C-PR ($\lambda = 0$)	0.131 [0.101, 0.162]	0.153 [0.120, 0.182]	0.029 [-0.003, 0.058]	0.012 [-0.014, 0.045]
C-PR ($\lambda = 0.25$)	0.128 [0.096, 0.158]	0.154 [0.122, 0.184]	0.025 [-0.007, 0.055]	0.007 [-0.019, 0.038]
C-PR ($\lambda = 0.5$)	0.121 [0.089, 0.152]	0.151 [0.120, 0.183]	0.024 [-0.009, 0.053]	0.007 [-0.019, 0.039]
C-PR ($\lambda = 0.75$)	0.111 [0.079, 0.140]	0.146 [0.114, 0.177]	0.023 [-0.009, 0.052]	0.009 [-0.019, 0.041]
S-PR	0.141 [0.114, 0.173]	0.120 [0.089, 0.150]	0.014 [-0.016, 0.047]	-0.008 [-0.034, 0.021]

On the two shares, which do not, no coefficient exceeds 0.03, and the in-approve degree is negative on both. That is what one should expect from a transfer score, which tracks activity, and from an allowance score that leaves most traders tied; none of the scores ranks how well a wallet trades. The Aave liquidation label of the same run covers only 174 wallets, 10 of them liquidated, which is too few to read; the GMX liquidation labels of this window are scored post hoc in Section 4.6.3. The registered window gives C-PR a trader label of its own.

4.6.2 Registered replication: June to August 2026

The registration of Section 3.7.2 was committed to the public repository on 1 October 2026, and the June–August logs were extracted later that day (Appendix 8.1): 163,553 approval logs, 1,013,023 transfer logs and 144,152 GMX closes, all of which decoded. The extraction manifest records the hashes of both registration files and no use of the testing override. Before any fresh number was computed, a check run of the evaluation code on the spring window reproduced exactly the sixteen spring coefficients of the eight scores it shares with Table 4.10 and the contrasts of the 14 September rule, and it gave the difference of +0.251 quoted above for C-PR against its transfer layer on new approval pairs. The registered evaluation then ran with 400 paired resamples and seed 42.

At the freeze on 31 May, 1,964 spenders held a positive latest allowance; 158 of them gained a new approval pair from June to August and 136 a new sender. Of the 5,521 matched wallets, 5,151 were in the freeze-date graph, and 1,402 of these closed at least three positions in the label window. Of the 1,402, 789 had at least one liquidation and 17 had every close liquidated. Table 4.14 gives every registered score on the three registered labels, and Table 4.15 the contrasts with their verdicts.

The rule is not met. F1 holds: C-PR ranks the spenders that later gained new approval pairs at $\tau = 0.238$ against 0.058 for its transfer layer walked alone, a difference of +0.179 (interval [0.132, 0.225]), so the spring gain on authorization replicates. F3 holds as well. On the liquidation-free close rate of the traders C-PR reaches 0.083 against 0.065 for the transfer walk (+0.018, [0.011, 0.024]). F2 fails. On new transfer senders C-PR reaches 0.246 against 0.257; the difference is -0.011 , and the lower end of its interval, -0.025 , lies below the margin of -0.02 . The

Table 4.14: Registered replication: Kendall τ between scores frozen at 31 May 2026 and the three registered labels, counted from 1 June to 31 August 2026. The spenders are every spender holding a positive latest allowance at the freeze; 158 of them gained a new approval pair and 136 a new sender. The traders are the 5,151 matched wallets in the freeze-date graph; the label is defined for the 1,402 with at least three closes in the label window, 789 of which had at least one liquidation. C-PR ($\lambda = 0$) is the comparator of the registered rule and AWP that of a registered sensitivity analysis (Table 4.15). CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Score at 31 May 2026	Spenders ($n = 1,964$)		Traders ($n = 1,402$)
	New approval pairs	New transfer senders	Liquidation-free close rate
In-approve degree at the freeze (baseline)	0.613 [0.564, 0.663]	0.371 [0.313, 0.429]	0.144 [0.110, 0.172]
Transfer in-degree at the freeze (baseline)	0.128 [0.064, 0.188]	0.435 [0.401, 0.468]	-0.006 [-0.046, 0.031]
AWP	0.118 [0.071, 0.161]	0.319 [0.291, 0.345]	0.045 [0.009, 0.079]
C-PR ($\lambda = 1$)	0.338 [0.292, 0.388]	0.289 [0.247, 0.332]	0.086 [0.037, 0.129]
C-PR ($\lambda = 0$)	0.058 [0.012, 0.103]	0.257 [0.229, 0.286]	0.065 [0.030, 0.102]
C-PR ($\lambda = 0.25$)	0.228 [0.190, 0.265]	0.245 [0.217, 0.273]	0.077 [0.042, 0.114]
C-PR ($\lambda = 0.5$)	0.238 [0.200, 0.274]	0.246 [0.218, 0.274]	0.083 [0.047, 0.119]
C-PR ($\lambda = 0.75$)	0.243 [0.206, 0.280]	0.248 [0.219, 0.275]	0.086 [0.051, 0.121]
S-PR	0.126 [0.076, 0.172]	0.254 [0.223, 0.281]	0.030 [-0.006, 0.066]

interval also contains zero, so the data show neither that C-PR is worse than the transfer walk on new senders nor that it is within 0.02 of it, which is what the rule asked. The rule needed all three hypotheses, so, as the plan said for this case, the thesis makes no claim that C-PR improves on a walk over the transfers alone.

The plan warned that F3 could pass with both coefficients near zero; both are small, but both intervals lie above zero (C-PR [0.047, 0.119], the transfer walk [0.030, 0.102]), so each walk carries a little information about which traders avoid liquidation. The in-approve degree at the freeze ranks this label better than any walk (Section 4.6.3).

The results reported beside the decision follow the same order. F4 holds: on the liquidation-free close rate C-PR is better than the transfer walk, and not only no worse. F5 fails, since C-PR and the allowance walk cannot be told apart on that label (-0.003 , $[-0.060, 0.051]$). F6, the primary part of the 14 September rule, fails again on both parts: C-PR is below the allowance walk on new approval pairs (-0.101 , $[-0.127, -0.076]$) and not above the transfer walk on new senders. C-PR's coefficients rise with λ on both spender labels, as in the spring. S-PR matches the transfer walk

Table 4.15: Registered replication: the paired contrasts F1–F6 with their verdicts, and the two sensitivity analyses added before the extraction. $\Delta\tau = \tau_a - \tau_b$, with C-PR as a . The registration documents call the two layers walked alone, C-PR ($\lambda = 1$) and C-PR ($\lambda = 0$), EndorseRank and AWP. Superiority holds when the interval lies above zero; non-inferiority holds when its lower end lies above -0.02 ; F6a, the first part of the 14 September rule, holds when the interval contains zero or lies above it. F1–F3 are on spenders (new approval pairs, new transfer senders) and traders (liquidation-free close rate) as in Table 4.14. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Contrast	Label	Test	τ_a	τ_b	$\Delta\tau$ [95% CI]	Verdict
<i>Registered rule: C-PR ($\lambda = 0.5$) against its transfer layer alone, C-PR ($\lambda = 0$)</i>						
F1	C-PR vs. C-PR ($\lambda = 0$) new approval pairs	superiority	0.238	0.058	+0.179 [0.132, 0.225]	holds
F2	C-PR vs. C-PR ($\lambda = 0$) new transfer senders	non-inferiority, -0.02	0.246	0.257	-0.011 [-0.025, 0.003]	fails
F3	C-PR vs. C-PR ($\lambda = 0$) liquidation-free close rate	non-inferiority, -0.02	0.083	0.065	+0.018 [0.011, 0.024]	holds
Decision: F1, F2 and F3 must all hold						not met
<i>Reported, not part of the decision</i>						
F4	C-PR vs. C-PR ($\lambda = 0$) liquidation-free close rate	superiority	0.083	0.065	+0.018 [0.011, 0.024]	holds
F5	C-PR vs. C-PR ($\lambda = 1$) liquidation-free close rate	superiority	0.083	0.086	-0.003 [-0.060, 0.051]	fails
F6a	C-PR vs. C-PR ($\lambda = 1$) new approval pairs	no worse	0.238	0.338	-0.101 [-0.127, -0.076]	fails
F6b	C-PR vs. C-PR ($\lambda = 0$) new transfer senders	superiority	0.246	0.257	-0.011 [-0.025, 0.003]	fails
<i>Sensitivity analysis: AWP as the comparator</i>						
F1	C-PR vs. AWP new approval pairs	superiority	0.238	0.118	+0.119 [0.075, 0.169]	holds
F2	C-PR vs. AWP new transfer senders	non-inferiority, -0.02	0.246	0.319	-0.073 [-0.092, -0.055]	fails
F3	C-PR vs. AWP liquidation-free close rate	non-inferiority, -0.02	0.083	0.045	+0.037 [0.011, 0.066]	holds
F4	C-PR vs. AWP liquidation-free close rate	superiority	0.083	0.045	+0.037 [0.011, 0.066]	holds
<i>Sensitivity analysis: wallets outside a graph kept as isolated nodes</i>						
F1	C-PR vs. C-PR ($\lambda = 0$) new approval pairs	superiority	0.238	0.053	+0.185 [0.137, 0.232]	holds
F2	C-PR vs. C-PR ($\lambda = 0$) new transfer senders	non-inferiority, -0.02	0.246	0.258	-0.012 [-0.027, 0.003]	fails
F3	C-PR vs. C-PR ($\lambda = 0$) liquidation-free close rate	non-inferiority, -0.02	0.083	0.065	+0.018 [0.010, 0.024]	holds
F4	C-PR vs. C-PR ($\lambda = 0$) liquidation-free close rate	superiority	0.083	0.065	+0.018 [0.010, 0.024]	holds
F5	C-PR vs. C-PR ($\lambda = 1$) liquidation-free close rate	superiority	0.083	0.155	-0.072 [-0.118, -0.028]	fails
F6a	C-PR vs. C-PR ($\lambda = 1$) new approval pairs	no worse	0.238	0.338	-0.101 [-0.127, -0.076]	fails
F6b	C-PR vs. C-PR ($\lambda = 0$) new transfer senders	superiority	0.246	0.258	-0.012 [-0.027, 0.003]	fails

on new senders (-0.003 , $[-0.017, 0.010]$) and falls far below the allowance walk on new approval pairs (-0.213).

Neither registered sensitivity analysis changes the verdict (Table 4.15). With AWP as the comparator, F1 and F3 still hold ($+0.119$ and $+0.037$) and F2 fails clearly. AWP ranks new senders at 0.319 , and C-PR's difference to it is -0.073 ($[-0.092, -0.055]$), an interval that lies entirely below the margin. Against AWP, then, C-PR gives up a measurable part of the ranking of later inflow in exchange for its gain on later approvals. With the wallets outside a graph kept as isolated nodes, F1 to F4 keep their verdicts. F5 changes from a null to a loss. The rule moves the matched traders outside the allowance graph from the bottom of the allowance walk's order into the block of wallets without in-edges (Section 4.4); the allowance walk's coefficient on the liquidation-free close rate then rises from 0.086 to 0.155 , and C-PR falls below it (-0.072 , $[-0.118, -0.028]$). F6 fails under this rule too.

Table 4.16: EndorseRank on the cohorts and labels of the registered replication, scored after that replication was evaluated; none of these rows enters its decision. The AWP, C-PR and degree rows repeat Table 4.14, and the contrasts use the same wallet resamples; the differences between EndorseRank and AWP are in Table 8.8. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Score at 31 May 2026	Spenders ($n = 1,964$)		Traders
	New approval pairs	New transfer senders	Liquidation-free close rate
In-approve degree at the freeze (baseline)	0.613 [0.564, 0.663]	0.371 [0.313, 0.429]	0.144 [0.110, 0.172]
Transfer in-degree at the freeze (baseline)	0.128 [0.064, 0.188]	0.435 [0.401, 0.468]	-0.006 [-0.046, 0.031]
EndorseRank	0.333 [0.306, 0.364]	0.255 [0.227, 0.284]	0.081 [0.032, 0.123]
EndorseRank, restarts by approving activity	0.325 [0.295, 0.358]	0.284 [0.252, 0.312]	0.077 [0.037, 0.116]
AWP	0.118 [0.071, 0.161]	0.319 [0.291, 0.345]	0.045 [0.009, 0.079]
C-PR ($\lambda = 0.5$)	0.238 [0.200, 0.274]	0.246 [0.218, 0.274]	0.083 [0.047, 0.119]
EndorseRank minus t1 in-approve degree, new approval pairs	$\Delta\tau = -0.280$ [-0.313, -0.247]		
AWP minus t1 in-degree, new transfer senders	$\Delta\tau = -0.116$ [-0.133, -0.095]		
C-PR minus EndorseRank, new approval pairs	$\Delta\tau = -0.096$ [-0.128, -0.065]		
C-PR minus EndorseRank, liquidation-free close rate	$\Delta\tau = +0.002$ [-0.053, 0.056]		

Table 4.16 adds EndorseRank, scored after the registered evaluation (Section 3.7.2); none of its rows enters the decision. The spring pattern holds for every PageRank in this window too. The in-approve degree at the freeze again ranks new approval pairs above every walk (0.613 ; EndorseRank

minus the degree -0.280 , $[-0.313, -0.247]$), and the transfer in-degree again ranks new senders above every walk (0.435 ; AWP minus the degree -0.116 , $[-0.133, -0.095]$). C-PR again sits below EndorseRank on new approval pairs (-0.096 , $[-0.128, -0.065]$). On the liquidation-free close rate of the traders EndorseRank reaches 0.081 , close to C-PR (0.083 ; $+0.002$, $[-0.053, 0.056]$).

Table 4.17: Registered replication, traders: Kendall τ between scores frozen at 31 May 2026 and the other GMX V2 labels counted from June to August 2026, on the 1,402 matched wallets with at least three closes in that window. No liquidation is 1 when none of the wallet's closes was a liquidation. Profitable closes and realized gain grow with the number of closes; the two shares do not. These labels are reported and do not enter the decision. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Score at 31 May 2026	No liquidation	Profitable closes	Realized gain	Profitable share	Non-loss share
In-approve degree at the freeze (baseline)	0.187 [0.150, 0.221]	-0.004 [-0.033, 0.028]	0.021 [-0.015, 0.054]	-0.074 [-0.105, -0.042]	-0.139 [-0.172, -0.101]
Transfer in-degree at the freeze (baseline)	-0.072 [-0.115, -0.030]	0.191 [0.150, 0.224]	0.127 [0.089, 0.166]	0.026 [-0.008, 0.060]	-0.001 [-0.036, 0.038]
AWP	-0.030 [-0.069, 0.011]	0.233 [0.193, 0.268]	0.182 [0.142, 0.217]	0.030 [-0.004, 0.060]	-0.035 [-0.065, -0.001]
C-PR ($\lambda = 1$)	0.069 [0.011, 0.116]	0.074 [0.036, 0.107]	0.016 [-0.024, 0.059]	-0.043 [-0.084, -0.000]	-0.130 [-0.166, -0.091]
C-PR ($\lambda = 0$)	0.014 [-0.026, 0.056]	0.140 [0.103, 0.177]	0.145 [0.110, 0.180]	0.033 [-0.001, 0.069]	-0.004 [-0.040, 0.035]
C-PR ($\lambda = 0.25$)	0.032 [-0.007, 0.072]	0.133 [0.094, 0.168]	0.145 [0.110, 0.182]	0.027 [-0.007, 0.062]	-0.011 [-0.047, 0.028]
C-PR ($\lambda = 0.5$)	0.043 [0.004, 0.082]	0.124 [0.084, 0.160]	0.142 [0.107, 0.179]	0.024 [-0.010, 0.058]	-0.010 [-0.045, 0.028]
C-PR ($\lambda = 0.75$)	0.053 [0.011, 0.091]	0.110 [0.068, 0.147]	0.135 [0.101, 0.171]	0.021 [-0.014, 0.056]	-0.008 [-0.041, 0.030]
S-PR	-0.030 [-0.070, 0.014]	0.176 [0.139, 0.211]	0.132 [0.094, 0.170]	0.040 [0.003, 0.075]	-0.023 [-0.057, 0.015]

Table 4.17 gives the registered trader labels that do not enter the decision, and they repeat the spring trader run. AWP ranks the number of profitable closes (0.233) and the realized gain (0.182) highest of the registered scores; the transfer in-degree reaches 0.191 and 0.127 , and C-PR 0.124 and 0.142 . No score exceeds 0.04 on the share of closes that were profitable, where the in-approve degree (-0.074) and C-PR ($\lambda = 1$) (-0.043) are negative. On the share that were not losses, only the intervals of the in-approve degree (-0.139), C-PR ($\lambda = 1$) (-0.130) and, just, AWP (-0.035 , $[-0.065, -0.001]$) lie below zero.

4.6.3 Outcomes built from neither edge type

The GMX trader labels are the only outcomes in these data built from neither edge type. The comparison of Section 3.7.3 was fixed after the registered labels and the single-score coefficients of Tables 4.17 and 4.16 were known, and everything below, every spring-window association included, was computed afterwards, so every result of this subsection is a pre-specified post hoc analysis of existing data. Intervals are 95% unless marked 97.5%. Table 4.18 gives the contrasts on the liquidation-free close rate, Table 4.19 the traders that receive approvals, and Table 8.9 (Appendix 8.5.4) all six trader labels.

Table 4.18: Allowance-side against transfer-side scores on the liquidation-free close rate of the matched traders with at least three closes in the label window, a label built from neither edge type. The analysis plan, contrasts and decision rules were committed on 9 October 2026, after the registered labels and the single-score coefficients of Tables 4.17 and 4.16 were known but before the contrasts, intervals, stratified, recipient and spring-window results were computed, so every row is a pre-specified post hoc result. Paired wallet bootstrap with 2,000 resamples (seed 42); the 97.5% intervals are the Bonferroni intervals of the two decision contrasts. The stratified rows combine Kendall's τ_b within strata of the number of label-window closes (3–4, 5–9, 10–24, 25 or more), weighted by the number of wallet pairs.

Contrast (a minus b)	τ_a	τ_b	$\Delta\tau$	95% CI	97.5% CI
<i>Registered window, June–August 2026 ($n = 1,402$ traders)</i>					
In-approve degree minus transfer in-degree	0.144	-0.006	0.150	[0.100, 0.201]	[0.091, 0.209]
EndorseRank (activity restarts) minus AWP	0.077	0.045	0.032	[-0.003, 0.065]	[-0.008, 0.071]
C-PR ($\lambda = 1$) minus C-PR ($\lambda = 0$)	0.086	0.065	0.021	[-0.035, 0.080]	–
EndorseRank minus AWP	0.081	0.045	0.035	[-0.018, 0.088]	–
C-PR ($\lambda = 0$) minus AWP	0.065	0.045	0.019	[-0.010, 0.047]	–
EndorseRank minus in-approve degree	0.081	0.144	-0.063	[-0.105, -0.022]	–
In-approve degree minus transfer in-degree, stratified by closes	0.155	-0.025	0.180	[0.125, 0.244]	–
EndorseRank (activity restarts) minus AWP, stratified by closes	0.064	0.043	0.021	[-0.016, 0.057]	–
<i>Spring window, March–May 2026 ($n = 1,860$ traders)</i>					
In-approve degree minus transfer in-degree	0.114	-0.040	0.153	[0.115, 0.193]	[0.108, 0.198]
EndorseRank (activity restarts) minus AWP	0.032	-0.002	0.034	[0.004, 0.064]	[-0.001, 0.068]
C-PR ($\lambda = 1$) minus C-PR ($\lambda = 0$)	0.011	0.035	-0.024	[-0.071, 0.022]	–
EndorseRank minus AWP	0.010	-0.002	0.012	[-0.027, 0.052]	–
C-PR ($\lambda = 0$) minus AWP	0.035	-0.002	0.037	[0.011, 0.063]	–
EndorseRank minus in-approve degree	0.010	0.114	-0.103	[-0.140, -0.065]	–
In-approve degree minus transfer in-degree, stratified by closes	0.128	-0.024	0.152	[0.108, 0.195]	–
EndorseRank (activity restarts) minus AWP, stratified by closes	0.063	0.026	0.037	[0.006, 0.068]	–

R1 holds. In June–August the in-approve degree at the freeze reaches $\tau = 0.144$ on the liquidation-

free close rate and the transfer in-degree -0.006 , so P, their difference, is $+0.150$ (97.5% interval $[0.091, 0.209]$). Stratified by the number of closes, P is $+0.180$ ($[0.125, 0.244]$), and in the spring window, where this label had never been scored out of window, $+0.153$ (97.5% interval $[0.108, 0.198]$). R2 fails. Q, EndorseRank with AWP's activity restarts minus AWP (the like-for-like PageRank pair), is $+0.032$ in June–August and $+0.034$ in the spring, but both 97.5% intervals include zero ($[-0.008, 0.071]$ and $[-0.001, 0.068]$). The PageRank versions point the same way and are not resolved, nor are C-PR's layers or EndorseRank against AWP in either window. C-PR ($\lambda = 0$) minus AWP, the raw transfer layer with uniform restarts against AWP on the same transfers, is $+0.019$ ($[-0.010, 0.047]$) in June–August and $+0.037$ ($[0.011, 0.063]$) in the spring, so part of C-PR's registered lead over AWP on this label (F3 against AWP, Table 4.15) comes from AWP's weights and restarts, not from allowance edges. EndorseRank again falls below the approval count, by 0.063 ($[-0.105, -0.022]$) in June–August and 0.103 in the spring.

The other labels do not enter the decision (Table 8.9). On the no-liquidation flag, P ($+0.259$, $[0.199, 0.320]$) and Q ($+0.075$, $[0.035, 0.116]$) lie above zero in June–August and stay above zero in the spring. On profitable closes and realized gain, which grow with activity, the transfer side ranks traders higher in all four pairs, every interval below zero in both windows; in June–August P is -0.194 and -0.107 and Q -0.136 and -0.124 . R3 applies. The in-approve degree ranks below the transfer in-degree on the profitable share (-0.101 , $[-0.149, -0.047]$) and the non-loss share (-0.138 , $[-0.191, -0.083]$), so the allowance signal is specific to liquidation avoidance and does not extend to trading success, while the transfer count ranks how much a trader trades but not how safely. Part of the non-loss gap is built in: the pipeline counts a liquidated close as a close that was not a loss, so a trader with many liquidations gains non-loss share, and R3 therefore rests mainly on the profitable share, which counts a liquidated close as not profitable.

The count-level lead rests on few traders. Only 44 of the 1,402 traders in June–August and 39 of the 1,860 in the spring had received an approval at the freeze, almost all with an in-approve degree of 1 (the maximum is 2), so P in effect compares these recipients with the rest. In June–August 95.5% of recipients had no liquidation, against 42.0% of the other traders (risk difference $+0.534$,

Table 4.19: Labelled traders with a positive in-approve degree at the freeze (recipients) against all other labelled traders (pre-specified post hoc analysis). Account types come from the code lookup of the spring spender cohort; a recipient that was not in that cohort is *unknown*. Bootstrap intervals use the resamples of Table 4.18.

	Recipients	Other traders
<i>Registered window, June–August 2026 (n = 1,402 labelled traders)</i>		
Traders	44	1,358
Share with no liquidation	0.955	0.420
Mean liquidation-free close rate	0.986	0.865
Mean profitable share	0.347	0.456
Mean non-loss share	0.435	0.668
Median closes in the label window	23.0	16.5
Risk difference in no liquidation: 0.534 [0.459, 0.593]; recipients by account type: 40 contract, 3 unknown, 1 EOA		
P without the 10 recipients of highest in-approve degree: 0.124 [0.073, 0.174]		
<i>Spring window, March–May 2026 (n = 1,860 labelled traders)</i>		
Traders	39	1,821
Share with no liquidation	0.974	0.494
Mean liquidation-free close rate	0.997	0.888
Mean profitable share	0.347	0.455
Mean non-loss share	0.398	0.647
Median closes in the label window	27.0	15.0
Risk difference in no liquidation: 0.481 [0.418, 0.523]; recipients by account type: 38 contract, 1 EOA		
P without the 10 recipients of highest in-approve degree: 0.133 [0.093, 0.172]		

[0.459, 0.593]); in the spring, 97.4% against 49.4%. Recipients also closed more positions (median 23 against 16.5) and had lower profitable and non-loss shares (mean non-loss share 0.435 against 0.668), the pattern behind R3. Without the 10 recipients of highest in-approve degree, P is +0.124 ([0.073, 0.174]) in June–August and +0.133 ([0.093, 0.172]) in the spring. Almost every recipient is a contract account (40 of 44 in June–August, with 3 of unknown type and 1 externally owned; 38 of 39 in the spring), so the check of R1 on externally owned recipients alone could not be run, and account type is an unresolved confound of this signal.

4.7 Claims (C1–C6)

The chapter’s results support the claims below, cross-referenced to the hypothesis and research question columns of Table 1.2; C1–C5 are the primary claims, and C6 rests on a pre-specified post hoc analysis.

- C1. On identical wallet sets, the cost of each score’s solver call follows the number of edges in its graph, because both matrix assembly and iteration scale with $|E|$. EndorseRank’s graph has about one-ninth of the transfer graph’s edges, and its call takes a matching fraction of AWP’s time and memory (Table 4.1). The ratio holds across the in-cohort and pool stages (Tables 4.2 and 8.1), although the pool stages do not test edge counts beyond those of the cohort graph. The low cost belongs to EndorseRank alone: C-PR and S-PR walk the transfer layer and take longer than AWP.
- C2. In the same window each score agrees with the proxies of the edge type it walks on, with intervals that exclude zero, and the mixtures follow their denser transfer layer (Table 4.3). These tests measure internal consistency, and part of the agreement is built in; external validity is outside their scope. EndorseRank’s coefficients are set by the rule for wallets outside the allowance graph (Table 4.5).
- C3. EndorseRank and AWP order the cohort differently. Their agreement has an interval that excludes zero and one, comes mainly from the wallets outside the allowance graph, and falls

slightly below zero when those wallets are kept as isolated nodes (Table 4.5, Figure 4.3).

- C4. Out of window no PageRank of this thesis ranks a spender label as well as the raw degree of the label's own edge type, in either window: every paired difference to the degree lies below zero (Tables 4.11 and 4.16), and the shortfall is not a matter of coverage (Table 4.12). On the matched traders the scores built on transfers rank counts of later profitable closes, and no score ranks the share of closes that were profitable (Table 4.13). Apart from the increment of C-PR ($\lambda = 1$), none of these comparisons was fixed in advance.
- C5. A single random walk over both edge types does not beat its two layers walked alone, and the registered replication does not confirm that it improves on the transfer walk. Under the rule of 14 September, C-PR falls between the two walks on new approval pairs at every λ tested, and S-PR reproduces the transfer walk (Tables 4.10 and 4.11). The spring gain over the transfer walk on new approval pairs, found after the labels were known, replicates in the registered window (F1, with F3 and F4), but the difference on new senders cannot be bounded within 0.02 (F2), so the registered rule is not met. Against AWP, in the registered sensitivity analysis, C-PR loses clearly on new senders (Table 4.15). Section 4.3.1 reports where the hybrids stand within the window.
- C6. On later liquidation avoidance, the one trader outcome built from neither edge type that bears on counterparty risk, the approval count at the freeze ranked traders above the transfer count ($\Delta\tau = +0.150$) in a pre-specified post hoc analysis; the lead survives stratification by the number of closes and points the same way in the spring window. The claim is count-level: the PageRank versions point the same way but are not resolved, and almost every approval-receiving trader is a contract account, so account type is an unresolved confound. The same signal is lower on the shares of closes that were profitable or not losing, so what it ranks is forced-liquidation avoidance, not trading success (Section 4.6.3).

The rank divergence answers RQ1, the same-window tables RQ2, the benchmark tables RQ3, and the holdouts RQ4 and RQ5; C6 adds no research question. The direct comparisons between En-

dorseRank and AWP are in Appendix 8.5 and Table 4.18, and none of them carries a claim of the thesis. Chapter 5 reads these findings against the research questions and objectives of Chapter 1.

5 Discussion

Chapter 4 reported the measurements taken on Arbitrum One. Here each research question and each objective of Chapter 1 is held to the criterion stated there, and every interpretation points to the table of Chapter 4 it rests on. Unless noted otherwise, same-window statements concern the matched cohort and holdout statements the spender cohort.

5.1 Findings by research question

5.1.1 RQ1: Reputation structure and rank divergence

RQ1 asks whether EndorseRank and AWP order the same wallets in different ways. They do: the interval of their agreement excludes both zero and one (Table 4.5). The agreement that exists comes from a rule of the pipeline more than from the graphs. EndorseRank ties most wallet pairs, and most of the pairs it does order set a wallet inside the allowance graph against one outside it; kept as isolated nodes instead of being scored zero, the wallets outside the graph join the common block and the agreement falls slightly below zero (Section 4.4).

The divergence follows from how the two graphs are built. The endorsement graph records the latest approvals and has about one ninth of the edges of the transfer graph, which records every transfer^[3], and the two methods restart differently. PageRank places stationary mass according to the edges it is given ^{[15],[23]}, so a wallet that receives many transfers from active senders but few approvals ranks high under AWP and modestly under EndorseRank, and the reverse holds for a contract that many owners approve without being a transfer hub. H1 is supported, largely by construction, since two walks over different graphs with different restarts were not expected to agree. In practice one score cannot stand in for the other: an operator who adopts EndorseRank for its speed gets a different ranking and not a faster approximation of AWP.

5.1.2 RQ2: Same-window alignment

RQ2 asks how closely each score agrees with contemporaneous allowance and transfer checks. Each score follows the edges it walks on: the walks over transfers agree with the transfer family and stay near zero on the allowance family, the walks over allowances agree with the allowance family, and every interval on a score's own family excludes zero (Table 4.3).

Two caveats bound H2. The first is circularity. Every proxy in a family is computed from the same events as the matching graph, so the agreement is construct validity in the sense of Cronbach and Meehl ^[22] and not external validation. The second is EndorseRank's tie structure (Section 4.4). Its allowance coefficient rests on the 131 wallets that receive approvals and sits at the ceiling the ties allow, and its transfer coefficient mostly reflects which wallets appear in the allowance graph at all; the two have different ceilings, so they say nothing about whether EndorseRank leans more on allowances than on transfers. The isolated-node rule changes both sharply (Table 4.5), but it was tried after the results were known, so H2 does not rest on it. In the narrow sense of agreement with the checks of a score's own edges, H2 is supported, largely by construction. None of these comparisons was fixed in advance, and the paired differences between EndorseRank and AWP, which carry no claim, are in Appendix 8.5.

5.1.3 RQ3: Computational cost

RQ3 asks what each score costs to compute on the same sample. The cost follows the edge count: EndorseRank is the cheapest score in time, memory and iterations, and the two hybrids, which walk the transfer layer, take longer than AWP (Table 4.1). Matrix assembly takes most of the timed call, and assembly and iteration are both linear in $|E|$ ^[23], so the time ratio tracks the edge ratio (Section 5.1.1) and the difference in iteration counts adds little.

The scaling tables qualify this result more than they extend it (Table 4.2). Pool addresses outside the cohort carry no extracted edges, so the stages never test a graph larger than the cohort graph. The damping and top-token checks leave the alignment pattern and the runtime ratio as they were

(Tables 4.6 and 4.7). H3 is supported on the matched cohort and its subgraphs, again largely by construction, since both scores run the same solver. H1–H3 are thus descriptive checks of the instruments; the tests whose outcome was open are RQ4, RQ5 and Section 5.1.6.

5.1.4 RQ4: Temporal holdout of future new approvals

RQ4 has two clauses. The first asks whether scores frozen at the end of February predict new owner–spender approvals and new transfer senders on the spender cohort over the following three months. Most scores do predict both labels, and the edge type decides how well (Table 4.10). On new approval pairs the in-approve degree leads, followed by the walks over allowances and the interior mixtures, while the scores built on transfers alone stay low. On new senders the transfer in-degree leads, but the spenders that many owners approve also gain many new senders, so the in-approve degree and the allowance walk rank this label better than any walk over transfers. The evidence concerns spenders; Section 5.1.6 turns to the traders.

The second clause asks whether any PageRank beats the raw degree it smooths. None does, in either window: every paired difference lies well below zero (Tables 4.11 and 4.16), and AWP’s shortfall holds on the spenders that had received a transfer before the freeze, so the spenders it scores zero do not explain it (Table 4.12). Known results lead one to expect this: in growing networks new links attach preferentially to nodes that already have many ^{[67],[68]}, which makes degree a standard link-prediction baseline ^[69], also on Ethereum transaction networks ^[74], and where degree correlations are weak, as on the Web, PageRank is on average proportional to in-degree ^[70]. On these data EndorseRank is almost exactly a reweighted approval count (Section 5.3.2), and AWP likewise divides each sender’s weight among all the addresses it paid. This does not undo the first clause, since the counts are statistics of the same edges; it rules out that propagation adds predictive power beyond the degree for these labels on this sample. Apart from the increment of C-PR ($\lambda = 1$), these contrasts were computed after the labels were known.

RQ4 departs from the research proposal, which validated the scores against trading success bounded by an outcome-based reference; that reference was circular and was withdrawn (Section 2.8). The

labels used instead lie outside the window but belong to the same construct, and neither contains any element of profit or repayment. RQ4 therefore shows the temporal predictive validity of each construct, with the raw degree as the stronger baseline, and leaves credit risk aside; the liquidation label of the registered window (Section 3.7.2), on which Section 5.1.6 compares the edge types, comes closest to it.

5.1.5 RQ5: Coupling the two edge types

Both halves of RQ5 get a negative answer, each under its own rule. The second half asks whether one walk over both layers ranks wallets better than either layer walked alone. Under the rule fixed on 14 September it does not. On the spring holdout C-PR falls between the two walks on new approval pairs and is not shown to beat the transfer walk on new senders (Table 4.11), and in the same window it tracks the transfer walk and stays near zero on the allowance family, so the secondary rule fails on its allowance part (Section 4.3.1). The first half asks whether the walk improves on a walk over the transfers alone. An exploratory comparison on the spring holdout said that it did on new approval pairs, with no detectable loss on new senders, but it was made after the labels were known. The registered replication decides the question, and its answer is no: C-PR again beat the transfer walk on new approval pairs and ranked liquidation-free traders slightly better, but non-inferiority on new senders was not shown, and the rule needed all three conditions (Section 4.6.2). Against AWP, the comparator of the registered sensitivity analysis, C-PR falls behind on new senders by a wider margin, and against EndorseRank it is below on new approval pairs in both windows. Its coefficients rise with the allowance weight on both spring labels, so out of window the transfer layer adds nothing for spenders beyond what the allowance layer supplies.

The ablation and the layer sizes explain both halves. S-PR keeps the transfer walk but restarts it from the allowance walk; it reproduces the transfer walk on the flow label and loses almost all of the endorsement signal, since a teleportation vector derived from endorsements changes little when transfer edges decide where the walk goes. In C-PR the allowance edges carry part of the walk, so part of the signal survives on the spender holdout, but the transfer layer is far denser and λ acts only

at the minority of nodes with out-edges in both layers (Section 4.3.1). The mixture kept enough of the allowance signal to beat the transfer walk and too little to match the allowance walk. Its layers carry raw amounts and restart uniformly, whereas AWP, which counts each transfer about once and restarts at active senders, ranks new senders better than C-PR in both windows. Other couplings were not tested (Section 5.9).

Both halves count as results. The rule for the second half was fixed before any coupled score existed, so λ could not have been tuned to it, and it limits an obvious next step: a convex mixture of the two edge types at the transition level yields no score that beats the better single layer on either construct. The rule for the first half was registered before its labels were extracted and fails on one of its three conditions. The replication establishes something narrower: adding allowance edges to a transfer walk gains information about authorization that replicates on new data, without a demonstrated bound on the loss in inflow.

5.1.6 Allowance against transfer on outcomes built from neither edge type

Unlike the spender labels of RQ4 and RQ5, which favor the scores built on their own edge type, the GMX trader outcomes are built from neither edge type and were measured after the scores were frozen. The comparison of the two edge types on them (Section 3.7.3) adds no research question and is reported as Claim C6 (Section 4.6.3).

This comparison bears on counterparty risk, the use that motivated the thesis (Chapter 1). For a protocol that extends leverage or credit, a counterparty that is often force-liquidated is a riskier one, and forced liquidation is the nearest outcome to default in these data. On later liquidation avoidance the approval count at the freeze ranked traders above the transfer count ($\Delta\tau = +0.150$ in June–August), a lead that survives stratification by the number of closes and recurs in the spring window, while the transfer count ranked this outcome near zero. The transfer scores rank instead the outcomes that grow with activity, the number of profitable closes and the realized gain (Table 4.17): they record how much a wallet trades, not how safely.

Five limits bound this reading. First, it is a pre-specified post hoc analysis of existing data. Second,

it holds at the count level only: the like-for-like PageRank pair, EndorseRank with AWP's restarts against AWP, points the same way but is not resolved in either window, and EndorseRank again falls below the approval count it smooths. Third, the lead rests on a few dozen approval-receiving traders per window, almost none of them ever liquidated and almost all of them contract accounts that other addresses authorized to spend their tokens. That pattern fits managed or pooled trading contracts, which the data do not identify, so account type is an unresolved confound that may explain the signal. Fourth, the same signal is lower on the share of closes that were profitable, so the construct it ranks is forced-liquidation avoidance, not trading success; its lower non-loss share is partly built in, because the pipeline counts a liquidated close as not a loss. Fifth, the registered rule for C-PR remains not met, and F4 stays a pre-specified secondary result outside it. Within these limits, on the outcome a margin or lending protocol screens for, the comparison favored allowance-edge over transfer-edge information at the count level.

5.2 Achievement of the research objectives

Chapter 1 attached a measurement and a criterion to each of its five objectives. Table 5.1 checks them against the evidence.

Six findings outside the objectives also shape the conclusions. Four limit the claims: EndorseRank's same-window coefficients are shaped by its ties (Section 5.1.2); the expanded-pool experiment never ran on a graph larger than the cohort graph (Section 5.1.3); on the matched traders no score ranks the share of profitable closes (Section 4.6.1); and on the spring spenders the raw-amount allowance walk ranked both labels better than EndorseRank, so how approvals are weighted matters (Section 5.8.1). Two add to the claims. In a pre-specified post hoc analysis the approval count ranked traders on later liquidation avoidance above the transfer count, at the count level only, with the PageRank pair unresolved, a possible account-type confound and no lead on trading success (Section 5.1.6). And on the depth-one graphs scored here EndorseRank is a reweighted approval count (Section 5.3.2). That the raw degrees out-predict every PageRank is not among them, because

Table 5.1: Research objectives, the criterion stated in Chapter 1, and the outcome.

Obj.	Criterion	Evidence	Outcome
O1	95% interval of inter-method τ contains neither 0 nor 1	Table 4.5, last row: the interval $[0.342, 0.379]$ lies strictly inside $(0, 1)$	Met (largely by construction)
O2	A score agrees with a construct when its interval on that family excludes 0; agreement with the construct of a score’s own edges is an intended-construct check	Table 4.3: every walk over transfers on the transfer family (0.494 to 0.612) and both walks over allowances on the allowance family (0.375 and 0.355); the proxies come from the same events as the graphs, and EndorseRank’s coefficients are set by its ties (Table 4.5)	Met (largely by construction)
O3	Runtime ratio given alongside the edge ratio, under a single fixed protocol	Tables 4.1, 4.2 and 8.1: mean, SD, memory, iterations, $ V $, $ E $ per configuration; one solver, so time follows $ E $	Met (largely by construction)
O4	A score predicts a label when its interval excludes 0; a PageRank improves on its degree only when the interval of the paired difference lies above 0	Tables 4.10 and 4.11: every score except C-PR ($\lambda = 0$) on new approval pairs predicts both labels; the differences to the degree are -0.233 (C-PR, $\lambda = 1$), -0.317 (EndorseRank) and -0.127 (AWP), all with intervals below 0, and again below 0 in June–August (Table 4.16)	Met (first clause); not met (second clause)
O5	(a) Rule of 14 September: on the holdout, C-PR does no worse than C-PR ($\lambda = 1$) on new approval pairs and better than C-PR ($\lambda = 0$) on new senders; in the same window, its point estimate lies within the leading layer’s interval on the transfer and allowance families, and it exceeds both layers on Sybil stability. (b) Rule written on 28 September and registered on 1 October: on the June–August window, C-PR beats C-PR ($\lambda = 0$) on new approval pairs and is not worse by more than 0.02 (lower 95% bound above -0.02) on new senders and on the traders’ liquidation-free close rate	(a) Table 4.11: both holdout conditions fail. Same window: transfer (0.527) lies inside the interval of C-PR ($\lambda = 0$) and allowance (-0.001) far outside that of C-PR ($\lambda = 1$) (Table 4.3); Sybil stability exceeds both layers (Table 4.4). The allowance part fails. (b) Table 4.15: new approval pairs $+0.179$ $[0.132, 0.225]$ and liquidation-free close rate $+0.018$ $[0.011, 0.024]$ hold; new senders -0.011 $[-0.025, 0.003]$, lower bound below -0.02 , fails	(a) Not met; (b) not met

it is the second clause of O4.

5.3 Theoretical implications

The results bear on three areas of theory: propagation semantics on financial graphs, the construct validity of on-chain scores, and the economics of trust under pseudonymity.

5.3.1 Propagation semantics beyond transfers

PageRank was first formulated for hyperlinks, read as a citation network ^[15]. Do and Do^[4] carried this reading over to Ethereum transactions, and AWP^[3] refined it with value and age weights and restarts at active senders. EndorseRank applies it to ERC-20 allowances, where an owner who lets a spender act for them issues a directed endorsement of spending authority ^[18]. The RQ1 divergence shows that changing what the edges mean changes the ranking even on the same nodes, so graph-based reputation is a family of methods, each valid relative to the relationships its edges encode. Peer-to-peer and web trust systems reached the same conclusion from the other side: in EigenTrust ^[16] and TrustRank ^[17] the seed set and the edge definitions largely decide the outcome. EndorseRank's edge is native to a token standard, where theirs belonged to a message-passing overlay or a hyperlink graph.

The holdouts add a qualification that the propagation metaphor hides. Changing the edge type changed what the score could anticipate: on new approvals the scores built on allowance edges led and those built on transfers alone stayed low, in both windows. Propagating along the edges did not add to that, since each raw count predicted its own arrival label better than every walk (Section 5.1.4). On these data the value lies in the choice of edge, and the walk reweights a count (Section 5.3.2). When one layer is much sparser than the other, coupling lets the dense layer set the route, and on the spender holdout the sparse layer's signal survives only in part (Section 5.1.5).

5.3.2 EndorseRank on a depth-one allowance graph

The allowance graphs keep only the approvals that a seed wallet grants or receives (Section 3.6), so almost every node is a pure owner, with out-edges only, or a pure spender, with in-edges only. On such a graph EndorseRank has a closed form.

Proposition. Let PageRank be computed by Equation (3.1) with damping d and uniform restarts, so that dangling mass is also spread uniformly. Then every node without in-edges has the same score c , and every node v whose in-neighbors all lack in-edges has

$$r_v = c(1 + d\delta(v)), \quad \delta(v) = \sum_u \frac{w(u, v)}{o(u)}, \quad (5.1)$$

where $\delta(v)$ is the owner-normalized in-strength of v : each owner's endorsement is divided over all the spenders it approves.

Derivation. At the fixed point, $r_v = (1 - d + dr_D)/|V| + d \sum_u r_u w(u, v)/o(u)$, where r_D is the mass held by the dangling nodes. A node without in-edges receives only the first term, so every such node has the score $c = (1 - d + dr_D)/|V|$, the constant b of Section 5.6.3. If every in-neighbor u of v is such a node, substituting $r_u = c$ gives Equation (5.1).

On the spring and registered freeze graphs, EndorseRank as computed in this thesis lies within 1% of Equation (5.1) for 96.4% and 97.4% of the spenders (the rest have an in-neighbor that also receives approvals), and over the spenders its Spearman correlation is 0.945 and 0.980 with δ but 0.584 and 0.568 with the in-approve degree (script in `analysis/closed-form`).

Two consequences follow. First, on this sample EndorseRank only reweights the approval count, counting an owner that approves many spenders for less at each; besides the time and token weights, that normalization is all that separates it from the count, which ranks the arrival labels better (Section 5.1.4). For EndorseRank the negative answer to the second clause of RQ4 thus has an analytic basis: there was no propagated information for the walk to add. Second, the walk has depth one: mass reaches a spender in one step from owners that share one score, so who the endorsing own-

ers are enters only through how many spenders they approve. That propagation adds nothing to the count therefore holds for this sample only; testing multi-hop propagation would need a deeper sample, such as the two-hop neighborhood of the seed wallets or a chain-wide allowance graph, or restarts seeded at vetted wallets.

5.3.3 Multi-construct domain alignment

The checks were designed to separate transfer centrality from allowance delegation, and each score aligns with the construct of the edges it walks on, the mixtures with their denser layer. Classical construct-validity theory would predict this pattern for instruments that measure different constructs ^[22], and Messick's ^[33] integrated view goes further, counting the consequences of using a score as part of its validity. Rank correlations ^{[20],[21]} put this logic into practice for ordinal reputation scores without requiring classification thresholds, while the paired bootstrap ^[34] makes the pattern testable. The same theory says what such a pattern cannot settle: when two instruments measure different constructs, the one that comes out ahead on a check is the one built for that check, so ordering the instruments by their coefficients does not show which is the better measure of reputation. Outcomes built from neither edge type are the exception, since neither instrument was built for them, which is why the thesis compares the edge types there (Section 5.1.6); even there, which edge type leads depends on the outcome. EndorseRank's ties add a practical point: two rank correlations can be compared only when they have the same attainable maximum.

5.3.4 Reputation under cheap pseudonyms

Identities are cheap to create on public ledgers ^[5], and under Ethereum's account model a new address needs no registration step at all ^[36]. Any reputation system that rewards inbound transfers or allowances is therefore exposed to Sybil strategies, in which an attacker creates edges among addresses it controls ^[11]. The empirical findings do not rule this out. Section 5.6 formalizes the cost of attacking an allowance graph, where a closed farm of addresses that approve one another needs, unlike transfer wash trading, no token balance and pays little beyond gas.

5.4 Practical implications and deployment scenarios

Protocols and infrastructure providers have to choose between transfer-hub detection and authorization-aware reputation. The measured results support choosing the score by scenario over recommending a single one.

5.4.1 Transfer-hub and flow monitoring

A protocol whose priority is describing current flow (for market-maker identification, routing analysis, or transfer-based compliance heuristics) can read that flow from in-degree and in-value directly. AWP^[3] adds propagation, time weights and restarts at active senders. It is the walk closest to present inflow (Table 4.3) and, among the walks over transfers, the closest to later new senders (Table 4.10). Out of window, though, the raw transfer in-degree ranked later new senders better than AWP in both windows. A protocol that wants to anticipate flow should start from the count.

5.4.2 Authorization-aware and low-latency refresh

Where explicit spending authorization matters most (token vaults, allowance-gated integrations, or risk engines that treat approvals as acts of trust), protocols have reason to use allowance edges. The cheapest form is the spender's in-approve degree, which predicted new approvals better than any PageRank (Table 4.11) and needs no solver at all. EndorseRank is its propagated version. Its allowance alignment (Table 4.3), its holdout result (Table 4.10) and its short solve time (Table 4.1) make it a reasonable choice where the operator wants a ranking that also reflects who the endorsing owners are. On this study's graphs, however, it reflects them only through how many spenders each owner approves (Section 5.3.2), a property not shown to improve any label, and under uniform teleportation the propagated score is open to cheap inflation by closed farms (Section 5.6). In return for authorization semantics and speed, operators accept a score that carries little transfer information.

5.4.3 Evaluation design and combined use

Whether to combine the two signals is a question about deployment and about method, and the evidence answers the two differently. An operator can run EndorseRank for continuous authorization-aware monitoring and AWP for periodic transfer-hub audits, at the cost of both solves. Folding the two edge types into one walk did not pay on these data: C-PR costs more than AWP (Table 4.1) and trails EndorseRank on new approvals and AWP on new senders in both windows (Section 5.1.5), so an operator free to choose should keep the two scores apart. For an operator tied to a single transfer walk weighted by raw amounts, adding allowance edges bought authorization information that replicated on new data, but a loss on inflow larger than 0.02 could not be ruled out. Because the pipeline is open, the analysis can be replicated on any ERC-20 chain that has allowance logs (Appendix 8.1); integration with live risk oracles is left for future work (Chapter 6).

5.4.4 Industry adoption scenarios

The scenarios below say for whom each score is useful and for whom it is not.

- DEX operators. A protocol that sets fee tiers or monitoring priority by wallet needs a score for its users, and its users are owners that grant approvals. EndorseRank gives almost all of them the same value (Section 4.4), so it cannot serve this purpose; transfer-based scores or plain activity counts can. EndorseRank can instead rank the routers and vaults that the DEX integrates with.
- Lending and margin protocols. Lending markets want to put more of their capital to use without cutting collateral ^[41], and the stress of 2020 showed the cost of updating risk parameters late ^[9]. Their screening question is whether a counterparty will be force-liquidated. On that outcome the approval count ranked GMX traders above the transfer count, but only in a post hoc, count-level result with the PageRank pair unresolved and a signal that is lower on trading success, carried by a few dozen traders per window, almost all of them contract accounts whose type may explain it (Section 5.1.6). A screen built on it would need a registered test

on new data, with account type controlled. The scores are not tools for predicting default.

- **Wallet and security tooling.** Users and integrators want to know whether a spender contract is widely authorized. A badge that shows how many owners have approved a spender, or its EndorseRank, makes that delegation explicit ^{[18],[19]}; it conveys the position of a graph node and is neither a security audit nor a guarantee against malicious contracts.
- **Liquidity providers and market makers.** To control inventory and spreads, these participants need to find transfer hubs and see how flow is structured. Transfer counts and AWP are the suitable tools for that goal^[3], and their benefit is lower screening cost, not higher profit and loss.
- **Airdrop and governance programs.** Sybil clusters routinely abuse token distributions and governance eligibility ^{[11],[32]}. EndorseRank as defined here offers little protection, because a closed farm collects several times its population share and can concentrate it on one target (Section 5.6). Programs can require diverse incoming endorsements from owners with independent histories, discount mass that circulates inside closed clusters, or send teleportation and dangling mass only to a vetted seed set; none of these stops an attacker who buys endorsements from honest owners.
- **Analytics and compliance providers.** Exchanges, funds and compliance teams need counterparty rankings that outsiders can recompute. EndorseRank and AWP can be published for defined windows and parameters and audited, unlike proprietary machine learning models ^[2], provided that their Sybil exposure is disclosed and they are not presented as profit predictions.

In none of these scenarios does reputation replace collateral.

5.4.5 Standardized scores as shared infrastructure

A score shared by several parties is useful only if each of them can recompute it and see how it might be manipulated. EndorseRank and AWP can be recomputed from public logs with the open pipeline (Appendix 8.1), and Sections 5.6 and 5.7 document how they can be manipulated.

5.5 Societal and practical impact

In the DeFi market, vetting a counterparty is expensive, and the capacity to do it is unevenly spread. Wealthy actors may be able to pay for proprietary risk tools, while retail participants work from anecdotal evidence or skip vetting altogether. A score built from the permissions users have already granted, and cheap to recompute (Table 4.1), lowers the barrier to vetting the contracts that ask for those permissions. An approval that a user granted earlier is one route by which tokens are drained (Section 2.17). A spender-centered endorsement score therefore conveys information about a risk users face directly and that transfer volume cannot measure.

For protocol developers, the results argue for replacing a single “reputation” construct with specific scores, each fitted to the decision it informs, which reduces the risk of using a flow score to decide authorization.

Where the decentralized credit scoring systems discussed by Packin and Lev-Aretz^[2] are opaque, a score built from public events with versioned parameters supplies the explanatory information that regulators suggest should be required when DeFi protocols are analyzed^[61]. It also brings a duty, described in Section 5.7, to ensure that an ordinal, graph-based statistic is not mistaken for an identity, a credit history, or proof of personhood.

Any reputation metric that affects how value is allocated is open to exploitation; the next section models what it costs to manipulate EndorseRank.

5.6 Sybil attack cost model on allowance graphs

A Sybil attack exploits cheap pseudonyms to create a false impression of trustworthiness^[11]. On a transfer graph, a malicious actor can inflate in-degree and in-flow artificially by moving assets among accounts it controls, paying for this in transaction fees and possible inventory loss^[29]. Allowance graphs expose a different attack surface, since each edge records an authorization, not an actual transfer. In this section we formulate a simple cost model for inflating EndorseRank with fake allowances. It builds on the optimal link farms of Gyöngyi and Garcia-Molina^[71] and on

Cheng and Friedman’s result that symmetric functions such as PageRank are not Sybil-proof ^[72] (Section 2.6), and adds the cost of an allowance edge and the size of the gain on this study’s graph.

5.6.1 Attack setup

Let the attacker control a set of addresses $\mathcal{S}$, and write $\mathcal{H} = V \setminus \mathcal{S}$ for the honest addresses in the endorsement graph $G_{\text{approve}} = (V, E, w)$. An edge (u, v) of weight $w_{uv} > 0$ records that u has authorized v to spend its tokens. The weight is that of Equation (3.7): for each token, the time weight of the latest in-window approval times the bounded value weight of its amount, summed over tokens. With damping parameter $d \in (0, 1)$, EndorseRank finds the PageRank vector $\mathbf{r}$ on G_{approve} as the solution of ^[15]:

$$\mathbf{r} = d P^\top \mathbf{r} + \frac{1-d}{|V|} \mathbf{1}, \quad (5.2)$$

where P is the transition matrix of Equation (2.1). A node with out-edges splits its row in proportion to its outgoing allowance weights. A dangling node, one without out-edges, has every entry of its row equal to $1/|V|$, so its mass is spread uniformly over all nodes ^[23]. The fixed point is the same as that of the iteration in Equation (3.1), which the implementation runs and which feeds dangling mass back through the uniform teleportation vector. The attacker’s objective is to maximize the mean EndorseRank of a target set $\mathcal{G} \subseteq \mathcal{S}$,

$$J(\mathcal{G}) = \frac{1}{|\mathcal{G}|} \sum_{i \in \mathcal{G}} r_i, \quad (5.3)$$

subject to a budget constraint on the cost of the attack, which is defined next.

5.6.2 Cost components

The attacker creates a set of edges $E_{\mathcal{S}}$, all with sources in $\mathcal{S}$, and may also buy a set $E_{\mathcal{H} \rightarrow \mathcal{S}}$ of edges from honest owners into $\mathcal{S}$. Costs are counted over q observation windows, the number of score refreshes for which the targets are meant to hold their rank. Gas is the only cost that every edge incurs. The other components arise only for particular edges or strategies.

1. Gas cost. Each approve or permit call consumes gas; write c_{gas} for the expected fee of one allowance update. EndorseRank reads only the Approval events inside its observation window, so an edge enters a window's graph only if its approval is emitted in that window. To keep E_S in place for q windows, the attacker has to emit every approval again in each window, which also keeps its time weight near the maximum $\sigma(0) = 0.86$, and the gas is paid q times:

$$C_{\text{gas}} = q |E_S| c_{\text{gas}}.$$

2. Exposure cost. An approval locks no tokens, and the owner need not hold any balance of the token when it approves. The amount hardly matters either. The value weight V gives an approval of a few base units almost the weight of an unlimited one, and after row normalization a single out-edge gives $P_{uv} = 1$ whatever its weight. An edge inside S therefore exposes nothing and costs nothing beyond gas. Exposure arises only when an attacker address that holds tokens approves a spender $v \notin S$, for instance to look like an ordinary user of known contracts, since v can then move those tokens. Such an edge also lets score mass leave S . With ℓ_{uv} the expected cost to the attacker of one such exposure over one window,

$$C_{\text{exp}} = q \sum_{(u,v) \in E_S, v \notin S} \ell_{uv},$$

which is zero for a closed farm.

3. Revocation and detection cost. Legitimate users revoke old allowances from time to time, and a farm whose approvals reappear unchanged in every window might be identifiable. Let c_{rev} be the expected cost per window of detection and of the revocations the attacker issues to avoid it. Over q windows,

$$C_{\text{rev}} = q c_{\text{rev}}.$$

4. Bought honest edges. The attacker can also pay honest owners to approve its addresses, for instance through an app that asks users for permission. A bought edge (h, a) , with $h \in \mathcal{H}$

and $a \in \mathcal{S}$, falls under the same window rule and costs, per window,

$$c_{\text{in}}(h, a) = c_{\text{gas}} + p_h + \rho_h, \quad (5.4)$$

where p_h is the payment to the honest owner and ρ_h is a risk premium the owner asks against the allowance being spent by the attacker.

Over the q windows, the total cost of the attack is

$$C = C_{\text{gas}} + C_{\text{exp}} + C_{\text{rev}} + q \sum_{(h,a) \in E_{\mathcal{H} \rightarrow \mathcal{S}}} c_{\text{in}}(h, a). \quad (5.5)$$

For a closed farm, $C_{\text{exp}} = 0$ and $E_{\mathcal{H} \rightarrow \mathcal{S}}$ is empty, so $C = q(|E_{\mathcal{S}}| c_{\text{gas}} + c_{\text{rev}})$.

5.6.3 Influence and a cost-effectiveness bound

By the usual sensitivity results for PageRank, an edge moves $\mathbf{r}$ in proportion to the stationary mass at its source node and to its normalized outgoing weight ^[23]. Consider first a closed farm: every node of $\mathcal{S}$ has at least one out-edge, all of these edges end in $\mathcal{S}$, and no honest owner approves an address in $\mathcal{S}$. Let D be the set of dangling nodes, all of which then lie in $\mathcal{H}$, and let $r_D = \sum_{j \in D} r_j$ be their stationary mass. By Equation (5.2), a node $i \in \mathcal{S}$ receives $d \sum_{j \in \mathcal{S}} r_j P_{ji}$ from the farm, $(1-d)/|V|$ from teleportation and $d r_D / |V|$ from the dangling rows. The rows of the farm's nodes sum to one inside $\mathcal{S}$, so summing over $\mathcal{S}$ gives $\sum_{i \in \mathcal{S}} r_i = d \sum_{i \in \mathcal{S}} r_i + |\mathcal{S}| (1-d + d r_D) / |V|$, that is,

$$\sum_{i \in \mathcal{S}} r_i = \frac{|\mathcal{S}|}{|V|} \cdot \frac{1-d + d r_D}{1-d}. \quad (5.6)$$

The first factor is the farm's population share, and the second says how many times that share the farm collects. The second factor equals one only when $r_D = 0$, that is, on a graph without dangling nodes; otherwise the dangling rows pass part of the honest mass to the farm. Since the dangling nodes lie outside $\mathcal{S}$, $r_D \leq 1 - \sum_{i \in \mathcal{S}} r_i$, and substituting this into Equation (5.6) gives a bound that

holds on any graph:

$$\sum_{i \in \mathcal{S}} r_i \leq \frac{|\mathcal{S}|}{(1-d)|V| + d|\mathcal{S}|}. \quad (5.7)$$

For a farm that is small against $|V|$, the bound is close to $1/(1-d)$ times the population share. On the EndorseRank graph of the matched cohort, the pipeline's own PageRank code gives $r_D = 0.532$, 0.553 and 0.572 at $d = 0.75$, 0.85 and 0.95 , so a closed farm on this graph collects 2.6, 4.1 and 11.9 times its population share. A direct check with the same code agrees: a ring of 20 addresses, each approving the next, added to the graph collects 4.09 times its population share at $d = 0.85$.

Equation (5.7) limits the farm's total mass, not the score of one target, and the attacker can concentrate that mass. Take a star in which m farm addresses each approve the target t and t approves each of them back. Every farm address a has t as its only out-neighbor, so $P_{at} = 1$, and the row of t sums to one over the farm. Write $b = (1-d + dr_D)/|V|$ for the mass that each node receives from teleportation and the dangling rows. The balance equations $r_a = b + dr_t P_{ta}$ and $r_t = b + d \sum_a r_a$ then give

$$r_t = \frac{b(1+dm)}{1-d^2}.$$

At $d = 0.85$, $b = 9.62 \times 10^{-5}$, which is the common score of every wallet that receives no allowance. With $m = 10$ (eleven addresses and twenty approvals per window), the target reaches $r_t = 3.27 \times 10^{-3}$ on the graph with the star added, 7.7 times the highest EndorseRank among the cohort wallets (4.23×10^{-4}), most of which are owners. The score grows linearly in m : each further farm address adds $db/(1-d^2)$ to r_t and costs two approvals per window.

Honest in-edges are therefore not needed to rise above every wallet in the cohort, but bought ones add to the gain. When the farm has no out-edges to honest addresses, an edge (h, a) bought from an honest owner h adds, to first order,

$$\Delta \sum_{i \in \mathcal{S}} r_i \approx \frac{dr_h P_{ha}}{1-d}$$

to the farm's mass, where P_{ha} is the share of h 's row that the new edge takes. The gain depends

on this share, not on the amount approved. Because V counts every approval of a few base units or more about once, a fresh approval bought from h takes $\sigma(0)/(o(h) + \sigma(0))$ of h 's row, roughly one over the number of h 's approvals plus one, whatever amount it authorizes; for the median owner in the graph that share is 0.30. The edge an attacker would buy is therefore an approval from an owner with few other approvals, preferably one with a high r_h , and the approval can be for a token amount too small to matter. Such an approval exposes almost nothing, so the risk premium ρ_h in Equation (5.4) can be close to zero. In the raw-amount allowance layer of C-PR the opposite holds: there only an unlimited approval takes a large share of an owner's row, and it puts the owner's entire balance of the token at the attacker's disposal (Section 5.8.1). The attack's cost-effectiveness is

$$\eta = \frac{\Delta J(\mathcal{G})}{C}, \quad (5.8)$$

with $\Delta J(\mathcal{G})$ the gain in the objective of Equation (5.3) that the attack edges produce, held over the q windows that C in Equation (5.5) pays for.

Equations (5.6)–(5.8) show that EndorseRank, as defined here, is open to cheap inflation by closed farms. Under uniform teleportation and uniform redistribution of dangling mass, a closed cluster is not held to its population share. The control that does bound closed clusters changes the walk itself: if teleportation and dangling redistribution go only to a vetted seed set, as in TrustRank^[17], a cluster outside that set gains mass only through edges from honest owners, which the attacker has to buy at the price of Equation (5.4). AWP's restart rule gives the transfer side no protection either: a farm whose addresses send small amounts to one another raises their activeness and so draws restart mass to itself. The Sybil-adjusted stability proxies (Section 3.9.3) flag repeated transfers from a few addresses but not a farm of fresh addresses, so they cannot replace an adversarial assessment, which we leave to future work.

5.7 Regulatory and ethical considerations

On-chain reputation metrics raise questions of transparency as well as equity. Earlier legal work has shown that decentralized finance markets remain bound by disclosure and consumer protection rules even though settlement happens on-chain ^[61]. Packin and Lev-Aretz^[2] add a warning of their own: decentralized credit scoring may perpetuate black-box harms when models cannot be interpreted or contested, or when they are linked to identity databases without due process. Both EndorseRank and AWP are computed deterministically from on-chain events with open-source algorithms, so anyone can recalculate them using the open-source software (Appendix 8.1). That openness partly answers the black-box criticism. It leaves the distributional concern untouched, because addresses with little transaction activity will score lower whatever their off-chain creditworthiness.

Ethically, these scores should never be used as proxies for identity, as creditworthiness measures in the traditional sense, or as evidence that an address belongs to a unique individual ^[11]. They are rank-based indicators of an address's position in the defined graph topology. If a deployment grants financial services on the strength of score thresholds, it should clearly document the graph definitions behind the scores, the timeframe, and the limitations, above all the exposure to Sybil attacks. A regulatory framework centered on interpretability would prefer this kind of ranking to proprietary machine learning models like those Kotb reviews ^[14], as long as providers make no exaggerated claims about how well they can predict loan repayment rates ^[2].

Privacy on an open blockchain is limited by the ledger's very nature ^[5], and computing the reputation scores draws on nothing beyond public data. The main ethical question is therefore how the results are put to use. Further attempts to uncover the identity behind each address are discouraged, and the scores should not be treated as permanent, fingerprint-like identifiers. This study works with wallet addresses alone and makes no attempt to link them to real-world individuals (Section 3.16).

5.8 Threats to validity

The findings reported here should be read against the limitations below, which are grouped under construct, internal, external and adversarial validity.

5.8.1 Construct validity

- These ratings are not credit ratings: no default labels from uncollateralized lending enter the analysis.
- Same-window checks rest on local allowance and transfer statistics.
- Holdout construct. A future new approval is still an endorsement event and a future new sender a flow event: both lie outside the score window, yet neither is an independent credit or trading construct. Being counts of arrivals, they also favor any score that tracks the current count of counterparties (Section 5.1.4).
- Unit of analysis. EndorseRank scores the receiving side of an approval. The matched cohort consists mostly of owners, which EndorseRank leaves tied, so its same-window coefficients there rest on few wallets and on the split between wallets inside and outside the allowance graph, and its out-of-window evidence comes from spenders, most of them contracts (Section 4.1.1). Claims about the reputation of traders rest on the exploratory trader run, the registered window and the post hoc analysis of Section 5.1.6.
- Account type. Almost every trader that receives approvals is a contract account, so the comparison of Section 5.1.6 may separate kinds of account and not endorsed from unendorsed traders.
- Own-construct agreement is partly mechanical. EndorseRank smooths spender in-approve degree over the whole graph, and AWP does the same for the number of incoming transfers, so each score is expected by construction to agree with its own family. The within-score contrasts of Table 8.4 were computed for this reason, but for EndorseRank the two coefficients in each contrast have different ceilings (Section 4.3), so a null result there is not informative;

the contrasts also depend on how wallets outside the graph are scored (Table 4.5).

- Value weights. EndorseRank and AWP pass each amount through $V(z)$ with $b = 1$ on raw base units, which is within 10^{-4} of one for every amount of ten base units or more. The weights therefore record how many approvals or transfers an edge carries and how recent they are, and almost nothing about value: an approval of a few base units of an 18-decimal token counts like an unlimited one. A value of b set on price-normalized amounts, which would let size matter, was not tried. The raw-amount layers of C-PR sit at the other extreme. Unlimited approvals, which most owners in the allowance graph have granted, take almost all of an owner's row, and amounts in tokens with different decimals are compared unit for unit (Section 3.6.3). On the spring spender holdout the raw allowance walk ranks both labels better than EndorseRank (Table 4.10), so how amounts are weighted matters for this construct, and neither weighting is shown to be the right one. Such weighting choices propagate into $\mathbf{r}$ and into the rank correlations ^[23].

5.8.2 Internal validity

- Shared observation window. Every method and proxy is computed over 1 December 2025 to 31 May 2026. Market regimes peculiar to that window (volatility, incentive programs, protocol changes) may push alignment up or down. The robustness checks on damping and top-token subgraphs lower sensitivity to hyperparameters and token concentration, but none of them varies the calendar window.
- Cohort selection. The matched cohort is the 5,521 GMX V2 wallets on Arbitrum One with at least three closes in the window, a list fixed from GMX data before any extraction, and results may not carry over to wallets outside this trading population. Nor do the graphs see every cohort wallet (Section 4.1.1): kept as isolated nodes, the wallets outside a graph change EndorseRank's same-window coefficients a great deal and AWP's not at all (Table 4.5), and the sample-definition sensitivity shows how strongly the coefficients react when the evaluation population changes (Table 4.8).

- **Baseline fidelity.** AWP follows the published method in its edge weights and its restarts (Section 3.6.1). The paper gives no parameter values, so k , t_0 and b are this study’s choices (see value weights above). The paper scored Ethereum transactions; here AWP scores ERC-20 transfers on Arbitrum One. EndorseRank takes AWP’s weights unchanged and restarts uniformly, a choice made after both restart rules had been compared on these data (Table 4.9).
- **Order of analysis.** EndorseRank and AWP were scored after the same-window checks and the spring labels were known, and EndorseRank also after the June–August labels, so every comparison involving them is reported with intervals, and none was fixed in advance except the registered sensitivity analysis that sets C-PR against AWP. The comparison of C-PR with its transfer layer alone was chosen after the spring labels had been seen; it is treated as exploratory there and decided on the June–August window, registered on 1 October 2026 before its logs were extracted (Section 3.7.2). The trader liquidation label was registered for that window and for contrasts involving C-PR only; the comparison of the two edge types on the trader outcomes was fixed on 9 October 2026 after the registered labels were known, so it is a pre-specified post hoc analysis (Section 3.7.3).
- **Timing environment.** All runtimes are wall-clock times from one machine (Appendix 8.1). The standard deviations in Table 4.1 cover only repeated solves of one configuration, and in the damping sweep each graph–damping pair is timed once. Variation between machines is not measured, so the comparison is stated in ratios instead of seconds.

5.8.3 External validity

- **Single chain and window.** The evidence comes from Arbitrum One alone, over six months. Extending the findings to other rollups, Layer-1 chains or lending-centered venues would need cross-protocol validation, since DeFi venues differ in protocol design^[8] and in market structure^[1].
- **Pool scaling versus alignment scaling.** The expanded pool supports only a statement about runtime on subgraphs of the matched-cohort graph (Section 4.2.1). Same-window alignment

remains anchored to the matched cohort, and the holdout to spenders. The efficiency gains should not be taken to preserve the same τ profile at arbitrary population sizes.

- Ego-network depth. Each graph keeps only the edges incident to the seed wallets, so the allowance walk has depth one (Section 5.3.2), and the findings about propagation do not carry over to graphs with longer endorsement paths.

5.8.4 Adversarial validity

- Sybil and wash trading. Rankings may be manipulated through adversarial allowance or transfer patterns ^[11]; the Sybil-adjusted proxies only partly address this, and the cost model of Section 5.6 is analytical, not an empirical red-team evaluation.
- Strategic approval behavior. Once EndorseRank is live, rational actors may grant allowances strategically to farm rank. An observational τ estimated before deployment may overstate how well the ranking resists such behavior.

5.9 Limitations

On this sample, EndorseRank’s strengths are agreement with the allowance checks, a ranking distinct from AWP’s, a low cost owed to graph sparsity and, on spenders, a ranking of new approval pairs in both windows, a label on which the scores built on transfers alone stay low. Its weaknesses are a ranking that carries little transfer information, same-window coefficients shaped by its own ties, a holdout coefficient below that of the raw in-approve degree it smooths, and little to say about traders who receive no approvals.

The coupled operator was tried at three fixed values of λ , with uniform teleportation and a per-node convex mixture at the transition level, on two layers weighted by raw amounts. Layers weighted as in EndorseRank and AWP, node-dependent or learned weights, mixtures of the stationary vectors instead of the transition rows, and other Markov chains went untested, so both answers to RQ5 apply only to that family of methods on this dataset. An exploratory check run after every result

was known shows how much the registered verdict depends on the raw amounts: with binary weights on both layers, C-PR would have met all three registered conditions in June–August (new approval pairs $+0.196$, $[0.151, 0.243]$; new senders -0.006 , $[-0.017, 0.008]$, inside the margin; liquidation-free close rate $+0.023$, $[0.014, 0.033]$), while with EndorseRank’s and AWP’s bounded weights it would have failed the margin narrowly (new senders -0.010 , $[-0.022, 0.002]$). The script and output are in `manuscripts/B-ieee-access/supplement` of the thesis repository; the registered verdict stands as registered. The longer-running baseline runs, other PageRank variants and a graph built around lending activity, are archived (Appendix 8.4). Deployment remains bound by the considerations of Section 5.7.

EndorseRank draws on no off-chain evidence. Unlike attestation tooling such as Human Passport, which uses identity checks, biometrics and vouching to establish that an address belongs to a distinct person ^[64], it gives no guarantee that two highly ranked addresses are controlled by different actors, so it should be read as an ordering of an already admitted population and not as a Sybil defense. Filtering addresses through an attestation gate before computing EndorseRank is a plausible deployment, but the gate’s own weaknesses, such as credential farming ^[66], would carry over to it, and evaluating the combination needs labeled Sybil data that this study did not collect.

These limitations narrow what can be concluded. For this cohort and time frame, EndorseRank is a cheap, allowance-specific reputation score for spenders; it is not a credit scoring method, nor a Sybil-resistant identity mechanism, nor a substitute for protocol-level collateralization ^[9]. The allowance edge it rests on is worth adding to on-chain reputation, and the approval count is the cheapest way to use it. Chapter 6 restates what has been shown, what that knowledge is worth and what it would take to extend these bounds.

6 Conclusion and Future Work

Chapter 5 interpreted the findings and judged whether the study met its objectives. The tables of Chapter 4 form the record.

6.1 Answers to the research questions

The thesis proposed EndorseRank, PageRank with the edge weights of Adaptive Weighted PageRank (AWP) on a latest-allowance endorsement graph, and evaluated it beside AWP itself^[3], a coupled walk over both edge types and the raw degrees these walks smooth, on GMX V2 wallets and spenders on Arbitrum One from December 2025 to May 2026, with a registered replication on June to August 2026.

RQ1 asked whether EndorseRank and AWP rank the wallets of one sample differently. They do, and H1 is supported, largely by construction (Section 4.4). Much of the difference, and of the agreement, comes from EndorseRank’s tie blocks, so the claim that the two reflect different kinds of reputation rests mainly on the holdouts of RQ4.

RQ2 asked how closely each score agrees with allowance and transfer statistics in the same window. Each score follows the edges it walks on (Table 4.3), which supports H2 only narrowly, because the proxies come from the same events as the graphs; since EndorseRank’s coefficients are set by its ties, no conclusion is drawn from the difference between its allowance and transfer coefficients.

RQ3 asked what each score costs to compute. The cost follows the edge count, so EndorseRank, on the sparsest graph, is the cheapest score, and the coupled walks cost more than AWP (Table 4.1); H3 is supported, largely by construction. The pool stages could not test edge growth, because the second-tier extraction was not completed (Table 4.2).

RQ4 asked whether scores frozen at t_1 predict new approvals and new transfer senders on the

spenders at t_2 , and whether any PageRank predicts them better than the raw degree it smooths. The first part holds, and the edge type decides which label a score anticipates (Table 4.10). The second does not: the raw in-approve degree predicts new approvals, and the raw transfer in-degree new senders, better than every walk in both windows (Tables 4.11 and 4.16). For these labels, on this depth-one sample, propagation adds no predictive value over the local count (Section 5.3.2). RQ5 asked whether a coupled PageRank over both edge types improves on a walk over the transfers alone, and whether it ranks wallets better than either layer walked alone. Both answers are no, each under a rule fixed in advance. Under the rule of 14 September, C-PR fails both conditions of the primary part and the allowance condition of the secondary part (Section 4.3.1; Table 4.11). Its exploratory gain over the transfer walk on new approvals in the spring replicated in the registered June–August window, but a loss on new senders larger than the registered margin could not be ruled out, so the first answer is no; against AWP the loss on new senders was clear (Table 4.15). The seeded ablation matches the transfer walk: allowance edges contribute when the walk moves along them, and not as a seed.

In a pre-specified post hoc analysis that adds no research question (Section 5.1.6), the approval count at the freeze ranked traders above the transfer count in both windows on later liquidation avoidance, the outcome built from neither edge type that bears on counterparty risk ($\Delta\tau = +0.150$ in June–August; Section 4.6.3). This lead holds at count level only: the like-for-like PageRank pair is not resolved, account type is an unresolved confound because most approval-receiving traders are contract accounts, and the signal is lower on the profitable share of closes, so it ranks forced-liquidation avoidance, not trading success.

Claims C1–C6 (Section 4.7) state these findings formally.

6.2 Original contribution to knowledge

In the terms of the literature reviewed in Chapter 2, the five contributions of Section 1.9 establish the following.

Before this work, graph reputation on public ledgers ranked wallets by their transfers ^{[3],[4]}, by how risk spreads through transaction networks ^[12], or by explicit ratings and stakes ^{[25],[26]}, and the reputation tooling deployed in DeFi scored the holder of an address from credentials ^[64]. None of it used the ERC-20 allowance, the on-chain record of a deliberate delegation of spending power ^[18], as the edge of a reputation ranking; approvals had been used to cluster the addresses of one entity ^[40] and studied as a security risk ^[73]. None evaluated on-chain reputation scores on one cohort against the checks of each construct and against the raw degrees they smooth, with confidence intervals, difference tests and a temporal holdout. This thesis supplies both; PageRank and AWP's weighting are taken from the literature ^{[3],[15]}, EndorseRank changes only the edge, and the coupled operator changes the walk.

The new knowledge has four parts. What an on-chain reputation score measures depends on the edges it uses, and this can be verified on one cohort with quantified uncertainty. The allowance edge yields a score that is cheaper to update, differs from the transfer-based score, and ranks later endorsements of spenders in two windows, a label on which the scores built on transfers alone stay low. Post hoc and at count level, the approval count also ranked traders' later avoidance of forced liquidation, though not their trading success, above the transfer count; the PageRank pair is not resolved, and account type may explain the lead. Two further steps that the literature suggested did not improve results on these data: propagation predicts the future growth of the graph no better than the raw degrees it smooths, and a weighted combination of the two edge types beats neither its better single layer nor, under the registered rule, the transfer-only walk. Finally, the first of these nulls has an analytic explanation for EndorseRank: on the depth-one allowance graphs of this sample it is, up to a constant, a linear function of the owner-normalized in-strength, which divides each owner's endorsement over all the spenders it approves (Section 5.3.2); multi-hop propagation remains untested. The construct-validity framing ^{[22],[33]} that makes these claims testable is part of the contribution, as is the open pipeline whose published summaries let others check every table, since whether decentralized systems miscalculate credit scores is a matter of public interest ^[2].

The scope is deliberately narrow. The thesis does not show that on-chain reputation can replace

collateral ^{[1],[9]}, that the results apply beyond Arbitrum One and the window studied, or that any score measures creditworthiness, default risk or trading skill, and it does not claim that EndorseRank resists Sybils: under uniform teleportation a closed farm of attacker addresses can inflate it cheaply (Section 5.6).

6.3 Future work

Each suggestion below corresponds to a limitation identified in Chapter 5. Any new score or weighting it leads to should be fixed before it is scored, replicated on registered new data and checked out of window against degree baselines.

- A registered test of the liquidation-avoidance lead. Claim C6 is post hoc and should be tested on a later window whose logs have not yet been extracted, with the count pair (in-approve degree against transfer in-degree) and one PageRank pair fixed in advance: AWP against a walk over AWP’s transfers plus an allowance layer with AWP’s weights and restarts, so that only the edge type changes. Account type, from a code lookup of every trader, would enter as a baseline, and the cohort would be sized to detect a lead half as large as the post hoc one.
- Propagation on a deeper sample. On the one-hop ego networks studied, EndorseRank reduces to a reweighted approval count (Section 5.3.2); a sample extracted two or more hops out would test whether any propagated score adds out-of-window value over the local count, on arrival labels and on labels that are not counts of arrivals.
- Complete the second-tier extraction. The expanded address pool has so far only varied the number of nodes; with the logs of its supplemental addresses extracted, we could measure how running time grows with the number of edges and turn the bounded claim for RQ3 into a general scaling claim.
- Validate the results in other contexts. Running the same analysis on other rollups and on Layer-1 networks would show whether the construct checks and holdout results extend beyond Arbitrum One; because DeFi is heterogeneous ^{[1],[8]}, this is the main threat to external validity.

- Extend the time window and use a credit-linked holdout. A longer window would show whether the Kendall τ coefficients stay stable across market conditions, and a credit-linked holdout on the owners, feasible where lending protocols cover enough of the cohort, whether either graph carries credit-risk information.
- Value weights. With $b = 1$ on raw base units, AWP's value weight counts every approval or transfer of a few base units or more about once. A value of b set on price-normalized amounts would let size matter again, and the spring holdout suggests that it might: the allowance layer weighted by raw amounts ranked both spender labels better than EndorseRank.
- Adversarial evaluation. The Sybil cost model is analytical, backed by computations with the thesis's own PageRank code on the observed EndorseRank graph, and no attack was carried out on chain (Section 5.6). Restricting teleportation and dangling redistribution to a vetted seed set, as TrustRank does^[17], would remove the uniform inflow that closed farms exploit and should be evaluated under the same attacks. Empirical red-teaming with wash approvals, coordinated spender farms and strategic revocation would measure the realized cost-effectiveness of attacks under live fee markets and detection heuristics^[11], and would test whether the Sybil-adjusted proxies remain informative under active adversaries.
- Coupling beyond a fixed transition-level mixture. Only a fixed λ with uniform teleportation on raw-amount layers was tried. Node-dependent or learned weights, per-layer damping, mixtures of the two stationary vectors, and layers weighted as in EndorseRank and AWP remain untested; any such design should report construct-specific checks and keep the auditability that sets graph propagation apart from opaque learned scores^{[2],[14]}.
- Richer authorization semantics. Where ERC-20 approvals are sparse, adding protocol-specific delegation logs to the edge definition may improve coverage.
- Composition with attestation tooling. Filtering addresses through a Passport-style gate^[64], which answers whether an address belongs to a distinct person, before EndorseRank orders them by the authority others delegate to them is a plausible deployment; it should be evaluated

on labeled Sybil data, with the gate's own weaknesses ^{[65],[66]} in view.

- Deployment studies. Integrating EndorseRank with risk oracles and governance tooling would test whether its runtime profile can sustain production refresh cadences. Controlled experiments on fee tiers or monitoring cadence would let evidence replace the adoption hypotheses of Section 5.4.4.

6.4 Closing reflection

In adversarial, pseudonymous markets on-chain reputation does not remove the need for collateral ^[9], but it can make behavioral structure readable, and which structure it reads depends on the edges used. A transfer records economic flow and an allowance records delegated authority. PageRank will rank either graph, yet one ranking does not serve both constructs, and a walk over both graphs did no better on either than the right single graph. For a protocol or researcher who needs a fast, auditable signal of which spenders owners trust, the allowance edge is the one to use on data like those studied here. The approval count is its cheapest form, and EndorseRank is the propagated form for those who want the endorsers to count as well, with its exposure to cheap inflation by closed farms in mind (Section 5.6). For describing transfer hubs and anticipating their inflow, the transfer edge is the one to use. For screening counterparties against forced liquidation, the approval count ranked traders in these data and the transfer count, which ranked how much traders trade, did not; this post hoc, count-level result concerns liquidation avoidance, not trading success, is unresolved at the PageRank level, may reflect account type, and awaits a registered test.

7 References

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8 Appendix

8.1 Evaluation Pipeline and Reproducibility

This appendix documents the evaluation pipeline behind Chapter 4: its stages, the extraction design, the default parameters, the proxy formulas and a glossary.

8.1.1 Pipeline stages

The evaluation has four stages. The first three run on Arbitrum One over the observation period 2025-12-01 to 2026-05-31 (UTC, inclusive at both ends); the fourth adds the months from June to August 2026.

1. Reputation graphs and scores: Fix the matched cohort ($n = 5,521$) from GMX V2 data before any approval or transfer extraction. Fetch the cohort's Approval and Transfer logs in monthly chunks, with a dry run before each query and an abort limit of 150 GiB. Build and score the latest-allowance graph (EndorseRank) and the transfer graph (AWP), then the two raw-amount layers (C-PR at $\lambda \in \{0, 0.25, 0.5, 0.75, 1\}$ and S-PR), all with the solver settings of Section 3.1.
2. Checks, alignment and benchmarks: Compute the checks of Section 3.9 and report Spearman ρ and Kendall τ for every score with 95% paired-bootstrap intervals (400 resamples, seed 42), together with the contrasts of Section 3.10. Time one solver call per method under the protocol of Section 3.11, and run the damping, top-token and sample-definition sweeps.
3. Temporal holdout: On the spender cohort ($n = 1,335$), freeze the scores at 28 February 2026 and count the new approval pairs and new transfer senders that appear in March–May 2026. Report every score and both raw-degree baselines against both labels with 400 paired resamples, along with the contrasts fixed before the coupled run and those computed afterwards (s

`cripts/run_holdout_eval.py--cohortspenders`).

4. Registered replication: With `config/fresh_holdout_2026q3.yaml` and its plan, `docs/fresh_holdout_2026q3_plan.md`, committed, first check the evaluation code against the spring holdout (`scripts/run_fresh_holdout.py--spring-check`). Then extract the June–August logs with the same SQL and wallet filter (`scripts/extract_fresh_window.py--dry-run`, then `--extract`) and evaluate both cohorts on scores frozen at 31 May 2026 (`scripts/run_fresh_holdout.py`). The extraction script will not run before the registration is committed with the status set to registered, and both scripts record the SHA-256 hashes of the two registration files.

The coupling weights, holdout labels and the first rule for RQ5 were written to the configuration and to the plan of 14 September 2026, before any coupled score was computed. On 30 September 2026 two inverse-risk proxies were found to carry the wrong sign and were corrected (Appendix 8.1.5). The same day `scripts/classify_spenders.py` checked on BigQuery which spenders emitted logs or sent transactions in the scoring window, and `scripts/check_spender_code.py` read each spender’s code from an Arbitrum node (`data/2-processed-tables-and-evaluations/spring-holdout-2026-03-to-2026-05/spender_account_types.csv`). On 1 October 2026 the registration gained its two sensitivity analyses (Section 3.7.2); AWP’s value parameter is `reputation.awp_paper` in the configuration. Later that day the status was set to registered (commit 4085ab9 of `phd_works`), the June–August logs were extracted in nine monthly queries billed at about 1.04 TB, and `scripts/run_fresh_holdout.py` evaluated them; `data/3-published-results-for-thesis/registered-replication-2026-06-to-2026-08/` holds the extraction manifest and the summary behind Tables 4.14, 4.15 and 4.17.

The other tables of Chapter 4 come from runs made on 2 October 2026 in a cloud container with four cores of an Intel Xeon processor at 2.8 GHz and 15 GB of memory, under Python 3.11 with NumPy 2.4, SciPy 1.17 and pandas 3.0: `scripts/run_dissertation_eval.py--real--robustness--benchmark-repeats5`; `scripts/run_holdout_eval.py` with `--cohortspenders`

and with `--cohortmatched`; `scripts/run_fresh_posthoc.py`, which scores `EndorseRank` on the cohorts and labels of the registered replication (Table 4.16); and `scripts/run_supplementary_checks.py`, behind Tables 4.5 and 4.12. The same day `scripts/run_fresh_holdout.py --spring-check` was run again on the spring window and reproduced the values it had given on 1 October. The tables were written to `results/tables/` by `scripts/export_latex_results.py` and the figures by `scripts/generate_dissertation_figures.py`. The summaries (`eval_summary.json`, `supplementary_checks.json`, `eval_summary_spenders.json` and `eval_summary_matched_traders.json` in the `spring-holdout` folder, and `fresh_holdout_summary.json` and `posthoc_endorserank.json` in the `registered-replication` folder) are committed under `data/3-published-results-for-thesis/` along with a copy of the configuration in force, so any number in a table can be checked against its summary without re-running the pipeline. The public repository https://github.com/abitocodes/phd_works contains the evaluation source in `1-Dissertation-Works/ERC20-Allowance-PageRank-Wallet-Reputation/dissertation/wallet-reputation-experiments` (the folder was `A-Skill-Programs/margin_rank` until 1 October 2026 and has that name in commit 4085ab9), the versioned configuration (`config/margin_config.yaml`), the extracted data and the summaries, at revision 7129981. The file `data/README.md` in that folder describes every data file. The neutral-label analysis of Sections 3.7.3 and 4.6.3 (plan, script and results) and the check of `EndorseRank`'s closed form (Section 5.3.2) are in the folder `analysis/` of the thesis repository, <https://github.com/abitocodes/phd-dissertation-draft-en>, where the tag `manuscripts-2026-11` fixes the state reported here; the plan was committed in 20e250f, before the analysis ran.

The implementation is checked offline before any real extraction: synthetic tests run the same scoring and alignment logic on fixture graphs, with no cloud access. Real extraction needs Google Cloud credentials and a billing project, and its scans are confined to the public Arbitrum One log dataset.

8.1.2 BigQuery extraction design

Each ERC-20 Approval or Transfer event is identified by its standard topic-0 hash. Queries are restricted to logs whose owner/spender or from/to address is on the evaluation wallet list, which ties scan cost to the size of the cohort instead of the whole chain. Batching by month keeps every query under the operational byte cap. Appendix 8.2 shows representative SQL patterns.

For auditing, the study’s acquisition log retains the extraction records (bytes processed, row counts, timestamps). This research reports only the resulting cohort sizes and edge counts, not the operational logs.

A second-tier extraction had been planned for the 93,559 supplemental addresses of the expanded wallet pool ($N = 99,080$; Section 3.12.1). For the six monthly approval and transfer queries over those addresses, the dry-run estimate came to more than 1.3 TiB, and the download stage failed repeatedly on the storage-backed result path. The extraction was therefore abandoned within the study period. As a result, the supplemental addresses enter the expanded-pool experiments only as sampled addresses. They have no approval or transfer edges of their own, and the solver graph takes its nodes from the endpoints of the retained edges, so they add no nodes either; that is why the full-pool row of Table 4.2 reproduces the matched-cohort graph. Chapter 6 lists the completion of that extraction as future work.

8.1.3 Scaling tables

Table 8.1 reports runtime scaling within the cohort for $n \leq 5,521$, measured under the timing protocol of Section 3.11. Its last row is the same measurement as the one in Table 4.1. Every in-cohort stage is a subset of the wallets whose edges were extracted, so $|V|$ and $|E|$ both grow with n . In the expanded-pool stages they grow as well, but only up to the size of the matched-cohort graph.

Table 8.1: In-cohort runtime scaling (SHA256 seed benchmark-scale-v1; $n \leq 5,521$ matched GMX cohort). Here both $|V|$ and $|E|$ grow with n because every stage is a subset of wallets whose edges were extracted. Runtime = mean wall-clock seconds over 5 timed PageRank solves after one untimed warm-up; SD over the same 5 runs.

n	ER (s)	AWP (s)	Speedup	ER $ V $	ER $ E $	AWP $ V $	AWP $ E $
1,000	0.096	0.821	$8.6\times$	1,302	2,584	6,463	23,058
2,000	0.185	1.755	$9.5\times$	2,362	5,147	11,849	47,031
4,000	0.392	3.645	$9.3\times$	4,645	10,557	22,591	97,105
5,521	0.547	5.029	$9.2\times$	6,442	14,727	30,791	137,087

8.1.4 Credit-delegation extract (deferred)

Aave V3 records credit delegation in BorrowAllowanceDelegated events, a second on-chain form of delegated spending power. This study did not extract them, so the delegation-based row of the archived matrix (Appendix 8.4) has no edges; adding these events to the allowance graph is future work.

8.1.5 Supplementary alignment summary table

The pipeline computes six proxy families, and Table 8.2 lists the family-mean τ of every score on each. Among them are the liquidation, inverse-risk and GMX trading-success families, which were withdrawn from the primary analysis along with the GF-PR reference (Section 2.8). Until 30 September 2026 the pipeline negated two of the three inverse-risk proxies, the summed loss and the worst close, so both ranked the largest losses highest. They are now oriented like every other proxy, a value nearer zero meaning smaller losses, and the table shows the corrected values; the correction changes no other entry. Every score is negative on this family, which fits the summed loss and the worst close growing with how much a wallet trades (Section 4.3). The transfer, allowance and Sybil-stability columns repeat Table 4.3 without intervals.

Table 8.2: Same-window mean Kendall τ by proxy family for every score ($n = 5,521$). The first three columns repeat Table 4.3 without intervals; the liquidation, inverse-risk and GMX trading-success families are archived.

Method	Transfer	Allowance	Sybil	Liq.	Inv.risk	GMX
EndorseRank	0.330	0.375	0.231	0.093	-0.055	0.096
AWP	0.612	-0.024	0.278	0.141	-0.127	0.236
C-PR ($\lambda = 1$)	0.333	0.355	0.233	0.093	-0.056	0.096
C-PR ($\lambda = 0$)	0.534	-0.030	0.286	0.113	-0.094	0.179
C-PR ($\lambda = 0.25$)	0.531	-0.009	0.288	0.114	-0.093	0.176
C-PR ($\lambda = 0.5$)	0.527	-0.001	0.290	0.113	-0.091	0.171
C-PR ($\lambda = 0.75$)	0.519	0.006	0.292	0.110	-0.086	0.164
S-PR	0.494	0.116	0.220	0.119	-0.099	0.181

Table 8.3: Default parameters used in the primary evaluation.

Parameter	Value	Role
PageRank damping d	0.85	Default teleport mixture
Convergence tolerance ε	10^{-8}	Power-iteration stop
Max iterations $I_{\max}$	300	Power-iteration cap
Logistic k	0.01	Decay slope (AWP, EndorseRank, transfer layer)
Logistic t_0	180 days	Decay midpoint (AWP, EndorseRank, transfer layer)
Value transform b	1 (raw base units)	Value weight $V(z)$ of AWP and EndorseRank
Restart distribution	$\propto X_u$; uniform	AWP; EndorseRank, C-PR
Coupling weight λ	0.5 (primary); 0.25, 0.75	C-PR allowance-layer weight, fixed before the run
S-PR restart vector	allowance walk, normalized, floor 0	Endorsement-seeded teleportation
Holdout freeze / outcome window	2026-02-28 / 2026-03-01 – 2026-05-31	Spender cohort $n = 1,335$
Registered replication freeze / labels	2026-05-31 / 2026-06-01 – 2026-08-31	Spender and trader cohorts; margin 0.02
Observation window	2025-12-01 – 2026-05-31	UTC inclusive
Bootstrap resamples / seed	400 / 42	Paired wallet bootstrap, percentile 95% CI
Timed repeats / warm-up	5 / 1	Runtime protocol
In-cohort subsample seed	benchmark-scale-v1	Deterministic SHA256 seed
Expanded pool subsample seed	benchmark-tier2-v1	Deterministic SHA256 seed

8.1.6 Parameter summary

8.1.7 Proxy operational definitions (reference)

The formulas below are those of Chapter 3.

- Transfer: $\text{in_degree}(w) = |\{u : \exists u \rightarrow w\}|$; $\text{in_value}(w) = \sum_{e:u \rightarrow w} \text{value}(e)$.
- Allowance: $\text{in_approve_degree}(w) = |\{o : \exists o \rightarrow w\}|$; $\text{in_approve_value}(w) = \sum_{o, \text{token}} a^*(o, w, \text{token})$, where w is the spender.
- Sybil-adjusted stability: $\text{inbound_counterparty_ratio}(w) = |\{u : u \rightarrow w\}| / |T^{\text{in}}(w)|$ (self-transfers excluded); tenure in days; active months.

8.1.8 Glossary

Activeness AWP's restart weight of an address: over the addresses it has sent to, the sum of the time weights of its latest transfer to each.

Allowance An ERC-20 authorization that lets a spender move tokens on an owner's behalf.

Alignment Rank agreement between reputation scores and a validation proxy, measured by Spearman's ρ on mid-ranks and by Kendall's τ ; every reported τ is the tie-corrected τ_b .

Construct overlap A ranking method and a proxy sharing a data source, which inflates alignment.

C-PR end points C-PR at $\lambda = 1$ and at $\lambda = 0$, which walk the raw-amount allowance layer or transfer layer alone with uniform restarts. The registration documents call them EndorseRank and AWP.

Endorsement graph Directed graph of latest allowances, used as the input to EndorseRank and as the allowance layer of C-PR.

Matched cohort Primary sample of $n = 5,521$ GMX V2 wallets on Arbitrum One with at least three closes between 2025-12-01 and 2026-05-31, fixed from GMX data before any approval

or transfer extraction. Membership does not depend on graph connectivity: 966 of these wallets touch no allowance edge and 381 touch no transfer edge.

Expanded wallet pool A larger address set ($N = 99,080$) for testing runtime sensitivity to the number of sampled wallets; its supplemental addresses have no extracted edges, so they add no nodes to the solver graph.

Paired bootstrap Resampling of wallets with replacement through a single index matrix shared by every coefficient, so that differences between coefficients can be tested on the same resamples.

PageRank Spectral ranking of nodes by the stationary distribution of a damped random walk.

Proxy Observable wallet metric for checking rankings against domain intuition.

Spender cohort Holdout sample of $n = 1,335$ wallets: every spender that holds at least one positive latest allowance at t_1 (28 February 2026, 23:59:59 UTC) in the extracted approval data.

Teleport The jump that PageRank makes with probability $1 - d$, to a node drawn from the restart distribution: uniform for EndorseRank and C-PR, weighted by sending activity for AWP.

8.2 SQL Templates, Configuration, and Extended Notes

To aid reproduction, this appendix gathers representative SQL patterns, configuration keys, and extended methodological notes.

8.2.1 Approval log extraction pattern

An ERC-20 Approval event is identified by its topic 0, which is the keccak256 hash of Approval (address,address,uint256). For one month, a simplified query is:

```
SELECT
    block_timestamp,
    block_number,
```

```

transaction_hash,
log_index,
address AS token,
topics[SAFE_OFFSET(1)] AS owner_topic,
topics[SAFE_OFFSET(2)] AS spender_topic,
data AS value_data
FROM `bigquery-public-data.goog_blockchain_arbitrum_one_us.logs`
WHERE block_timestamp >= TIMESTAMP('2025-12-01')
      AND block_timestamp <  TIMESTAMP('2026-01-01')
      AND topics[SAFE_OFFSET(0)] =
      ↪ '0x8c5be1e5ebec7d5bd14f71427d1e84f3dd0314c0f7b2291e5b200ac8c7c3b925'
      AND (
        LOWER(CONCAT('0x', SUBSTR(topics[SAFE_OFFSET(1)], 27))) IN
        ↪ UNNEST(@wallet_list)
        OR LOWER(CONCAT('0x', SUBSTR(topics[SAFE_OFFSET(2)], 27))) IN
        ↪ UNNEST(@wallet_list)
      );

```

With rows filtered against @wallet_list, scan volume is proportional to the cohort instead of the full chain. Monthly batches keep every query below the per-query byte caps documented in the study's acquisition record.

8.2.2 Transfer log extraction pattern

Transfer events are selected by the topic 0 of Transfer(address,address,uint256):

```

SELECT
  block_timestamp,
  block_number,
  transaction_hash,

```

```

log_index,
address AS token,
topics[SAFE_OFFSET(1)] AS from_topic,
topics[SAFE_OFFSET(2)] AS to_topic,
data AS value_data
FROM `bigquery-public-data.goog_blockchain_arbitrum_one_us.logs`
WHERE block_timestamp >= TIMESTAMP('2025-12-01')
      AND block_timestamp <  TIMESTAMP('2026-01-01')
      AND topics[SAFE_OFFSET(0)] =
      ↪ '0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef'
      AND (
          LOWER(CONCAT('0x', SUBSTR(topics[SAFE_OFFSET(1)], 27))) IN
          ↪ UNNEST(@wallet_list)
          OR LOWER(CONCAT('0x', SUBSTR(topics[SAFE_OFFSET(2)], 27))) IN
          ↪ UNNEST(@wallet_list)
      );

```

Before PageRank, post-processing keeps, for each (token, owner, spender) triple, the value of the last Approval event in (block_number, log_index) order and drops zero (revoked) values. It then weights each latest approval and each transfer by its time weight and bounded value weight, builds the raw-amount layers of C-PR beside them, and keeps every edge with at least one endpoint in the matched cohort. Amounts stay in raw token base units, the uint256 value cast to a float, and no decimal scaling is applied before the value weight or, in the C-PR layers, before allowances or transfer values are summed across tokens.

8.2.3 Configuration keys

All of the pipeline's parameters are kept in a single versioned configuration file, whose principal keys include:

- start and end timestamps of the observation window;
- the PageRank damping factor, tolerance, and maximum number of iterations;
- logistic decay parameters k and t_0 and the value-weight parameter b ;
- subsample seeds and the number of benchmark repetitions;
- the locations to which intermediate artifacts and table fragments are written (results/tables/).

Changing the damping factor or the decay parameter requires a rerun of the entire evaluation, including the robustness sweeps and the table export. All figures must then be regenerated from the same artifacts so that the reported numbers match the pipeline’s machine-readable summary.

8.2.4 Complexity notes

Write $m = |E|$. Building the latest-allowance edges sorts the R_a approval rows in (block_number, log_index) order, in $O(R_a \log R_a)$ time, and the transfer edges take $O(R_t)$ over the transfer rows. PageRank over I iterations costs $O(m + mI)$, the first term for building the sparse operator, which dominates the measured calls (Appendix 8.3.3). The edge lists and score vectors account for most of the memory, so the peak-memory ratio in Table 4.1 is of the order of the edge ratio.

8.2.5 Conventions for the result tables

The tables of Chapter 4 follow these conventions:

1. Every τ is Kendall’s τ_b on raw scores and proxies, reported with its 95% paired-bootstrap interval. Differences between coefficients come from the paired contrast tables, not from subtracting point estimates.
2. On the matched cohort, EndorseRank’s coefficients are capped by its ties (Section 4.4), so they are compared with other coefficients only through the contrasts and with that cap in mind.
3. A runtime is the time of one PageRank call on edge lists built in advance, covering matrix

assembly and power iteration but not extraction or preprocessing. The same graph under the same settings shows the same numbers in every table (Section 3.11).

4. The sample-definition sweep (Table 4.8) is a sensitivity analysis, and its absolute values are not compared with the matched-cohort tables.

8.2.6 Archived extended baselines

The archived matrices, and the reasons they stay out of the main comparison, are described in Appendix 8.4.

8.2.7 Limitations

The SQL above was run only on Arbitrum One. All reported numbers come from the pipeline’s machine-readable summaries for the matched cohort ($n = 5,521$), the pool subsamples, the spender cohort ($n = 1,335$) and the exploratory trader run; the registered June–August results come from the summary of that run (Section 4.6.2).

8.3 Elementary Properties of PageRank and Rank Correlation

Chapters 2–4 rely on the elementary properties summarized here. The material is standard: the PageRank properties follow Page et al.^[15] and Langville and Meyer^[23], and the rank correlations Spearman^[20] and Kendall^[21].

8.3.1 Existence and uniqueness of PageRank

Take P to be a row-stochastic matrix on $|V|$ nodes, possibly after the dangling-node repair of Equation (2.1). For $d \in (0, 1)$ the Google matrix of Equation (2.2),

$$G_{\text{PR}} = dP^\top + \frac{1-d}{|V|} \mathbf{1}\mathbf{1}^\top,$$

is positive and column-stochastic. By the Perron–Frobenius theorem for positive matrices, G_{PR} has a single eigenvalue of modulus one, and the corresponding eigenvector $\mathbf{r}$ is strictly positive; we can scale it so that $\sum_i r_i = 1$. It follows that, once the damping and the dangling-node rules are fixed, every directed graph has unique PageRank scores^[23].

Two restart vectors in this thesis are not uniform. AWP’s, proportional to activeness, is zero at every address that has sent nothing, and S-PR’s, the normalized score of the allowance walk, is zero at every transfer-layer wallet without an allowance score, so neither Google matrix is positive. The fixed point is still unique: for two probability vectors the restart term cancels in their difference, and the operator multiplies their $\|\cdot\|_1$ distance by at most d . In general it is not strictly positive, however, because a wallet with no restart mass that no path from a restart wallet reaches keeps a score of zero.

Uniqueness guarantees a well-defined comparison between EndorseRank and AWP. Once its parameters are fixed, each method assigns exactly one score vector to a given edge set, so any difference between $\mathbf{s}_{\text{ER}}$ and $\mathbf{s}_{\text{AWP}}$ must arise from their edges and restart distributions and cannot be traced to multiple solvers.

8.3.2 Continuity in edge weights

For fixed $d \in (0, 1)$, PageRank depends continuously on the entries of P . If $\tilde{P}$ is a second row-stochastic matrix on the same nodes, with PageRank vector $\tilde{\mathbf{r}}$, then

$$\|\mathbf{r} - \tilde{\mathbf{r}}\|_1 \leq \frac{d}{1-d} \max_i \|P_{i\cdot} - \tilde{P}_{i\cdot}\|_1,$$

a bound that does not depend on $|V|$ ^[23].

Continuity in P is not the same as continuity in the edge weights. A row $P_{i\cdot} = w_{i\cdot}/o_i$ moves continuously with the weights only while the out-strength o_i stays positive. When a node gains its first out-edge or loses its last one, its row jumps between the uniform dangling row and a row concentrated on its out-neighbors, however small the weight involved. In the worked example of

Section 2.13, adding an edge $D \rightarrow A$ of weight 10^{-12} moves $\mathbf{r}$ by 0.37 in the $\|\cdot\|_1$ norm and raises r_A from 0.1375 to 0.3202.

The damping sweep is covered by continuity: it keeps P fixed and changes d smoothly, and for $d < 1$ the PageRank vector varies continuously with d . The top-token filter is not covered. It removes edges, which can leave a wallet without out-edges, so the stability of that check rests on the data alone (Table 4.7). Empirically, EndorseRank’s allowance family mean is unchanged in the first three decimal places across $d \in \{0.75, 0.85, 0.95\}$ (Table 4.6); with most of the cohort tied, the ranking could hardly have moved.

8.3.3 Sparsity and per-iteration cost

With $m = |E|$ nonzero edges, the sparse product $P^\top \mathbf{x}$ costs $O(m)$ arithmetic operations, and building the sparse operator from the edge list is also $O(m)$, so power iteration over I iterations is $O(m + mI)$. On the matched cohort the transfer graph has about 9.3 times as many edges as the endorsement graph, so even at identical I the transfer method would run roughly an order of magnitude longer. In the measured calls the assembly of the operator takes most of the time and the iterations little (Section 4.2), so the measured time ratio of 9.2 tracks the edge ratio.

8.3.4 Properties of Spearman’s ρ

Spearman’s ρ is Pearson’s correlation computed on ranks. It lies in $[-1, 1]$ and does not change under strictly increasing transformations of either variable. A value of $\rho = 1$ means perfect monotone agreement, and $\rho = -1$ perfect reverse agreement. Tied values receive mid-ranks, and the evaluation pipeline relies on standard tie-aware implementations. Because reputation scores and proxies are continuous and heavy-tailed, rank-based association suits them better than Pearson correlation on the raw values. The AWP baseline paper likewise evaluates its ranking with rank statistics ^[3].

8.3.5 Properties of Kendall's τ

Kendall's τ compares the numbers of concordant and discordant pairs, n_c and n_d . Without tie correction, $\tau_a = (n_c - n_d)/n_0$ with $n_0 = n(n-1)/2$ is the probability that two randomly drawn wallets are ordered the same way by both rankings minus the probability that they are ordered oppositely ^[21]. Every reported τ is the tie-corrected τ_b of Equation (2.6). The two coincide without ties, and here ties are heavy. EndorseRank takes 99 distinct values on the matched cohort, so 67.3% of all wallet pairs are tied on it, against 0.5% on AWP, and for EndorseRank against AWP the reported $\tau_b = 0.360$ corresponds to $\tau_a = 0.206$. Ties also cap τ_b . EndorseRank's largest attainable value is about 0.38 against the allowance proxies, which are positive for exactly the 131 wallets it separates, and about 0.57 against a proxy without ties, so its 0.375 on the allowance family (Table 4.3) is nearly complete agreement on the pairs the rankings separate and cannot be compared directly with a coefficient that has a different ceiling.

When n is large, τ and ρ usually agree in sign, but they differ in magnitude. Across the pairs of EndorseRank or AWP with a proxy where $|\tau| > 0.02$, the ratio ρ/τ lies between 1.01 and 1.49, and ρ is always the larger of the two. We report both, as the AWP baseline does ^[3]. Our main summary statistic is the family-mean τ with a paired-bootstrap interval ^[34].

8.3.6 Construct validity note

Cronbach and Meehl ^[22] distinguish criterion-related validity from construct validity. Since no definitive gold standard default label exists, this research takes a construct-oriented approach. Each score is tested against validation checks that embody its conceptual definition (e.g. receiving allowances for EndorseRank; receiving inbound transfers for AWP), and also against a holdout period with two labels, new approval pairs and new transfer senders. High within-construct τ supports the interpretation. A low cross-construct τ , by contrast, does not automatically signal failure; it may indicate discriminant validity.

8.3.7 Where each primary quantity is reported

For each primary quantity, the index below names the table where it is established.

Quantity	Reported in
Matched-cohort size, $ V $, $ E $, runtime, SD, memory, iterations (four methods)	Table 4.1
In-cohort scaling stages	Table 8.1
Pool subsample stages ($N = 99,080$)	Table 4.2
Family-mean τ with 95% CI, every score	Table 4.3
Paired $\Delta\tau$ contrasts of C-PR and S-PR	Table 4.4
Inter-method τ ; wallets outside a graph as isolated nodes	Table 4.5
Damping, top-token, sample-definition sweeps	Tables 4.6, 4.7, 4.8
Spender holdout ($n = 1,335$)	Tables 4.10, 4.11 and 4.12
Exploratory trader run	Table 4.13
Registered June–August replication	Section 4.6.2
Comparisons between EndorseRank and AWP, per-proxy results	Appendix 8.5
Observation window	2025-12-01 – 2026-05-31 (Table 8.3)

Each quantity listed appears both in the evaluation summary and in a table of Chapter 4 or of the appendix.

8.4 Archived Extended Baselines

Further PageRank variants were scored on the same matched cohort and proxy families. They are not among the scores evaluated in Chapter 4.

8.4.1 Seven-method alignment matrix

The evaluation archive keeps the seven-method matrices (LP-PR, CW-AWP, LF-PR, and RiskProp-style variants), which are not drawn here. Their rows were computed with the two raw-amount layers of C-PR walked alone as the allowance and transfer references. Among these methods, the

allowance walk leads on the allowance family, and the transfer walk, together with its closely related variants, leads on transfer.

The archive matrices show the following patterns:

- Methods that use transfer-like edges cluster around the transfer walk on the transfer and Sybil families.
- When a method has very few edges, its rankings can be unstable or degenerate, and a low runtime may simply reflect a tiny m .
- RiskProp-style rows that match the transfer walk numerically indicate shared transfer structure; they do not constitute an independent discovery.

8.4.2 Six-method Aave-oriented matrix

An Aave-oriented preset is archived as well. Within the study window, several of the lending-event graphs on Arbitrum are sparse, and the delegation-based rows may be degenerate when BorrowAllowanceDelegated events were not extracted (Appendix 8.1).

That preset remains in the archive and is not drawn here.

Although these matrices motivate future work on credit-delegation edges and cross-protocol lending labels, they leave Claims C1–C6 unchanged.

8.4.3 Why these variants are archived

Three reasons explain why the scores evaluated are EndorseRank and AWP, with C-PR and S-PR built from the same two layers:

1. Allowance-based endorsement is the new construct, and evaluating it beside the established transfer method^[3] isolates the research question.
2. Variants that share transfer edges with AWP contribute extra rows but no independent construct.
3. The outcome-native GF-PR reference from the research proposal is derived from the same

GMX events as the trading-success proxies, so any alignment between the two counts as construct overlap and not as validation (Section 2.8). The GF-PR three-method matrix is archived alongside the seven-method and six-Aave presets.

8.4.4 Reproducing archive tables

Running the evaluation pipeline’s table export with the seven-method and six-Aave presets regenerates the archive tables. CSV copies are written out together with the table fragments.

8.4.5 Notes on archived method identifiers

The archive tables and CSV exports use the identifiers below.

LP-PR The raw-amount transfer layer, with AWP’s time decay, with each edge weight multiplied by one minus the GMX liquidation rate of its sender and one minus that of its receiver. It tracks the transfer walk closely on the transfer families. Its weights use the same liquidation outcomes as the liquidation proxies, so it shares the circularity of GF-PR.

CW-AWP The raw-amount transfer layer restricted to transfers of a fixed list of collateral tokens. Its transfer τ is slightly lower than that of the full layer.

LF-PR A lending-flow graph from Aave events, with borrows as edges from the pool to the user and repayments and liquidations as edges from the user to the pool, all under AWP’s time decay. Few cohort wallets used Aave on Arbitrum in the window, so the graph has very few edges and near-zero runtimes, and its rankings cannot be compared with the transfer walks.

RiskProp The raw-amount transfer layer with each edge weight multiplied by one minus the GMX liquidation rate of its sender. It borrows the idea of propagating risk from Lin et al. ^[12] but is not their method. On this cohort it mirrors the transfer walk numerically on several families, so the liquidation weighting changed little.

LiqCall-PR / Borrow-PR / Repay-PR Aave-oriented wallet-to-wallet graphs derived from liquidation calls, borrows, and repays, where applicable. On Arbitrum in the study window they have only tens to hundreds of edges, which leaves alignment weak and runtimes negligible.

Delegation-PR Intended to use credit-delegation allowances. With zero delegation edges extracted, its scores are degenerate and its τ cannot be interpreted.

No archive row is a peer-reviewed baseline.

8.4.6 Criteria for the scores evaluated

Chapters 4–5 concentrate on EndorseRank, AWP and the walks that couple their two layers; the other variants fall short of the following criteria:

1. Construct independence: a baseline should implement different edge semantics instead of merely reweighting transfers.
2. Data adequacy: there should be enough edges for PageRank to be stable on the matched cohort.
3. Scope: the contribution is allowance-based endorsement; it is not meant to survey every possible PageRank variant.

By these criteria, EndorseRank (allowance) and AWP (transfer), with the walks that couple their layers, make up a minimal sufficient set. The seven-method and six-Aave matrices remain archived and do not enter into Claims C1–C6.

8.5 EndorseRank and AWP Side by Side

Chapter 4 reports every score of the thesis together and reads each PageRank against the raw degree it smooths. This appendix collects the direct comparisons between the two single-layer scores, EndorseRank and AWP: their family means and paired differences in the same window, their per-proxy coefficients, and their paired differences out of window. Apart from the trader grid at its

end, none of these comparisons was fixed in advance, and none carries a claim of the thesis. The two scores walk on different edges and are built for different constructs, so on each check the score ahead is, almost every time, the one whose edges the check counts. Outcomes built from neither edge type are the exception, since neither score was built for them, and are reported in Section 4.6.3. Their coverage differs as well. No allowance edge touches 966 of the matched wallets and no transfer edge touches 381, and 318 of the 1,335 holdout spenders have no transfer edge before the freeze. EndorseRank’s ties decide most of its same-window ordering (Section 4.4). In both holdout windows each of the two falls below the raw degree it smooths (Section 4.6), so a difference between them does not show that either adds anything to its degree.

8.5.1 Same window

Table 4.3 gives the family means of the two scores with their intervals, and the last row of Table 4.5 their agreement. The two patterns cross. AWP tracks the transfer family most closely (0.612) and is slightly negative on the allowance family, while EndorseRank tracks the allowance family most closely (0.375) and the transfer family moderately (0.330). Table 8.4 tests five paired differences on the same wallet resamples (Section 3.10), numbered i to v in the order of its rows.

Table 8.4: Paired bootstrap tests of differences between Kendall τ coefficients (matched cohort, $n = 5,521$). Family entries use the family-mean τ ; the same wallet resamples are used for a and b , so the interval is on the paired difference. An interval excluding 0 indicates a difference not attributable to sampling variation at the 5% level. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Contrast (a minus b)	τ_a	τ_b	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
EndorseRank allowance minus EndorseRank transfer	0.375	0.330	+0.045	[-0.001, 0.084]	0.970
AWP transfer minus EndorseRank transfer	0.612	0.330	+0.282	[0.263, 0.299]	1.000
EndorseRank allowance minus AWP allowance	0.375	-0.024	+0.398	[0.356, 0.432]	1.000
AWP sybil-stability minus EndorseRank sybil-stability	0.278	0.231	+0.047	[0.031, 0.062]	1.000
EndorseRank in-approve degree minus EndorseRank in-degree	0.376	0.341	+0.035	[-0.015, 0.079]	0.925

In three of the five contrasts the interval excludes zero. AWP’s transfer alignment is higher than EndorseRank’s (+0.282), EndorseRank’s allowance alignment is higher than AWP’s (+0.398), and AWP leads EndorseRank on the Sybil-stability family (+0.047). The remaining two compare En-

dorseRank with itself, and both are null: its allowance alignment is not shown to be higher than its transfer alignment (contrast i, $+0.045$, interval $[-0.001, 0.084]$), and its in-approve-degree alignment is not shown to be higher than its in-degree alignment (contrast v, $+0.035$, $[-0.015, 0.079]$). These two nulls cannot be read as evidence either way. EndorseRank's ties cap τ_b at about 0.38 for the allowance proxies and at about 0.57 for a proxy without ties (Section 4.3), so the coefficients are not on one scale. With the wallets outside the allowance graph kept as isolated nodes instead of being scored zero, contrasts i and v become $+1.061$ and $+1.090$, with intervals far above zero. The opposite signs of contrasts ii and iii follow from the constructs. Each score is ahead on the family built from its own edges, and the main proxies of each family include the in-degree that the corresponding PageRank smooths. EndorseRank's lead on the allowance family rests on the 131 wallets that receive approvals, all of which it lifts above its common value, while AWP spreads those 131 over most of its ranking. AWP's lead on the transfer family is measured against a transfer coefficient of EndorseRank's that turns negative once the missing wallets are kept as isolated nodes. Neither difference says which score is the better measure of reputation.

On the staged pool subsamples of Table 4.8, whose values are not comparable with those of the matched cohort, the order of the two scores within each family is the same at every stage: AWP is ahead on the transfer family and EndorseRank on the allowance family.

8.5.2 Proxy detail

Each family mean condenses several per-proxy coefficients. Tables 8.5 to 8.7 report them one proxy at a time for EndorseRank and AWP, as Spearman ρ , Kendall τ and the 95% interval of τ . Before correlation, every proxy is oriented so that a higher value means higher expected reputation, and the transfer proxies are the ones AWP's source paper used to validate it^[3].

8.5.2.1 Transfer family

In Table 8.5, AWP's agreement with in-degree is the largest per-proxy Kendall τ on the matched cohort, and its agreement with in-value is much lower, for the reason given in Section 4.3. En-

Table 8.5: Domain alignment with transfer proxies (inbound in-degree and in-value, the validation proxies used for the AWP baseline). Kendall τ and Spearman ρ between reputation scores and proxy rankings. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Proxy	Spearman ρ	Kendall τ	95% CI of τ
<i>EndorseRank</i>			
In-degree	0.418	0.341	[0.319, 0.363]
In-value (sum inbound)	0.396	0.319	[0.300, 0.340]
<i>AWP</i>			
In-degree	0.884	0.728	[0.718, 0.737]
In-value (sum inbound)	0.669	0.495	[0.481, 0.510]

dorseRank’s transfer coefficients are moderate (0.341 and 0.319), and their intervals overlap neither zero nor AWP’s intervals. They mainly reflect that wallets in the allowance graph also receive more transfers than wallets outside it (Section 4.4). A protocol that equates reputation with inbound flow can read that flow from the in-degree itself, and out of window AWP ranks new senders below the in-degree (Section 4.6).

8.5.2.2 Allowance family

Table 8.6: Domain alignment with allowance proxies (spender-side in-approve degree and value). CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Proxy	Spearman ρ	Kendall τ	95% CI of τ
<i>EndorseRank</i>			
In-approve degree	0.380	0.376	[0.344, 0.403]
In-approve value (sum)	0.380	0.373	[0.341, 0.398]
<i>AWP</i>			
In-approve degree	-0.029	-0.024	[-0.043, -0.002]
In-approve value (sum)	-0.029	-0.023	[-0.043, -0.002]

In Table 8.6, EndorseRank’s two coefficients are almost equal and sit at the ceiling that its ties allow (Section 4.3). AWP’s are slightly negative (-0.024 and -0.023), with intervals that exclude zero: among the cohort’s wallets, transfer centrality does not pick out those that are approved as spenders. The comparison rests on the 131 wallets that receive approvals and is partly built in, since

EndorseRank is computed from the same approvals.

8.5.2.3 Sybil-adjusted stability family

Table 8.7: Domain alignment with Sybil-adjusted stability proxies from transfer activity. CI = 95% percentile bootstrap over wallets (400 paired resamples, seed 42).

Proxy	Spearman ρ	Kendall τ	95% CI of τ
<i>EndorseRank</i>			
Inbound counterparty ratio	0.091	0.072	[0.045, 0.095]
Transfer tenure (days)	0.363	0.294	[0.274, 0.316]
Active months	0.377	0.327	[0.307, 0.348]
<i>AWP</i>			
Inbound counterparty ratio	-0.269	-0.228	[-0.250, -0.207]
Transfer tenure (days)	0.685	0.507	[0.493, 0.521]
Active months	0.704	0.555	[0.540, 0.569]

Table 8.7 shows AWP ahead of EndorseRank on tenure (0.507 against 0.294) and active months (0.555 against 0.327), as one would expect from a transfer score that rewards activity spread over the window; the paired difference on the family mean excludes zero (Table 8.4, contrast iv). On the inbound-counterparty ratio AWP is clearly negative (-0.228) and EndorseRank weakly positive (0.072). The family describes how much longitudinal transfer information each score carries.

On the archived trading families of Table 8.2, AWP is ahead of EndorseRank on liquidation (0.141 against 0.093) and on trading success (0.236 against 0.096). On the inverse-risk family both are negative, -0.055 for EndorseRank and -0.127 for AWP, and on the non-loss close rate, a share that does not grow with trading volume, the two coefficients are -0.031 and -0.006 .

8.5.3 Out of window

Table 8.8 collects the paired differences between the two scores in both holdout windows. On the full spender cohorts, a is the score built from the edge type that the label counts; on the subset of the second block both rows take EndorseRank as a .

Table 8.8: Paired bootstrap differences between EndorseRank and AWP out of window. None was fixed in advance: the spring rows were computed after the spring labels were known, and EndorseRank was scored on the June–August cohorts after the registered evaluation (Table 4.16). The second block is the subset of Table 4.12. Within each block a and b share the wallet resamples (400, seed 42).

Contrast (a minus b)	Label	τ_a	τ_b	$\Delta\tau$	95% CI of $\Delta\tau$	$P(\Delta\tau > 0)$
<i>Spring holdout, spenders ($n = 1,335$)</i>						
EndorseRank minus AWP	new approval pairs	0.394	0.123	+0.271	[0.207, 0.332]	1.000
AWP minus EndorseRank	new transfer senders	0.344	0.283	+0.061	[0.026, 0.098]	1.000
<i>Spring holdout, spenders that had received a transfer before the freeze ($n = 977$)</i>						
EndorseRank minus AWP	new approval pairs	0.340	0.298	+0.042	[0.003, 0.085]	0.978
EndorseRank minus AWP	new transfer senders	0.358	0.388	-0.029	[-0.070, 0.011]	0.085
<i>June–August, spenders ($n = 1,964$)</i>						
EndorseRank minus AWP	new approval pairs	0.333	0.118	+0.215	[0.171, 0.270]	1.000
AWP minus EndorseRank	new transfer senders	0.319	0.255	+0.064	[0.036, 0.088]	1.000
<i>June–August, traders with at least three closes ($n = 1,402$)</i>						
EndorseRank minus AWP	liquidation-free close rate	0.081	0.045	+0.035	[-0.017, 0.087]	0.925

On the spring spenders EndorseRank ranks new approval pairs at $\tau = 0.394$ (interval [0.360, 0.431]) against 0.123 for AWP, a paired difference of +0.271 ([0.207, 0.332]). On new transfer senders the order reverses: AWP reaches 0.344 against 0.283 for EndorseRank, a difference of +0.061 ([0.026, 0.098]). Every label falls after all scored events, so on this cohort each edge type carries information about its own construct that the other lacks. Coverage explains much of the first gap. AWP scores zero for the 318 spenders without a transfer edge before the freeze, and these gained new approval pairs more often than the rest. On the 977 spenders that had received a transfer, EndorseRank’s lead on new approval pairs shrinks to +0.042 ([0.003, 0.085]), and the two scores are level on new senders (−0.029, [−0.070, 0.011]). The in-approve degree, which uses no transfer information, is also ahead of AWP on new senders (0.404 against 0.344, without a paired test). In June–August, with EndorseRank scored after the registered evaluation, the pattern repeats. EndorseRank ranks new approval pairs at 0.333 against 0.118 for AWP (+0.215, [0.171, 0.270]), and AWP ranks new senders at 0.319 against 0.255 (+0.064, [0.036, 0.088]). On the liquidation-free close rate of the traders EndorseRank reaches 0.081 and AWP 0.045, a difference that cannot be

told from zero (+0.035, $[-0.017, 0.087]$).

On the matched traders of the exploratory spring run (Table 4.13), AWP ranks the number of profitable closes at 0.210 (interval $[0.181, 0.238]$) against 0.072 ($[0.041, 0.111]$) for EndorseRank, and the realized gain at 0.191 ($[0.159, 0.219]$) against 0.048 ($[0.017, 0.082]$). On the share of closes that were profitable neither score shows anything: 0.016 ($[-0.015, 0.044]$) for AWP and -0.016 ($[-0.053, 0.024]$) for EndorseRank. AWP's lead is on counts and sums, which grow with how much a wallet trades, and the transfer in-degree does about as well on the same two labels.

8.5.4 All trader labels in both windows

Table 8.9 sets the allowance side against the transfer side on all six trader labels in both windows; its contrasts were fixed before they were computed but after the labels and the single-score coefficients of the registered window were known (Section 3.7.3), and only P on the liquidation-free close rate enters a claim (C6, Section 4.6.3). The counts (P) favor the allowance side on both liquidation labels in both windows; Q, the PageRank pair with the same weights and restarts, points the same way, with intervals above zero on the no-liquidation flag but not, at 97.5%, on the liquidation-free close rate. Every pair favors the transfer side, with intervals below zero, on profitable closes and realized gain, which grow with activity, and the counts favor it on the two shares as well.

8.6 Claim-to-Evidence Map

This appendix maps each hypothesis, research question, and empirical claim to the relevant evidence in Chapter 4.

8.6.1 H1 / RQ1: Rank divergence

- Statement: on the same cohort, EndorseRank and AWP rank wallets differently.
- Evidence: Section 4.4; Table 4.5, whose last row gives the inter-method τ_b with its interval, also with the wallets outside each graph kept as isolated nodes; Figures 4.3 and 4.2.

Table 8.9: Allowance-side minus transfer-side scores, $\Delta\tau$ with 95% paired bootstrap interval, on all six trader labels of the registered window and of the spring window (pre-specified post hoc analysis; Section 4.6.3). P: in-approve degree minus transfer in-degree. Q: EndorseRank with activity restarts minus AWP (same edge weights and restart rule). C-PR: $\lambda = 1$ minus $\lambda = 0$ (same raw-amount build). ER: EndorseRank minus AWP. Profitable closes and realized gain grow with the number of closes; the other four labels do not.

Label	P (counts)	Q (PageRanks)	C-PR layers	ER minus AWP
<i>Registered window, June–August 2026</i>				
Liquidation-free close rate	0.150 [0.100, 0.201]	0.032 [-0.003, 0.065]	0.021 [-0.035, 0.080]	0.035 [-0.018, 0.088]
No liquidation	0.259 [0.199, 0.320]	0.075 [0.035, 0.116]	0.055 [-0.012, 0.125]	0.094 [0.034, 0.155]
Profitable closes	-0.194 [-0.247, -0.139]	-0.136 [-0.168, -0.101]	-0.066 [-0.114, -0.016]	-0.157 [-0.201, -0.112]
Realized gain	-0.107 [-0.160, -0.052]	-0.124 [-0.155, -0.092]	-0.128 [-0.182, -0.075]	-0.165 [-0.210, -0.119]
Profitable share	-0.101 [-0.149, -0.047]	-0.020 [-0.053, 0.012]	-0.075 [-0.130, -0.019]	-0.074 [-0.123, -0.028]
Non-loss share	-0.138 [-0.191, -0.083]	-0.007 [-0.040, 0.026]	-0.126 [-0.181, -0.072]	-0.093 [-0.140, -0.045]
<i>Spring window, March–May 2026</i>				
Liquidation-free close rate	0.153 [0.115, 0.193]	0.034 [0.004, 0.064]	-0.024 [-0.071, 0.022]	0.012 [-0.027, 0.052]
No liquidation	0.237 [0.192, 0.286]	0.060 [0.026, 0.095]	-0.007 [-0.060, 0.045]	0.051 [0.006, 0.095]
Profitable closes	-0.206 [-0.252, -0.158]	-0.104 [-0.133, -0.077]	-0.058 [-0.098, -0.017]	-0.137 [-0.171, -0.102]
Realized gain	-0.156 [-0.203, -0.109]	-0.112 [-0.139, -0.086]	-0.105 [-0.144, -0.062]	-0.143 [-0.177, -0.109]
Profitable share	-0.074 [-0.126, -0.016]	-0.010 [-0.038, 0.017]	-0.044 [-0.088, 0.001]	-0.032 [-0.070, 0.007]
Non-loss share	-0.113 [-0.164, -0.060]	0.001 [-0.026, 0.028]	-0.028 [-0.075, 0.017]	0.004 [-0.036, 0.040]

- Interpretation: the rankings are not equivalent, and their agreement comes from the wallets that EndorseRank scores zero; the O1 criterion is met (Table 5.1).

8.6.2 H2 / RQ2: Same-window alignment

- Statement: in the same window, each score agrees with the checks built from the edges it walks on.
- Evidence: the family means of every score with their CIs (Table 4.3); the same-window coefficients with the wallets outside a graph kept as isolated nodes (Table 4.5).
- Note: the agreement is partly built in, since the proxies come from the same events as the graphs. EndorseRank’s allowance and transfer coefficients have different ceilings (Section 4.3), so the claim says nothing about which of the two it follows more closely, and the isolated-node rule was tried after the results were known. None of these comparisons was fixed in advance. The paired differences between EndorseRank and AWP and their per-proxy results are in Appendix 8.5.

8.6.3 H3 / RQ3: Computational cost

- Statement: at the same n , a score’s computation time follows the number of edges in its graph, so EndorseRank, on the sparsest graph, runs fastest.
- Evidence: Table 4.1 (runtime, SD, memory, iterations, $|V|$, $|E|$); Tables 8.1 and 4.2; Figure 4.1.
- Note: the expanded-pool stages add sampled addresses but no edges, and hence no solver-graph nodes, beyond the matched-cohort graph (Section 4.2.1); the claim is confined to edge count.

8.6.4 RQ4: Temporal holdout

- Statement: with scores fixed at t_1 , the scores predict the new owner–spender approvals and new transfer senders that appear after t_1 and up to t_2 on the spender cohort ($n = 1,335$), with

the edge type deciding which label a score ranks well, but no PageRank predicts either label better than the raw t_1 degree of its own edge type.

- Evidence: Tables 4.10 and 4.11 (Section 4.6); the June–August window, Table 4.16; the spenders that had received a transfer before the freeze, Table 4.12.
- Qualification: both labels are the constructs as observed in a later period (new approvals, new senders), not credit or trading outcomes, and both are counts of arrivals. RQ4 replaced the proposal’s trading-success validation (Section 2.8). Apart from the increment of C-PR ($\lambda = 1$), none of these comparisons was fixed in advance. The differences between EndorseRank and AWP are in Appendix 8.5.

8.6.5 RQ5: Coupled operator

- Statement: under the rule of 14 September, a per-node convex mixture of the raw-amount allowance and transfer layers (C-PR) does not beat both layers walked alone, and the seeded ablation (S-PR) reproduces the transfer walk. The registered replication does not confirm that C-PR improves on the transfer walk alone, as it did in an exploratory comparison on the spring holdout: F1 and F3 hold, F2 fails. With AWP as the comparator, F2 fails by a wider margin.
- Evidence, rule of 14 September: Table 4.11 for the primary part, and Tables 4.3 and 4.4 for the secondary part (Section 4.3.1).
- Evidence, transfer walk alone: the exploratory spring contrast in Table 4.11, and the registered decision (F1–F3, with F4–F6 reported beside it) in Table 4.15 (Section 4.6.2).
- Evidence, AWP: the registered sensitivity analysis in Table 4.15, and the spring contrasts computed after the labels were known in Table 4.11.
- Evidence, cost: runtime rows of Table 4.1.
- Qualification: tested with fixed $\lambda \in \{0.25, 0.5, 0.75\}$, raw-amount layers, uniform teleportation, transition-level mixing and one chain. The rule of 14 September and the values of λ were set in the configuration before any coupled score was computed (Section 3.7); the rule

for the comparison with the transfer walk alone was written on 28 September 2026 for the June–August window and registered on 1 October 2026 together with two sensitivity analyses, one with AWP as the comparator, before the extraction (Section 3.7.2). The registration documents call the two layers walked alone EndorseRank and AWP.

8.6.6 Claims C1–C5

- C1: the runtime evidence listed under H3/RQ3; the profiling run and the edge ratio are in Section 4.2 and Appendix 8.3.3. The claim is specific to EndorseRank, since C-PR costs at least as much as AWP.
- C2: the same-window agreement of every score with the checks of its own edges (Table 4.3), partly built in and, for EndorseRank, set by the rule for wallets outside the graph (Table 4.5).
- C3: the rank divergence listed under H1/RQ1 (Table 4.5, Figure 4.3).
- C4: the comparison of every PageRank with the raw degree of its own edge type on the spender holdout (Tables 4.10 and 4.11), with the coverage check (Table 4.12), the exploratory trader run as its limit on the traders (Table 4.13) and the June–August window as its repetition (Table 4.16); the robustness checks and the sample-definition sensitivity are reported in Tables 4.6–4.8, Table 4.9 and Figure 4.4.
- C5: coupling does not beat its two layers walked alone, a bounded negative result, and the registered replication does not confirm its exploratory gain over the transfer walk alone (Table 4.15); see RQ5 above and Section 4.3.1.

8.6.7 Explicit non-claims

This research does *not* claim:

- that any of the scores is the better measure of reputation in general, or that AWP has become obsolete (the comparisons between EndorseRank and AWP in Appendix 8.5 follow the constructs and depend on coverage);

- that EndorseRank aligns more closely with the allowance family than with the transfer family (the two coefficients have different ceilings);
- that any PageRank of this thesis forecasts the future growth of its own construct better than the raw t_1 degree does;
- that allowance edges lead transfer edges on liquidation avoidance at the PageRank level, or that EndorseRank beats AWP there (the PageRank contrasts of Section 4.6.3 are not resolved);
- that any score measures creditworthiness, default risk or trading skill (the allowance signal is lower on the profitable share, and the data contain no default label);
- that the liquidation-avoidance signal belongs to the allowance edge and not to account type (almost every approval-receiving trader is a contract account, and account type is unresolved);
- that the value parameter $b = 1$, or EndorseRank's uniform restart, is the best choice (both were fixed by this study, the restart after comparing the two rules on these data);
- that C-PR improves on a walk over the transfers alone (the registered rule failed on new senders) or on AWP (it falls behind AWP on new senders in both windows);
- that no combination of the two edge types can beat the single-layer methods, only that the per-node convex mixture tested here, with fixed λ , does not;
- that runtime scales favorably when the edge population grows beyond the cohort graph;
- that, without replication, the results generalize beyond Arbitrum One and the study window;
- that on-chain scores substitute for collateral or legal identity systems.

8.6.8 Replication artifacts

1. A merged wallet-ranking table with the scores of EndorseRank, AWP, C-PR ($\lambda \in \{0, 0.25, 0.5, 0.75, 1\}$) and S-PR.
2. The machine-readable holdout summaries for the spender cohort and the matched traders, with the paired contrasts and primary-criterion verdict (committed under data/3-publish ed-results-for-thesis/spring-holdout-2026-03-to-2026-05/).

3. The machine-readable evaluation summary: authoritative same-window numbers, per-run timings and the bootstrap protocol (committed under `data/3-published-results-for-thesis/`).
4. The registered replication: the registration file and plan as committed on 1 October 2026, the extraction manifest with the hashes of both files, the evaluation summary with every coefficient, contrast and verdict, and the post hoc scores of EndorseRank on the same cohorts (committed under `config-copies/` and `registered-replication-2026-06-to-2026-08/` in `data/3-published-results-for-thesis/`).
5. The exported table fragments (`results/tables/`).
6. The exported vector figures (`Figures/generated/`).
7. The extracted logs and the tables built from them (`data/1-raw-blockchain-logs/` and `data/2-processed-tables-and-evaluations/`); files over 100 MiB are left out and can be rebuilt from the monthly files.
8. The archived extended-baseline matrices (`data/4-archived-extended-baselines/`, optional).

The tables and figures are rebuilt by running the evaluation pipeline on real data with the robustness sweeps, exporting the tables, and then redrawing the figures from the same artifacts (Appendix 8.1.1).

8.6.9 C6 — Outcomes built from neither edge type

- Statement: on the trader outcome built from neither edge type that bears on counterparty risk, later liquidation avoidance, the approval count at the freeze ranked traders above the transfer count in the registered window, after stratification by the number of closes, and in the spring window.
- Evidence: Tables 4.18 and 4.19 (Section 4.6.3); all six trader labels in both windows, Table 8.9; the protocol, Section 3.7.3; the plan and claim ledger, `analysis/neutral-label/P`

LAN.md, committed in 20e250f before the analysis ran, with its script and results in the same folder.

- **Qualification:** a pre-specified post hoc analysis of existing data that adds no research question. The claim is count-level, since the like-for-like PageRank pair points the same way but is not resolved. Most recipients are contract accounts, so account type is an unresolved confound, and the same signal is lower on the profitable share, so it ranks forced-liquidation avoidance, not trading success. Any use in screening would first need a registered test on new data with account type controlled (Section 6.3).

8.7 Ethics Clearance Certificate

8.7.1 Certificate

This study uses public blockchain data only and involves no human participants (Section 3.16). The ethics clearance certificate issued by the Unisa School of Computing / College of Agriculture and Environmental Sciences research ethics review committee is reproduced below.

Ethics clearance certificate Reference: [Unisa SoC/CAES ethics clearance ref: TBD] Date of issue: [TBD] (Scanned certificate to be inserted here.)
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